"Brilliant analysis."
—*Wall Street Journal*

"A triumph of critical thinking."
—*Washington Post*

DANIEL T. WILLINGHAM

WHY DON'T STUDENTS *Like* SCHOOL?

SECOND EDITION



A **COGNITIVE SCIENTIST** ANSWERS QUESTIONS ABOUT HOW **THE MIND** WORKS AND WHAT IT MEANS FOR **THE CLASSROOM**

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—Patrice M. Bain, EdS, educator and author of *Powerful Teaching*

Why Don't Students *Like* School?

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WILLINGHAM



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Second Edition

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For Trisha



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Introduction

Arguably the greatest mysteries in the universe lie in the three-pound mass of cells, approximately the consistency of oatmeal, that reside in the skull of each of us. It has even been suggested that the brain is so complex that our species is smart enough to fathom everything except what makes us so smart; that is, the brain is so cunningly designed for intelligence that it is too stupid to understand itself. We now know that is not true. The mind is at last yielding its secrets to persistent scientific investigation. We have learned more about how the mind works in the last 25 years than we did in the previous twenty-five hundred.

It would seem that greater knowledge of the mind would yield important benefits to education – after all, education is based on change in the minds of students, so surely understanding the student's cognitive equipment would make teaching easier or more effective. Yet the teachers I know don't believe they've seen much benefit from what psychologists call "the cognitive revolution." We all read stories in the newspaper about research breakthroughs in learning or problem solving, but it is not clear how each latest advance is supposed to change what a teacher does on Monday morning.

The gap between research and practice is understandable. When cognitive scientists study the mind, they intentionally isolate mental processes (for example, learning or attention) in the laboratory in order to make them easier to study. But mental processes are not isolated in the classroom. They all operate simultaneously, and they often interact in difficult-to-predict ways. To provide an obvious example, laboratory studies show that repetition helps learning, but any teacher knows that you can't take that finding and pop it into a classroom by, for example, having students repeat long-division

problems until they've mastered the process. Repetition is good for learning but terrible for motivation. With too much repetition, motivation plummets, students stop paying attention, and no learning takes place. The classroom application would not duplicate the laboratory result.

Why Don't Students Like School? began as a list of nine principles that are so fundamental to the mind's operation that they do *not* change as circumstances change. They are as true in the classroom as they are in the laboratory* and therefore can reliably be applied to classroom situations. Many of these principles likely won't surprise you: factual knowledge is important, practice is necessary, and so on.

What may surprise you are the implications for teaching that follow. You'll learn why it's more useful to view the human species as *bad* at thinking rather than as cognitively gifted. You'll discover that authors routinely write only a fraction of what they mean, which I'll argue implies very little for reading instruction but a great deal for the factual knowledge your students must gain. You'll explore why you remember the plot of *Star Wars* without even trying, and you'll learn how to harness that ease of learning for your classroom. You'll follow the brilliant mind of television doctor Gregory House as he solves a case, and you'll discover why you should *not* try to get your students to think like real scientists. You'll see how people like American politician Julian Castro and actress Scarlett Johansson have helped psychologists analyze the obvious truth that kids inherit their intelligence from their parents – only to find that it's not true after all, and you'll understand why it is so important that you communicate that fact to your students.

Why Don't Students Like School? ranges over a variety of subjects in pursuit of two goals that are straightforward but far from simple: to tell you how your students' minds work and to clarify how to use that knowledge to be a better teacher.

Note

*There actually were three other criteria for inclusion: (i) using versus ignoring a principle had to have a big impact on student learning; (ii) there had to be an enormous amount of data, not just a few studies, to support the principle; and (iii) the principle had to suggest classroom applications that teachers might not already know. The first edition offered nine principles; in this second edition I've added a tenth chapter on technology and education.

1

Why Don't Students Like School?

Question: Most of the teachers I know entered the profession because they loved school as children. They want to help their students feel the same excitement and passion for learning that they felt.

They are understandably dejected when they find that some of their pupils don't like school much, and that they, the teachers, have trouble inspiring them. Why is it difficult to make school enjoyable for students?

Answer: Contrary to popular belief, the brain is not designed for thinking. It's designed to save you from having to think, because the brain is actually not very good at thinking. Thinking is slow and unreliable. Nevertheless, people enjoy mental work if it is successful. People like to solve problems but not to work on unsolvable problems. If schoolwork is always just a bit too difficult (or too easy) for a student, it should be no surprise that she doesn't like school much. The cognitive principle that guides this chapter is:

> People are naturally curious, but we are not naturally good thinkers; unless the cognitive conditions are right, we will avoid thinking.

The implication of this principle is that teachers should reconsider how they encourage their students to think, in order to maximize the likelihood that students will get the pleasurable rush that comes from successful thought.

The Mind Is Not Designed for Thinking

What is the essence of being human? What sets us apart from other species? Many people would answer that it is our ability to reason – birds fly, fish swim, and humans think. (By thinking I mean solving problems, reasoning, reading something complex, or doing any mental work that requires some effort.) Shakespeare extolled our cognitive ability in Hamlet: "What a piece of work is man! How noble in reason!" Some three hundred years later Henry Ford more cynically observed, "Thinking is the hardest work there is, which is the probable reason why so few people engage in it"* (Figure 1.1).

Both Shakespeare and Ford had a point. Humans are good at certain types of reasoning, particularly in comparison to other animals, but we exercise those abilities infrequently. A cognitive scientist would add another observation: Humans don't think very often because our brains are designed not for thought but for the avoidance of thought.



FIGURE 1.1: Kanye West is one the most successful and respected songwriters and performers, as well as a highly successful businessman. But he has said, "I actually don't like thinking. I think people think I like to think a lot. And I don't. I do not like to think at all."[1] Source: © Getty Images/ Brad Barket.

Your brain has many capabilities, and thinking is not the one it does best. Your brain also supports the ability to see and to move, for example, and these functions operate much more efficiently and reliably than your ability to think. It's no accident that most of your brain's real estate is devoted to these activities. The extra brain power is needed because seeing is actually more difficult than playing chess or solving calculus problems.

You can appreciate the power of your visual system by comparing human abilities to those of computers. When it comes to math, science, and other traditional "thinking" tasks, machines beat people, no contest. Calculators that can perform simple calculations faster and more accurately than any human have been cheaply available for 40 years. With $50 you can buy chess software that can defeat more than 99% of the world's population. But we're still struggling to get a computer to drive a truck as well as a human. That's because computers can't see, especially not in complex, ever-changing environments like the one you face every time you drive. And in fact, the self-driving vehicles in development typically use radar, lasers, and other sensors to supplement information from visible light.

Robots are similarly limited in how they move. Humans are excellent at configuring our bodies for tasks, even if the configuration is unusual, such as when you twist your torso and contort your arm in an effort to dust behind books on a shelf. Robots are not very good at figuring out novel ways to move and are most useful in repetitive work such as spray painting automotive parts or moving pallets or boxes at an Amazon fulfillment center – jobs in which the objects to be grasped and the locations to move them are predictable. Tasks that you take for granted – for example, walking on a rocky shore where the footing is uncertain – are much more difficult than playing top-level chess (Figure 1.2).



FIGURE 1.2: Hollywood robots (left), like humans, can move in complex environments, but that's true only in the movies. Most real-life robots (right) move in predictable environments. Our ability to see and move is a remarkable cognitive feat. Source: Hollywood robots © Getty Images/Koichi Kamoshida; factory robots © Getty Images/Christopher Furlong.

Compared to your ability to see and move, thinking is slow, effortful, and uncertain. To get a feel for why I say this, try solving this problem:

> In an empty room are a candle, some matches, and a box of tacks. The goal is to have the lit candle about 5 ft off the ground. You've tried melting some of the wax on the bottom of the candle and sticking it to the wall, but that wasn't effective. How can you get the lit candle 5 ft off the ground without having to hold it there?[2]

Twenty minutes is the usual maximum time allowed, and few people are able to solve it by then, although once you hear the answer you will realize it's not especially tricky. You dump the tacks out of the box, tack the box to the wall, and use it as a platform for the candle.

This problem illustrates three properties of thinking. First, thinking is slow. Your visual system instantly takes in a complex scene. When you enter a friend's backyard you don't think to yourself, "Hmmm, there's some green stuff. Probably grass, but it could be some other ground cover – and what's that rough brown object sticking up there? A fence, perhaps?" You take in the whole scene – lawn, fence, flowerbeds, gazebo – at a glance. Your thinking system does not instantly calculate the answer to a problem the way your visual system immediately takes in a visual scene. Second, thinking is effortful; you don't have to try to see, but thinking takes concentration. You can perform other tasks while you are seeing, but you can't think about something else while you are working on a problem. Finally, thinking is uncertain. Your visual system seldom makes mistakes, and when it does you usually think you see something similar to what is actually out there – you're close, if not exactly right. Your thinking system might not even get you close. In fact, your thinking system may not produce an answer at all, which is what happens to most people when they try to solve the candle problem.

If we're all so bad at thinking, how does anyone get through the day? How do we find our way to work or spot a bargain at the grocery store? How does a teacher make the hundreds of decisions necessary to get through her day? The answer is that when we can get away with it, we don't think. Instead we rely on memory. Most of the problems we face are ones we've solved before, so we just do what we've done in the past. For example, suppose that next week a friend gives you the candle problem. You would immediately say, "Oh, right. I've heard this one. You tack the box to the wall." Just as your visual system takes in a scene and, without any effort on your part, tells you what is in the environment, so too your memory system immediately and effortlessly recognizes that you've heard the problem before and provides the answer. You may think you have a terrible memory, and it's true that your memory system is not as reliable as your visual or movement system – sometimes you forget, sometimes you think you remember when you don't – but your memory system is much more reliable than your thinking system, and it provides answers quickly and with little effort.

We normally think of memory as storing personal events (memories of my wedding) and facts (the seat of the Coptic Orthodox Church is in Egypt). Our memory also stores strategies to guide what we should do: where to turn when driving home, how to handle a minor dispute when monitoring recess, what to do when a pot on the stove starts to boil over (Figure 1.3). For the vast majority of decisions we make,



FIGURE 1.3: Your memory system operates so quickly and effortlessly that you seldom notice it working. For example, your memory has stored away information about what things look like (Gandhi's face) and how to manipulate objects (turn the left faucet for hot water, the right for cold) and strategies for dealing with problems you've encountered before (such as a pot boiling over). Source: Gandhi © Getty Images/Dinodia Photos; faucet © Shutterstock/RVillalon; pot © Shutterstock/Andrey_Popov.

we don't stop to consider what we might do, reason about it, anticipate possible consequences, and so on. For example, when I decide to make spaghetti for dinner, I don't scour the Internet for recipes, weighing each for taste, nutritional value, ease of preparation, cost of ingredients, visual appeal, and so on – I just make spaghetti sauce the way I usually do. As two psychologists put it, "Most of the time what we do is what we do most of the time."[3] When you feel as though you are "on autopilot," even if you're doing something rather complex, such as driving home from school, it's because you are using memory to guide your behavior. Using memory doesn't require much of your attention, so you are free to daydream, even as you're stopping at red lights, passing cars, watching for pedestrians, and so on.

Of course you could make each decision with care and thought. When someone encourages you to "think outside the box" that's usually what he means – don't go on autopilot, don't do what you (or others) have always done. Consider what life would be like if you always strove to think outside the box. Suppose you approached every task afresh and tried to see all of its possibilities, even daily tasks like chopping an onion, entering your workplace, or sending a text message. The novelty might be fun for a while, but life would soon be exhausting (Figure 1.4).

You may have experienced something similar when traveling, especially if you've traveled where you don't speak the local language. Everything is unfamiliar and even trivial actions demand lots of thought. For example, buying a soft drink from a vendor requires figuring out the flavors from the exotic packaging, trying to communicate with the vendor, working through which coin or bill to use, and so on. That's



FIGURE 1.4: "Thinking outside the box" for a mundane task like selecting bread at the supermarket would probably not be worth the mental effort. Source: © Shutterstock/B Brown.

one reason that traveling is so tiring: all of the trivial actions that at home could be made on autopilot require your full attention.

So far I've described two ways in which your brain is set up to save you from having to think. First, some of the most important functions (for example, vision and movement) don't require thought: you don't have to reason about what you see; you just immediately know what's out in the world. Second, you are biased to use memory to guide your actions rather than to think. But your brain doesn't leave it there; it is capable of changing in order to save you from having to think. If you repeat the same thought-demanding task again and again, it will eventually become automatic; your brain will change so that you can complete the task without thinking about it. I discuss this process in more detail in Chapter 5, but a familiar example here will illustrate what I mean. You can probably recall that learning to drive a car was mentally very demanding. I remember focusing on how hard to depress the accelerator, when to apply the brake as I approached a red light, how far to turn the steering wheel to execute a turn, when to check my mirrors, and so forth. I didn't even listen to music while I drove, for fear of being distracted. With practice, however, the process of driving became automatic, and now I don't need to think about those small-scale bits of driving any more than I need to think about how to walk. I can drive while simultaneously chatting with friends, gesturing with one hand, and eating French fries – an impressive cognitive feat, if not very attractive to watch.[†] Thus a task that initially takes a great deal of thought becomes, with practice, a task that requires little thought.

The implications for education sound rather grim. If people are bad at thinking and try to avoid it, what does that say about students' attitudes toward school? Fortunately, the story doesn't end with people stubbornly refusing to think. Despite the fact that we're not that good at it, we actually like to think. We are naturally curious, and we look for opportunities to engage in certain types of thought. But because thinking is so hard, the conditions have to be right for this curiosity to thrive, or we quit thinking rather readily. The next section explains when we like to think and when we don't.

People Are Naturally Curious, But Curiosity Is Fragile

Even though the brain is not set up for very efficient thinking, people actually enjoy mental activity, at least in some circumstances. We have hobbies like solving crossword puzzles or scrutinizing maps. We watch information-packed documentaries. We pursue careers – such as teaching – that offer greater mental challenge than competing careers, even if the pay is lower. Not only are we willing to think, we intentionally seek out situations that demand thought.

Solving problems brings pleasure. When I say "problem solving" in this book, I mean any cognitive work that succeeds; it might be understanding a difficult passage of prose, planning a garden, or sizing up an investment opportunity. There is a sense of satisfaction, of fulfillment, in successful thinking. Neuroscientists have found overlap between the brain areas that are important in learning and those that are important in perception of pleasure, and many neuroscientists suspect that the two systems are related. Rats in a maze learn better when rewarded with cheese. When you solve a problem or satisfy your curiosity, your brain may reward itself with a small burst of a naturally occurring chemical in the brain's pleasure system. Even though the neurochemistry is not completely understood, it seems undeniable that people take pleasure in solving problems.

It's notable too that the pleasure is in the solving of the problem. Working on a problem with no sense that you're making progress is not pleasurable. In fact, it's frustrating. Then too, there's not great pleasure in simply knowing the answer. I told you the solution to the candle problem; did you get any fun out of it? Think how much more fun it would have been if you had solved it yourself – in fact, the problem would have seemed more clever, just as a joke that you get is funnier than a joke that has to be explained. Even if someone doesn't tell you the answer to a problem, once you've had too many hints you lose the sense that you've solved the problem, and getting the answer doesn't bring the same mental snap of satisfaction.

Mental work appeals to us because it offers the opportunity for that pleasant feeling when it succeeds. But not all types of thinking are equally attractive. People choose to work crossword puzzles but not

algebra problems. A biography of Taylor Swift is more likely to sell well than a biography of Keats. What characterizes the mental activity that people enjoy (Figure 1.5)?

The answer that most people would give may seem obvious: "I think crossword puzzles are fun and Taytay is cool, but math is boring and so is Keats." In other words, it's the content that matters. We're curious about some stuff but not about other stuff. Certainly that's the way people describe our own interests – "I'm a stamp collector" or "I'm into medieval symphonic music." But I don't think content drives interest. We've all attended a lecture or watched a video (perhaps only after being prevailed upon to do so) about a subject we thought we weren't interested in, only to find ourselves fascinated. And it's easy to find yourself bored even when you usually like the topic. I'll never forget my eagerness for the day my middle school teacher was to talk about sex. As a teenage boy in a staid 1970s suburban culture, I fizzed with anticipation of any talk about sex, anytime, anywhere. But when the big day came, my friends and I were bored senseless. It's not that the teacher talked about flowers and pollination – he really did talk about human sexuality – but somehow it was still dull. I actually wish I could remember how he did it; boring a bunch of hormonal teenagers with a sex talk is quite a feat.

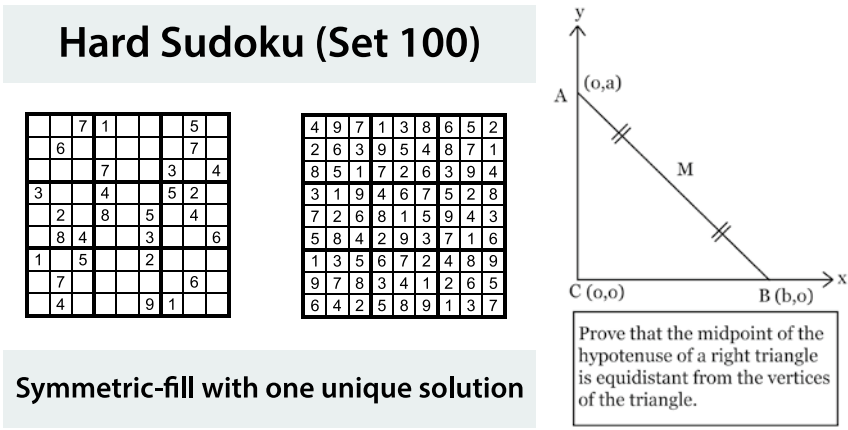


FIGURE 1.5: Why are many people fascinated by problems like the one shown on the left, but very few people willingly work on problems like the one on the right? Source: Sudoku © Shutterstock/Heather Wallace; geometry © Anne Carlyle Lindsay.

I once made this point to a group of teachers when talking about motivation and cognition. About five minutes into the talk I presented a slide depicting the model of motivation shown in Figure 1.6. I didn't prepare the audience for the slide in any way; I just displayed it and started describing it. After about 15 seconds I stopped and said to the audience, "Anyone who is still listening to me, please raise your hand." One person did. The other 59 were also attending voluntarily; it was a topic in which they were presumably interested, and the talk had only just started – but in 15 seconds their minds were somewhere else. To be clear, I'm not blaming them; the content of a problem – whether it's about sex or human motivation – may be sufficient to prompt your interest, but it won't maintain it.

So, if content is not enough to keep your attention, when does curiosity have staying power? The answer seems to lie in our judgment of how much we are likely to learn. Curiosity is maintained when we think we'll learn a lot.

That judgment – will I learn? – is closely related to our perception of the difficulty of the problem. If it's that little burst of pleasure from solving a problem that we look forward to, then there's no point in working on a problem that is too easy – there'll be no pleasure when it's solved because it didn't feel like much of a problem in the first place. Then too, when you size up a problem as very difficult, you are judging that you're unlikely to solve it, and are therefore unlikely to get the satisfaction that comes with the solution. A crossword puzzle

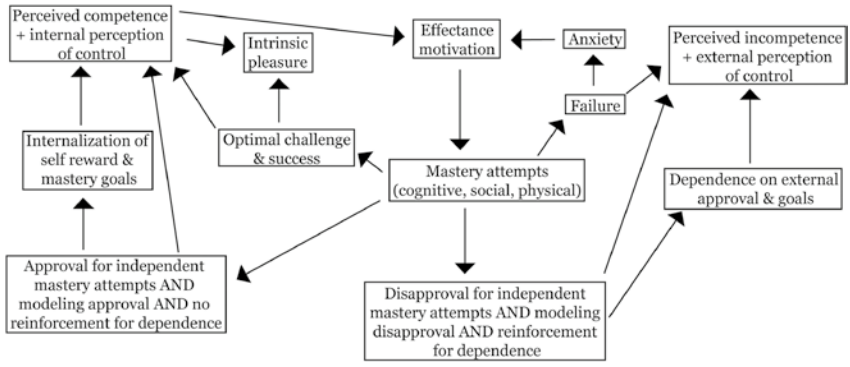


FIGURE 1.6: A difficult-to-understand figure that will bore most people unless it is adequately introduced. Source: © Anne Carlyle Lindsay.

that is too easy is just mindless work: you fill in the squares, scarcely thinking about it, and there's no gratification, even though you're getting all the answers. But you're unlikely to work long at a crossword puzzle that's too difficult. You know you'll solve very little of it, so it will just be frustrating. The slide in Figure 1.6 is too detailed to be absorbed with minimal introduction; my audience quickly concluded that it was overwhelming and mentally checked out of my talk.

To summarize, I've said that thinking is slow, effortful, and uncertain. Nevertheless, people like to think – or more properly, we like to think if we judge that the mental work will pay off with the pleasurable feeling we get when we learn something new. So there is no inconsistency in claiming that people avoid thought and in claiming that people are naturally curious – curiosity prompts people to explore new ideas and problems, but when we do, we quickly evaluate how much mental work it will take to solve the problem or understand what's described. If it's too much work or too little, we stop thinking about the problem if we can.

This analysis of the sorts of mental work that people seek out or avoid also provides one answer to why more students don't like school. Working on problems that are of the right level of difficulty is rewarding, but working on problems that are too easy or too difficult is unpleasant. Students can't opt out of these problems the way adults often can. If the student routinely gets work that is a bit too difficult, it's little wonder that he doesn't care much for school. I wouldn't want to work on the Sunday New York Times crossword puzzle for several hours each day.

So what's the solution? Give the student easier work? You could, but of course you'd have to be careful not to make it so easy that the student would be bored. And anyway, wouldn't it be better to boost the student's ability a little bit? Instead of making the work easier, is it possible to make thinking easier?

How Thinking Works

Understanding a bit about how thinking happens will help you understand what makes thinking hard. That will in turn help you understand how to make thinking easier for your students, and therefore help them enjoy school more.

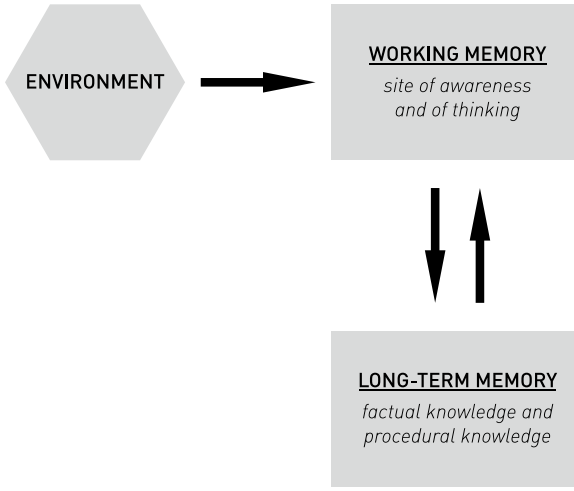


FIGURE 1.7: Just about the simplest model of the mind possible. Source: © Greg Culley.

Let's begin with a very simple model of the mind. On the left of Figure 1.7 is the environment, full of things to see and hear, problems to be solved, and so on. On the right is one component of your mind that scientists call working memory. For the moment, consider it to be synonymous with consciousness; it holds the stuff you're thinking about. The arrow from the environment to working memory shows that working memory is the part of your mind where you are aware of what is around you: the sight of a shaft of light falling onto a dusty table, the sound of a dog barking in the distance, and so forth. Of course you can also be aware of things that are not currently in the environment; for example, you can recall the sound of your mother's voice, even if she's not in the room (or indeed no longer living). Long-term memory is the vast storehouse in which you maintain your factual knowledge of the world: that leopards have spots, that your favorite flavor of ice cream is chocolate, that your three-year-old surprised you yesterday by mentioning kumquats, and so on. Factual knowledge can be abstract; for example, it would include the idea that triangles are closed figures with three sides and your knowledge of what a dog generally looks like. All of the information in long-term memory resides outside of awareness. It lies quietly until it is needed and then enters working memory and so enters consciousness. For example, if I asked you, "What color is a polar bear?" you would say, "white" almost immediately. That information was in long-term memory 30 seconds ago, but you weren't aware of it until I posed the question that made it relevant to ongoing thought, whereupon it entered working memory.

Thinking occurs when you combine information (from the environment and long-term memory) in new ways. That combining happens in working memory. To get a feel for this process, read the problem depicted in Figure 1.8 and try to solve it. (The point is not so much to solve it as to experience what is meant by thinking and working memory.)

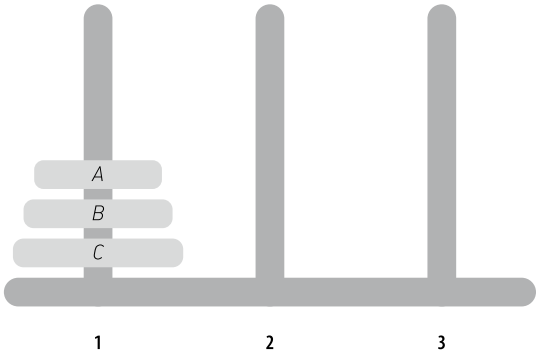


FIGURE 1.8: The figure depicts a playing board with three pegs. There are three rings of decreasing size on the leftmost peg. The goal is to move all three rings from the leftmost peg to the rightmost peg. There are just two rules about how you can move rings: you can move only one ring at a time, and you can't place a larger ring on top of a smaller ring. Source: © Greg Culley.

With some diligence you might be able to solve this problem,‡ but the real point is to feel what it's like to have working memory absorbed by the problem. You begin by taking information from the environment – the rules and the configuration of the game board – and then imagine moving the discs to try to reach the goal. Within working memory you must maintain your current state in the puzzle – where the discs are – and imagine and evaluate potential moves. At the same time you have to remember the rules regarding which moves are legal, as shown in Figure 1.9.

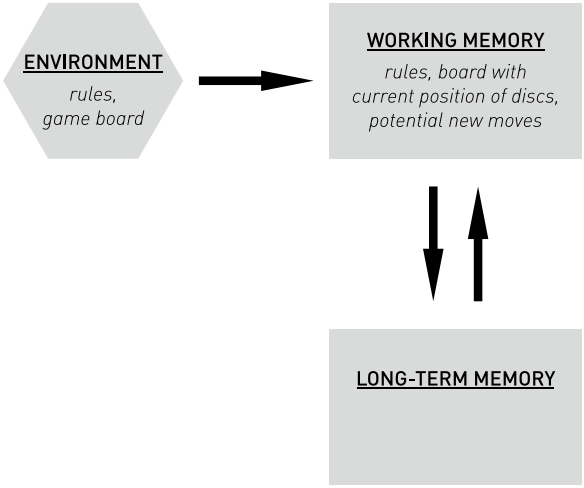


FIGURE 1.9: A depiction of your mind when you're working on the puzzle shown in Figure 1.8. Source: © Greg Culley.

The description of thinking makes it clear that knowing how to combine and rearrange ideas in working memory is essential to successful thinking. For example, in the discs and pegs problem, how do you know where to move the discs? If you hadn't seen the problem before, you probably felt like you were pretty much guessing. You didn't have any information in long-term memory to guide you, as depicted in Figure 1.9. But if you have had experience with this particular type of problem, then you likely have information in long-term memory about how to solve it, even if the information is not foolproof. For example, try to work this math problem in your head:

$$18 \times 7$$

You know just what to do for this problem. The sequence of your mental processes was likely something close to this:

1. Multiply 8 and 7.
2. Retrieve the fact that $8 \times 7 = 56$ from long-term memory.
3. Remember that the 6 is part of the solution, then carry the 5.
4. Multiply 7 and 1.
5. Retrieve the fact that $7 \times 1 = 7$ from long-term memory.
6. Add the carried 5 to the 7.
7. Retrieve the fact that $5 + 7 = 12$ from long-term memory.
8. Put the 12 down, append the 6.
9. The answer is 126.

Your long-term memory contains not only factual information, such as the color of polar bears and the value of 8×7, but it also contains what we'll call procedural knowledge, which is your knowledge of the mental procedures necessary to execute tasks. If thinking is combining information in working memory, then procedural knowledge is a list of what to combine and when – it's like a recipe to accomplish a particular type of thought. You might have stored procedures for the steps needed to calculate the area of a triangle, or to duplicate a computer file using Windows, or to drive from your home to your workplace.

It's pretty obvious that having the appropriate procedure stored in long-term memory helps a great deal when we're thinking. That's why it was easy to solve the math problem and hard to solve the discs-and-pegs problem. But how about factual knowledge? Does that help you think as well? It does, in several different ways, which are discussed in Chapter 2. For now, note that solving the math problem required the retrieval of factual information, such as the fact that $8 \times 7 = 56$. I've said that thinking entails combining information in working memory. Often the information provided in the environment is not sufficient to solve a problem, and you need to supplement it with information from long-term memory.

There's a final necessity for thinking, which is best understood through an example. Have a look at this problem:

> In the inns of certain Himalayan villages is practiced a refined tea ceremony. The ceremony involves a host and exactly two guests, neither more nor less. When his guests have arrived and seated themselves at his table, the host performs three services for them. These services are listed in the order of the nobility the Himalayans attribute to them: stoking the fire, fanning the flames, and pouring the tea. During the ceremony, any of those present may ask another, "Honored Sir, may I perform this onerous task for you?" However, a person may request of another only the least noble of the tasks that the other is performing. Furthermore, if a person is performing any tasks, then he may not request a task that is nobler than the least noble task he is already performing. Custom requires that by the time the tea ceremony is over, all the tasks will have been transferred from the host to the most senior of the guests. How can this be accomplished?[4]

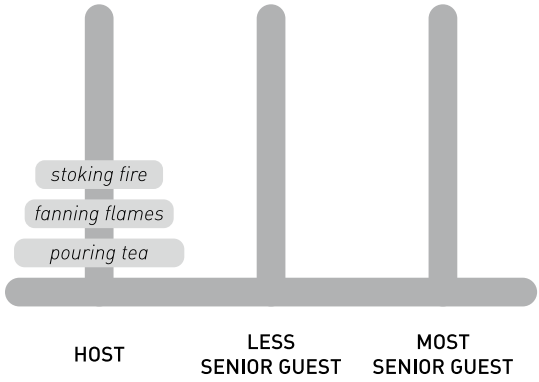


FIGURE 1.10: The tea-ceremony problem, depicted to show the analogy to the discs-and-pegs problem. Source: © Greg Culley.

Your first thought upon reading this problem was likely "Huh?" You could probably tell that you'd have to read it several times just to understand it, let alone begin working on the solution. It seemed overwhelming because you did not have sufficient space in working memory to hold all of the aspects of the problem. Working memory has limited space, so thinking becomes increasingly difficult as working memory gets crowded.

The tea-ceremony problem is actually the same as the discs-and-pegs problem presented in Figure 1.8. The host and two guests are like the three pegs, and the tasks are the three discs to be moved among them, as shown in Figure 1.10. (The fact that very few people see this analogy and its importance for education is taken up in Chapter 4.)

This version of the problem seems much harder because some parts of the problem that are laid out in Figure 1.8 must be juggled in your head in this new version. For example, Figure 1.8 provides a picture of the pegs that you can use to help maintain a mental image of the discs as you consider moves, whereas the tea ceremony version provides no such support. And in the tea version, the description of the rules that govern moves is longer and therefore occupies so much space in working memory that it's difficult to plan a solution.

Aristotle said, "The pleasures arising from thinking and learning will make us think and learn all the more."[5] You've seen that this view is too optimistic. It's *successful* learning that's pleasurable and that will keep students coming back for more. We've seen that one of

the factors in successful learning is having the right information in long-term memory. In the next chapter, we examine that need more closely.

Summary

People's minds are not especially well suited to thinking; thinking is slow, effortful, and uncertain. For this reason, deliberate thinking does not guide people's behavior in most situations. Rather, we rely on our memories, following courses of action that we have taken before. Nevertheless, we find successful thinking pleasurable. We like solving problems, understanding new ideas, and so forth. Thus, we will seek out opportunities to think, but we are selective in doing so; we choose problems that pose some challenge but that seem likely to be solvable, because these are the problems that lead to feelings of pleasure and satisfaction. For problems to be solved, the thinker needs adequate information from the environment, room in working memory, and the required facts and procedures in long-term memory.

Implications for the Classroom

Let's turn now to the question that opened this chapter: Why don't students like school, or more accurately, why don't more of them like it? Any teacher knows that there are lots of reasons that a student might or might not enjoy school. (My wife loved it, but primarily for social reasons.) From a cognitive perspective, an important factor is whether or not a student consistently experiences the pleasurable rush of learning something new, of solving a problem. What can teachers do to ensure that each student gets that pleasure?

Be Sure That There Are Problems to Be Solved

By problem I don't necessarily mean a question addressed to the class by the teacher, or a mathematical puzzle. I mean cognitive work that poses moderate challenge, including such activities as understanding a poem or thinking of novel uses for recyclable materials. This sort

of cognitive work is of course the main stuff of teaching – we want our students to think. But without some attention, a lesson plan can become a long string of teacher explanations, with little opportunity for students to solve problems. So scan each lesson plan with an eye toward the cognitive work that students will be doing. How often does such work occur? Is it intermixed with cognitive breaks? Is it real cognitive work that can lead to the feeling of discovery and not just retrieval from memory? (Think especially about questions posed during whole-class instruction – research shows it's easy for teachers to slip into a pattern of asking lots of fact-retrieval questions.) When you have identified the challenges, consider whether they are open to negative outcomes such as students failing to understand what they are to do, or students being unlikely to solve the problem, or students simply trying to guess what you would like them to say or do.

Respect Students' Cognitive Limits

When trying to develop effective mental challenges for your students, bear in mind the cognitive limitations discussed in this chapter. For example, suppose you began a history lesson with a question: "You've read that 35 nations united to expel Iraq from Kuwait in the First Gulf War, the largest coalition since World War II. Why do you suppose so many nations joined?" Do your students have the necessary background knowledge in memory to consider this question? What do they know about the relationship of Iraq and neighboring countries that ended up joining the coalition prior to the war? Do they know about how Iraq brought their dispute with Kuwait to the Arab League before the invasion? Do they know about the significance of oil to the world economy and the forecast economic consequences of the invasion? Could they generate reasonable alternative courses of action for those countries leading the invasion? If they lack the appropriate background knowledge, the question you pose will quickly be judged as "boring." If students lack the background knowledge to engage with a problem, save it for another time when they have that knowledge.

Equally important is the limit on working memory. Remember that people can keep only so much information in mind at once, as

you experienced when you read the tea-ceremony version of the discs-and-pegs problem. Overloads of working memory are caused by such things as multistep instructions, lists of unconnected facts, chains of logic more than two or three steps long, and the application of a just-learned concept to new material (unless the concept is quite simple). The solution to working memory overloads is straightforward: slow the pace, and use memory aids such as writing on the whiteboard to save students from keeping too much information in working memory.

Clarifying the Problems to Be Solved

How can you make the problem interesting? A common strategy is to try to make the material "relevant" to students. This strategy sometimes works well, but it's hard to use for some material, and your struggle to make it relevant to students is usually obvious. Another difficulty is that a teacher's class may include two football fans, a doll collector, a NASCAR enthusiast, a horseback riding competitor – you get the idea. Mentioning a popular singer in the course of a history lesson may give the class a giggle, but it won't do much more than that. I have emphasized that our curiosity is provoked when we perceive a problem that we believe we can solve. What is the question that will engage students and make them want to know the answer?

One way to view schoolwork is as a series of answers. We want students to know Boyle's law, or three causes of World War I, or why Poe's raven kept saying, "Nevermore." Sometimes I think that we, as teachers, are so eager to get to the answers that we do not devote sufficient time to developing the question. That probably happens because the question is obvious to us. But of course it's not obvious to students, and as the information in this chapter indicates, it's the question that piques people's interest. Being told an answer doesn't do anything for you. You may have noted that I could have organized this book around principles of cognitive psychology. Instead I organized it around questions that I thought teachers would find interesting.

When you plan a lesson, you start with the information you want students to know by its end. As a next step, consider what the key

question for that lesson might be and how you can frame that question so it will have the right level of difficulty to engage your students and so you will respect your students' cognitive limitations.

Reconsider When to Puzzle Students

Teachers often seek to draw students into a lesson by presenting a problem that we believe will interest the students. For example, asking, "Why is there a law that you have to go to school?" could introduce the process by which laws are passed. Another strategy is to conduct a demonstration or present a fact that we think students will find surprising. In either case, the goal is to puzzle students, to make them curious. This is a useful technique, but it's worth considering whether these strategies might be used not only at the beginning of a lesson but also after the basic concepts have been learned. For example, a classic science demonstration is to put a burning piece of paper in a milk bottle and then put a boiled egg over the bottle's opening. After the paper burns, the egg appears to be sucked into the bottle. Students will no doubt be astonished, but if they don't know the principle behind it, the demonstration is like a magic trick – it's a momentary thrill, but their curiosity to understand may not be long-lasting. Another strategy would be to conduct the demonstration after students know that warm air expands and cooling air contracts, potentially forming a vacuum. Every fact or demonstration that would puzzle students before they have the right background knowledge has the potential to be an experience that will puzzle students momentarily and then lead to the pleasure of problem solving. It is worth thinking about when to use a marvelous device like the egg-in-the-bottle trick.

Accept and Act on Variation in Student Preparation

As I describe in Chapter 8, I don't accept that some students are "just not very bright" and ought to be tracked into less demanding classes. But it's naïve to pretend that all students come to your class equally prepared to excel; they have had different preparations, as well as different levels of support at home, and they will therefore differ in their abilities, as well as their perception of themselves as students. Those factors, in turn, affect therefore their persistence and

resilience to failure. If that's true, and if what I've said in this chapter is true, it is self-defeating to give all of your students the same work. The less capable students will find it too difficult and will struggle against their brain's bias to mentally walk away from schoolwork. To the extent that you can, it's smart, I think, to assign work to individuals or groups of students that is appropriate to their current level of competence. Naturally, you will want to do this in a sensitive way, minimizing the extent to which some students will perceive themselves as behind others. But the fact is that they are behind the others, and giving them work that is beyond them is unlikely to help them catch up and is likely to make them fall still further behind.

Change the Pace

We all inevitably lose the attention of our students, and as this chapter has described, it's likely to happen if they feel somewhat confused. They will mentally check out. The good news is that it's relatively easy to get them back. Change grabs attention, as you no doubt know. When there's a bang outside your classroom, every head turns to the windows. When you change topics, start a new activity, or in some other way show that you are shifting gears, virtually every student's attention will come back to you, and you will have a new chance to engage them. So plan shifts and monitor your class's attention to see whether you need to make them more often or less frequently.

Keep a Diary

The core idea presented in this chapter is that solving a problem gives people pleasure, but the problem must be easy enough to be solved yet difficult enough to take some mental effort. Finding this sweet spot of difficulty is not easy. Your experience in the classroom is your best guide – whatever works, do again; whatever doesn't, discard. But don't expect that you will really remember how well a lesson plan worked a year later. Whether a lesson goes brilliantly well or down in flames, it feels at the time that we'll never forget what happened; but the ravages of memory can surprise us, so write it down. Even if it's just a quick scratch on a sticky note, try to make a habit of recording your success in gauging the level of difficulty in the problems you pose for your students.

One of the factors that contributes to successful thought is the amount and quality of information in long-term memory. In Chapter 2 I elaborate on the importance of background knowledge to effective thinking.

Notes

*A more eloquent version comes from eighteenth-century British painter Sir Joshua Reynolds: "There is no expedient to which a man will not resort to avoid the real labor of thinking."

†And in fact people's driving is more impaired than they realize when they multitask. Don't try this at home!

‡If you couldn't solve it, here's a solution. As you can see, the rings are marked A, B, and C, and the pegs are marked 1, 2, and 3. The solution is A3, B2, A2, C3, A1, B3, A3.

Further Reading

Less Technical

Coalition for Psychology in Schools and Education. (2015). Top 20 principles from psychology for preK-12 teaching and learning. https://www.apa.org/ed/schools/teaching-learning/top-twenty-principles.pdf (accessed 13 July 2020). A brief, easy introduction to applying knowledge from psychology to classrooms, and a free download.

Csikszentmihalyi, M. (1990). *Flow: The Psychology of Optimal Experience*. New York: Harper Perennial. The author describes the ultimate state of interest, when one is completely absorbed in what one is doing, to the point that time itself stops. The book does not tell you how to enter this state, but it is an interesting read in its own right.

Didau, D., & Rose, N. (2016). *What Every Teacher Needs to Know About Psychology*. Melton, UK: John Catt. Brief chapters on a broad sweep of topics, including evolution, creativity, motivation, and more, very much from a teacher's perspective.

National Academies of Sciences, Engineering, and Medicine. (2018). *How People Learn II: Learners, Contexts, and Cultures*. National Academies Press. https://www.nap.edu/catalog/24783/how-people-learn-ii-learners-contexts-and-cultures (accessed 13 July 2020). Intended as an overview of cognition applied to education, this book sometimes overreaches, moving into peripheral topics, but it's worth the read, and it's a free download.

Willingham, D.T. (2019a). The high price of multitasking. *New York Times* (15 July), p. A21. https://www.nytimes.com/2019/07/14/opinion/multitasking-brain.html. A quick review of evidence that we cannot cope with working memory overload as well as we think we can.

Willingham, D. T. (2019b). Why aren't we curious about the things we want to be curious about? *New York Times* (20 October), p. SR9. https://www.nytimes.com/2019/10/18/opinion/sunday/curiosity-brain.html. This op-ed considers what triggers curiosity and how we might make ourselves curious about things that align with our long-term interests.

More Technical

Baddeley, A. (2018). *Exploring Working Memory: Selected Works of Alan Baddely*. Oxford, UK: Routledge. A retrospective of the most important articles by Alan Baddeley, commonly considered the central figure in the development of working memory theory over the last 50 years.

Berridge, K. C., & Kringelbach, M. L. (2015). Pleasure systems in the brain. *Neuron*, 86(3), 646–664. Reviews evidence that various types of pleasure we feel—from using addictive drugs, listening to music, experiencing romantic love, or eating delicious food—all have a common anatomic basis in the brain.

Kidd, C., & Hayden, B. Y. (2015). The psychology and neuroscience of curiosity. *Neuron*, 88(3), 449–460. An overview of contemporary theories of curiosity, focusing on the idea that curiosity evolved to ensure that animals, including humans, learn about their environment. We are maximally curious when we think the environment offers the greatest opportunity to learn.

Long, N. M., Kuhl, B. A., & Chun, M. M. (2018). Memory and attention. *Stevens' Handbook of Experimental Psychology and Cognitive Neuroscience*, 1–37. Hoboken, NJ: Wiley. You probably didn't need to be convinced that paying attention is a prerequisite for learning. This chapter offers much more detail and, as you'd expect, some caveats and complications.

Willingham, D. T., & Riener, C. (2019). *Cognition: The Thinking Animal*, 4. Cambridge, UK: Cambridge University Press. This is a college-level textbook on cognitive psychology that can serve as an introduction to the field. It assumes no background, but it is a textbook, so although it is thorough, it might be a bit more detailed than you want.

Discussion Questions

1. We like to think only when we believe we'll be successful. If you want to engage in thinking more often, how then might you change your environment so that you more often encounter the right sort of mental challenge? Or what might you say to yourself to get you to try more often?
2. When are your students on autopilot? It's easy to say there's value in trying to go off autopilot and *think* more often, but what are the obstacles to doing so? Can you think of ways of picking problems that they currently solve on autopilot that actually seem promising to think about?
3. Just how much reward does a student need for the work of thinking? There's no firm answer to this question, and we'd certainly guess that it would vary by age and would vary within a classroom, probably based on students' concept of themselves as students and some inherent aspect of each student's persistence. Nevertheless it's worth considering: on average, how often would you like students to feel the pleasure of success? Equally important, how do you know when they do? If they solve a problem, are you sure they feel success? If not, is there anything you can do to prompt that feeling?
4. Think of some assignments that your students consistently enjoy, and view them through the cognitive-work-that-succeeds lens. Does that work share any common characteristics?

5. I suggested that students don't fully understand or appreciate the question that the to-be-learned content is meant to answer. How often do you think that applies to your context? How easy or difficulty might it be to get students to both understand the question at stake and to engage with it?
6. I was a little casual in the implications section when I said that it's straightforward to deal with working memory overload. I said you should slow down and break things into smaller chunks, which is true, as far as it goes. What's trickier to deal with are the differences among students in what makes them feel overloaded. What can be done about that?

2

How Can I Teach Students the Skills They Need When Standardized Tests Require Only Facts?

Question: Much has been written about fact learning, and much of it has been negative. The narrow-minded schoolmaster demanding that students parrot facts they do not understand has become a cliché of American education, although the stereotype is neither new nor especially American – Dickens used it in *Hard Times*, published in 1854. Concern has increased since the early 2000s as the importance of standardized tests have increased; they typically offer little opportunity for students to analyze, synthesize, or critique and instead demand the recall of isolated facts. Many teachers feel that time for teaching skills is crowded out by preparation for standardized tests. Just how useful or useless is fact learning?

Answer: There is no doubt that having students memorize lists of isolated facts is not enriching. It is also true (though less often appreciated) that trying to teach students skills such as analysis or synthesis in the absence of factual knowledge is impossible. Research from cognitive science has shown that the sorts of skills that teachers want for students – such as the ability to analyze and to think critically – require extensive factual knowledge. The cognitive principle that guides this chapter is:

> Thinking skills depend on factual knowledge.

The implication is that students must have the opportunity to learn factual knowledge. That should be done in the context of practicing thinking skills and will ideally start in preschool and even before.

> There is a great danger in the present day lest science-teaching should degenerate into the accumulation of disconnected facts and unexplained formulae, which burden the memory without cultivating the understanding.[1]
>
> *–J. D. Everett, writing in 1873*

When I was a freshman in college a guy down the hall from me had a poster depicting Einstein and a quotation from the brilliant, frowsy-haired physicist: "Imagination is more important than knowledge." I thought this was very deep. Perhaps I was anticipating what I might say to my parents if my grades were poor: "Sure, I got Cs, but I have imagination! And according to Einstein . . ."

Some 40 years later teachers have a different reason to be wary and weary of "knowledge." Governments seek some assurance of quality in their schools. That usually means that children will be tested, and those tests are often heavy on multiple-choice questions that require straightforward recall of facts because those questions are the easiest to score. Here are two examples from my home state of Virginia, one from the eighth-grade science test and one from the third-grade history test.[2]

Sea anemones are poisonous. However, the clownfish has developed an outer layer of mucus that provides protection from the stinging cells of the sea anemone. The mucus is best described as

A. *an adaptation*
B. a relationship
C. an energy requirement
D. a social hierarchy

George Washington Carver invented new ways to use

A. paper
B. electricity

C. *peanuts*
D. bananas

It's easy to see why a teacher, parent, or student would protest that knowing the answer to a lot of these questions doesn't prove that one really *knows* science or history. We want our students to think, not simply to memorize. When someone shows evidence of thinking critically, we consider her smart and well educated. When someone spouts facts without context, we consider her boring and a show-off.

That said, there are obvious cases in which everyone would agree that factual knowledge is necessary. When a speaker uses unfamiliar vocabulary, you may not understand what he means. For example, if a friend texted you that she thought your daughter was dating a "yegg," you'd certainly want to know the definition of the word (Figure 2.1). Similarly, you may know all of the vocabulary words but lack the conceptual knowledge to knit the words together into something comprehensible. For example, a recent copy of the technical journal *Science* contained an article titled "Measuring magnetic field texture in correlated electron systems under extreme conditions."[3] I know what each of these words means, but I don't know enough about magnetic fields to understand what's meant by their texture, let alone why one might want to measure it.

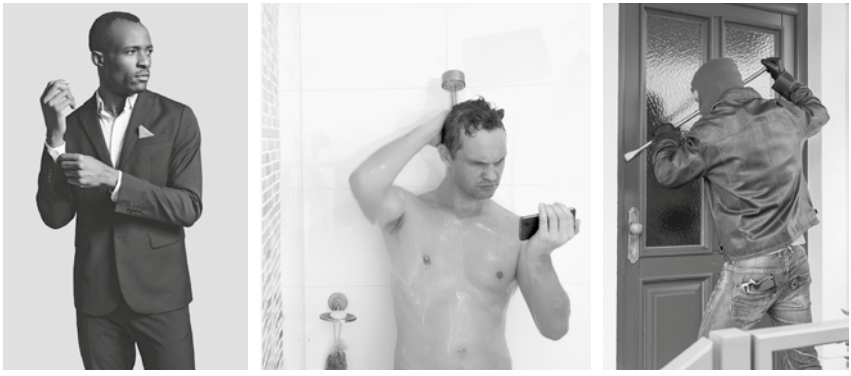


FIGURE 2.1: If someone said your daughter is dating a *yegg,* you'd certainly want to know whether the word meant "nice-looking fellow," "Internet addict," or "burglar." Source: Nice fellow © Shutterstock/G-Stock Studio; addict © Shutterstock/Kopytin Georgy; burglar © Shutterstock/Lisa_S.

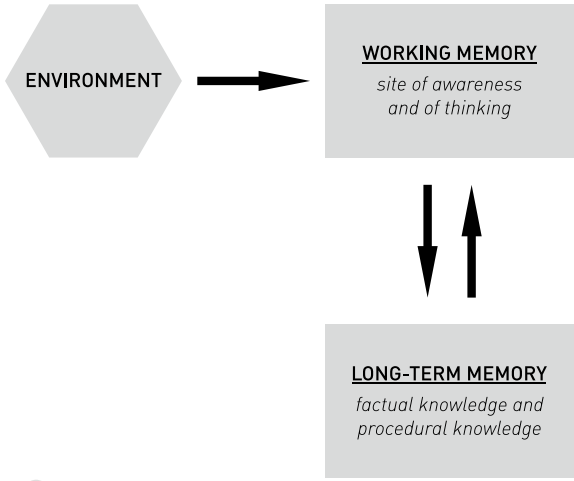


FIGURE 2.2: Our simple figure of the mind.
Source: © Greg Culley.

The necessity of background knowledge for comprehension is pretty obvious, at least as I've described it so far. You could summarize this view by noting that *to think* is a transitive verb. You need something to think *about*. But you could counter (and I've heard the argument often) that you don't need to have this information memorized – you can always look it up. Recall the figure of the mind in Chapter 1 (Figure 2.2).

I defined *thinking* as combining information in new ways. The information can come from long-term memory – facts you've memorized – or from the environment. In today's world, why memorize anything? You can find any factual information you need in seconds via the Internet – including the definition of *yegg*. Then too, things change so quickly that half of the information you commit to memory will be out of date in five years – or so the argument goes. Perhaps instead of learning facts, it's better to practice critical thinking, to have students work at *evaluating* all the information available on the Internet rather than trying to commit some small part of it to memory.

In this chapter I show that this argument is false. (In Chapter 9 we'll examine looking things up.) Data from the last 40 years lead to a conclusion that is not scientifically challengeable: thinking well requires knowing facts, and that's true not simply because you need something to think *about*. The very processes that teachers care about most – critical thinking processes such as reasoning and problem solving – are intimately intertwined with factual knowledge that is stored in long-term memory (not just found in the environment).

It's hard for many people to conceive of thinking processes as intertwined with knowledge. Most people believe that thinking processes

are akin to the functions of a calculator. A calculator has available a set of procedures (addition, multiplication, and so on) that can manipulate numbers, and these procedures can be applied to *any set of numbers*. The data (the numbers) and the operations that manipulate the data are separate. Thus, if you learn a new thinking operation (for example, how to critically analyze historical documents), that operation should be applicable to all historical documents, just as a fancier calculator that computes sines can do so for all numbers.

But the human mind does not work that way. When we learn to think critically about, say, the geopolitics of Europe resulting from World War II, it does not mean we can also think critically about a chess game or about the current situation in the Middle East or even about geopolitics of Europe resulting from the French revolution. Critical thinking processes are tied to background knowledge (although they become less so when we gain significant experience, as I describe in Chapter 6). The conclusion from this work in cognitive science is straightforward: we must ensure that students acquire background knowledge in parallel with practicing critical thinking skills (Figure 2.3).

In this chapter I describe how cognitive scientists know that thinking skills and knowledge are bound together.



FIGURE 2.3: The human mind is more like this astrolabe than like a calculator. Among other functions, it can be used to measure the altitude of a celestial body above the horizon, but that function is *not* independent of knowledge. An astrolabe's knowledge representations – the numbers you see engraved, and the way the parts move in relation to one another – are part of its construction. Of course, you need processing – you need to know *how* to use the astrolabe – but the knowledge representations are also essential. Source: © Getty Images/Science & Society Picture Library.

Knowledge Is Essential to Reading Comprehension

Background knowledge helps you understand what someone is saying or writing. In the last section I gave a couple of rather obvious examples: if a vocabulary word (for example, *yegg*) or a concept (for example, *magnetic field texture*) is missing from your long-term memory, you'll likely be confused. But the need for background knowledge is deeper than the need for definitions.

Suppose a sentence contains two ideas – call them A and B. Even if you know the vocabulary and you understand both A and B, you still might need background knowledge to understand the sentence. For example, suppose you read the following sentence in a novel:

> *"I shouldn't use my new barbecue when the boss comes to dinner," Mark said.*

You could say that idea A is Mark using his new barbecue, and idea B is that he shouldn't do it when his boss comes to dinner. Obviously, you're meant to understand more than just A and B; the writer expects you to understand the relationship of these two ideas, that B (boss coming) is the cause of A (shouldn't use new barbecue). But the writer has omitted two pieces of information that would help you bridge A and B: that people often make mistakes the first time they use a new appliance and that Mark would like to impress his boss. Putting these facts together would help you understand that Mark is afraid he'll ruin the food the first time he uses his new barbecue, and he doesn't want that to be the meal he serves to his boss.

Reading comprehension depends on combining the ideas in a passage, not just comprehending each idea on its own. But writers seldom provide the reader all of the information needed to bridge ideas. Writers assume the reader has that knowledge in long-term memory. In the example just given, the writer assumed that the reader would know the relevant facts about new appliances and about bosses.

Why do writers leave gaps? Don't they run the risk that the reader *won't* have the right background knowledge and so will be confused? That's a risk, but writers can't include all the factual details. If

they did, prose would be impossibly long and tedious. For example, imagine reading this:

> *"I shouldn't use my new barbecue when the boss comes to dinner," Mark said. Then he added, "Let me make clear that by* boss *I mean my immediate supervisor. Not the president of the company, nor any of the other supervisors intervening. And I'm using* dinner *in the local vernacular, not to mean 'noontime meal,' as it is used in some parts of the United States. And when I said* barbecue*, I was speaking imprecisely, because I really meant grill, because* barbecue *generally refers to slower roasting, whereas I plan to cook over high heat. Anyway, my concern, of course, is that my inexperience with the barbecue (that is, grill) will lead to inferior food, and I hope to impress the boss. Because I believe impressing my boss will improve my standing at the company. And bad food is not impressive."*

We've all known someone who talks that way (and we try to avoid him or her), but not many; most writers and speakers feel safe omitting some information.

How do writers and speakers decide what to omit? It depends on their audience. Suppose you were sitting in your office, working at your laptop, when someone came in and asked, "What are you doing?" How would you respond?

It depends on who asked. If it was bring-your-child-to-work day and the questioner were a colleague's two-year-old you might say, "I'm typing on a computer." But that would be a ridiculous answer for an adult. Why? Because you should assume that the adult knows you're typing. A more appropriate response might be, "I'm filling out a form."

Thus, we calibrate our answers, providing more or less (or different) information depending on our judgment of what the other person knows, thereby deciding what we can safely leave out and what needs to be explained (Figure 2.4).*

What happens when the knowledge is missing? Suppose you read the following sentence:

> *I believed him when he said he had a lake house, until he said it's only 40 feet from the water at high tide.*

If you're like me, you're confused. When I read a similar passage, my mother-in-law later explained to me that lakes don't have appreciable tides. I didn't have that bit of background knowledge that the author



FIGURE 2.4: Consider the difference in how you would describe a planned lesson to your spouse or friend, versus a fellow teacher. Source: © Shutterstock/Monkey Business Images.

assumed I had, so I didn't understand the passage.

So, background knowledge in the form of vocabulary helps us understand a single idea (call it A), but background knowledge is also often necessary to understand the connection between two ideas (A and B). In fact, writers often present multiple ideas in rapid succession – A, B, C, D, E, and F – expecting that the reader will knit them together into a coherent whole. Have a look at this sentence from chapter 35 of *Moby-Dick:*

> *Now, it was plainly a labor of love for Captain Sleet to describe, as he does, all the little detailed conveniences of his crow's-nest; but though he so enlarges upon many of these, and though he treats us to a very scientific account of his experiments in this crow's-nest, with a small compass he kept there for the purpose of counteracting the errors resulting from what is called the "local attraction" of all binnacle magnets; an error ascribable to the horizontal vicinity of the iron in the ship's planks, and in the Glacier's case, perhaps, to there having been so many broken-down blacksmiths among her crew; I say, that though the Captain is very discreet and scientific here, yet, for all his learned "binnacle deviations," "azimuth compass observations," and "approximate errors," he knows very well, Captain Sleet, that he was not so much immersed in those profound magnetic meditations, as to fail being attracted occasionally towards that well replenished little case-bottle, so nicely tucked in on one side of his crow's-nest, within easy reach of his hand.*[4]

Why is this sentence so hard to understand? You run out of mental room. It has a lot of ideas in it, and because it's one sentence, you try to keep them all in mind at once and to relate them to one another. But there are so many ideas, you can't keep them all in

mind simultaneously. To use the terminology from Chapter 1, you don't have sufficient capacity in working memory. In some situations, background knowledge can help with this problem.

To understand why, let's start with a demonstration. Read the following list of letters once, then cover the list and see how many letters you can remember.

X C N
N P H
D L O
L G M
T F A
Q X

Okay, how many could you remember? If you're like most people, the answer would perhaps be seven. Now try the same task with this list:

X
C N N
P H D
L O L
G M T
F A Q
X

You probably got many more letters correct with this second list, and you no doubt noticed that it's easier because the letters form acronyms that are familiar. But did you notice that the first and second lists are the same? I just changed the spacing to make the acronyms more apparent in the second list.

This is a working memory task. You'll remember from Chapter 1 that working memory is the part of your mind in which you combine and manipulate information – think of it as synonymous with consciousness. Working memory has a limited capacity, so you can't maintain in your working memory all of the letters from list one. But you can for list two. Why? Because the amount of space in working

memory doesn't depend on the number of letters; it depends on the number of meaningful objects. If you can remember seven individual letters, you can remember seven (or just about seven) meaningful acronyms or words. The letters *F, A,* and *Q* together count as only one object because combined they are meaningful to you.

The phenomenon of mentally tying together separate pieces of information from the environment is called *chunking*. The advantage is obvious: you can keep more stuff in working memory if it can be chunked. The trick, however, is that chunking works only when you have applicable factual knowledge in long-term memory. You will see *CNN* as meaningful only if you already know what CNN is. In the first list, one of the three-letter groups was *NPH*. If you're a neurologist you may have treated this group as a chunk, because *NPH* stands for a brain disorder, normal pressure hydrocephalus. If you don't have that knowledge in your long-term memory, you would not treat *NPH* as a chunk. This basic effect – using background knowledge to group things in working memory – doesn't work only for letters. It works for anything. Bridge players do it with hands of cards, dancing experts do it with dance moves, and so forth.

So factual knowledge in long-term memory allows chunking, and chunking increases space in working memory. What does the ability to chunk have to do with reading comprehension? Well, I was saying before that if you read ideas A, B, C, D, E, and F, you would need to relate them to one another in order to comprehend their meaning. That's a lot of stuff to keep in working memory. But suppose you could *chunk* A through F into a single idea? Comprehension would be much easier. For example, consider this passage:

> *Ashburn hit a ground ball to Wirtz, the shortstop, who threw it to Dark, the second baseman. Dark stepped on the bag, forcing out Cremin, who was running from first, and then threw it to Anderson, the first baseman. Ashburn failed to beat the throw.*

If you're like me this passage is hard to comprehend. There are a number of individual actions, and they are hard to tie together. But for someone who knows about baseball, it's a familiar pattern, like *CNN*. The sentences describe a frequently encountered sequence of events called a double play.

A number of studies have shown that people understand what they read much better if they already have some background knowledge about the subject. Part of the reason is chunking. A clever study on this point was conducted with junior high school students.[5] Half were good readers and half were poor readers, according to standard reading tests. The researchers asked the students to read a story that described half an inning of a baseball game. As they read, the students were periodically stopped and asked to show that they understood what was happening in the story by using a model of a baseball field and players. The interesting thing about this study was that some of the students knew a lot about baseball and some knew just a little. (The researchers made sure that everyone could comprehend individual actions, for example, what happened when a player got a double.) The dramatic finding, shown in Figure 2.5, was

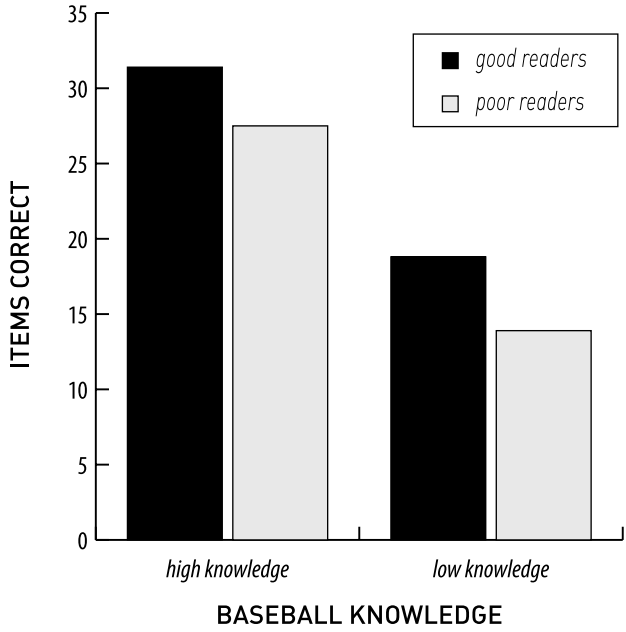


FIGURE 2.5: Results from a study of reading. As you would predict, the good readers (shaded bars) understood more than the poor readers (unshaded bars), but this effect is modest compared to the effect of knowledge. The people who knew a lot about baseball (leftmost columns) understood the passage much better than the people who didn't know a lot, regardless of whether they were "good" or "poor" readers, as measured by standard reading tests. Source: Based on data from "Effect of prior knowledge on good and poor readers' memory of text" by D. R. Recht and L. Leslie in *Journal of Educational Psychology* 80: 16-20. Copyright © 1988 by the American Psychological Association.

that the students' knowledge of baseball determined how much they understood of the story. Whether they were "good readers" or "bad readers" didn't matter nearly as much as what they knew.

Thus, background knowledge allows chunking, which makes more room in working memory, which makes it easier to relate ideas, and therefore to comprehend.

Background knowledge also clarifies details that would otherwise be ambiguous and confusing. In one experiment illustrating this effect,[6] subjects read the following passage:

> *The procedure is actually quite simple. First, you arrange items into different groups. Of course one pile may be sufficient depending on how much there is to do. If you have to go somewhere else due to lack of facilities, that is the next step; otherwise, you are pretty well set. It is important not to overdo things. That is, it is better to do too few things at once than too many.*

The passage went on in this vein, vague and meandering, and therefore very difficult to understand. The problem is not that you're missing vocabulary. Rather, everything seems really ambiguous. Not surprisingly, people couldn't remember much of this paragraph when asked about it later. They remembered much more, however, if they had first been told that the passage's title is "Washing Clothes." Have another look at the passage now that you know the title. The title tells you which background knowledge is relevant, and you recruit that knowledge to clarify ambiguities. For example, "Arrange items into groups" is interpreted as sorting darks, bright colors, and whites. This experiment indicates that we don't take in new information in a vacuum. We interpret new things we read in light of other information we already have on the topic. In this case, the title, "Washing Clothes," tells the reader which background knowledge to use to understand the passage. Naturally, most of what we read is not so vague, and we usually know which background information is relevant. Thus, when we read an ambiguous sentence, we seamlessly use background knowledge to interpret it, and likely don't even notice the potential ambiguities (Figure 2.6).

I've listed four ways that background knowledge is important to reading comprehension: (i) it provides vocabulary; (ii) it allows you



FIGURE 2.6: Most people would not even notice that the word "it" is ambiguous because background knowledge tells you what ought to be picked up and what need not be. Source: Man holding dog @istock/Aleksandr Zotov; dog waste station © Daniel Willingham; Photoshop © nyretouch/Nihal Organ.

to bridge logical gaps that writers leave; (iii) it allows chunking, which increases room in working memory and thereby makes it easier to tie ideas together; and (iv) it guides the interpretation of ambiguous sentences. There are in fact other ways that background knowledge helps reading, but these are some of the highlights.

Some observers believe that the importance of knowledge to reading is the main driver of the fourth-grade slump. If you're unfamiliar with that term, it refers to the fact that students from underprivileged homes often read at grade level through the third grade, but then suddenly in the fourth grade they fall behind. The interpretation is that reading instruction through third grade focuses mostly on decoding – figuring out how to sound out words using the printed symbols – so that's what reading tests emphasize. By the time the fourth grade rolls around, most students are good decoders, so reading tests start to emphasize *comprehension*. As described here, comprehension depends on background knowledge, and that's where kids from privileged homes have an edge. They come to school

with a bigger vocabulary and more knowledge about the world than underprivileged kids. And because knowing things makes it easier to learn new things (as described in the next section), the gap between privileged and underprivileged kids widens.

Background Knowledge Is Necessary for Cognitive Skills

Not only does background knowledge make you a better reader, it also is necessary to be a good thinker. The processes we most hope to engender in our students – thinking critically and logically – are not possible without background knowledge.

First, you should know that much of the time when we see someone apparently engaged in logical thinking, he or she is actually engaged in memory retrieval. As I described in Chapter 1, memory is the cognitive process of *first* resort. When faced with a problem, you will first search for a solution in memory, and if you find one, you will very likely use it. Doing so is easy and fairly likely to be effective; you probably remember the solution to a problem because it worked the last time, not because it failed. To appreciate this effect, first try a problem for which you *don't* have relevant background knowledge, such as the one shown in Figure 2.7.[7]

The problem depicted in Figure 2.7 is more difficult than it first appears. In fact, only about 15% or 20% of college students get it right. The correct answer is to turn over the A card and the 3 card. Most people get A – it's clear that if there is not an even number on the other side, the rule has been violated. Many people incorrectly think they need to turn over the 2 card. The rule does not, however, say what must be

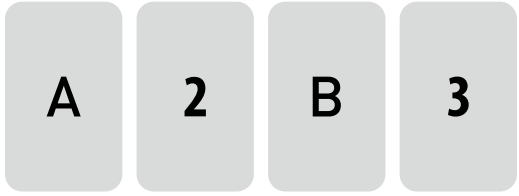


FIGURE 2.7: Each card has a letter on one side and a digit on the other. There is a rule: If there is a vowel on one side, there must be an even number on the other side. Your job is to verify whether this rule is met for this set of four cards and to turn over the minimum number of cards necessary to do so. Which cards would you turn over? Source: © Greg Culley.

on the other side of a card with an even number. The 3 card must be flipped because if there is a vowel on the other side, the rule has been violated.



FIGURE 2.8: You are to imagine that you are a doorman in a bar. Each card represents a patron, with the person's age on one side and their drink on the other. You are to enforce this rule: If you're drinking beer, then you must be 21 or over. Your job is to verify whether this rule is met for this set of four people. You should turn over the minimum number of cards necessary to do so. Which cards would you turn over? Source: © Greg Culley.

Now let's look at another version of the problem, shown in Figure 2.8.[8]

If you're like most people, this problem is relatively easy: you flip the beer card (to be sure this patron is over 21) and you flip the 17 card (to be sure this kid isn't drinking beer). Yet logically the 17 card has the same role in the problem that the 3 card did in the previous version, and it was the 3 card that everyone missed. Why is it so much easier this time? One reason (but not the only one) is that the topic is familiar. You have background knowledge about the idea of a drinking age, and you know what's involved in enforcing that rule. Thus you don't need to reason logically. You have experience with the problem and you remember what to do rather than needing to reason it out.

In fact, people draw on memory to solve problems more often than you might expect. For example, it appears that much of the difference among the world's best chess players is *not* their ability to reason about the game or to plan the best move; rather, it is their memory for game positions. Here's a key finding that led to that conclusion. Chess matches are timed, with each player getting an hour to complete his or her moves in the game. There are also so-called blitz tournaments in which players get just five minutes to make all of their moves in a match (Figure 2.9). It's no surprise that everyone plays a little bit worse in a blitz tournament. What's surprising is that the best players are still the best, the nearly best are still nearly best, and so on.† This finding indicates that whatever makes the best players better than everyone else is still present in blitz tournaments;



FIGURE 2.9: A device used to time a chess match. The black hand on each clock counts down the minutes remaining. After making a move, the player pushes the button above his clock, which stops it and causes his opponent's clock to restart. Players set identical amounts of time to elapse on each clock – just five minutes in a blitz tournament – representing the total time the player can take for all moves in the game. The flag near the 12 on each clock is pushed aside by the black hand as it approaches 12. When the flag falls, the player has exceeded his allotted time, and so forfeits the match. Source: © Shutterstock/Gavran333.

whatever gives them their edge is *not* a process that takes a lot of time, because if it were they would have lost their edge in blitz tournaments.

It seems that it is memory that creates the differences among the best players. When tournament-level chess players select a move, they first size up the game, deciding which part of the board is the most critical, the location of weak spots in their defense and that of their opponents, and so on. This process relies on the player's memory for similar board positions and, because it's a memory process, it takes very little time, perhaps a few seconds. This assessment greatly narrows the possible moves the player might make. Only then does the player engage slower reasoning processes to select the best among several candidate moves. This is why top players are still quite good even in a blitz tournament. Most of the heavy lifting is done by memory, a process that takes very little time. On the basis of this and other research, psychologists estimate that top chess players may have fifty thousand board positions in long-term memory. Thus background knowledge is decisive even in chess, which we might think is the prototypical game of reasoning.

That's not to say that all problems are solved by comparing them to cases you've seen in the past. You do, of course, sometimes reason,

but even then background knowledge can help. Earlier in this chapter I discussed chunking, the process that allows us to think of individual items as a single unit (for example, when *C, N,* and *N* become *CNN*), thereby creating more room in working memory. I emphasized that in reading, the extra mental space afforded by chunking can be used to relate the meaning of sentences to one another. This extra space is also useful when reasoning.

Here's an example. Do you have a friend who can walk into someone else's kitchen and rapidly produce a nice dinner from whatever food is around, usually to the astonishment of whoever's kitchen it is? When your friend looks in a cupboard, she doesn't see ingredients, she sees recipes. She draws on extensive background knowledge about food and cooking. For example, have a look at the pantry in Figure 2.10.

A food expert will have the background knowledge to see many possibilities here, for example, crushing pecans with the Stove Top stuffing to bread the chicken, or using the tea to scent the rice. The necessary ingredients will then become a chunk in working memory, so the expert will have room in working memory to devote to other aspects of planning, for example, to consider other dishes that might complement this one, or to begin to plan the steps of cooking.

Chunking applies to classroom activities as well. For example, take two algebra students. One is still a little shaky on the distributive property, the other knows it cold. When the first student is trying to solve a problem and sees a(b + c), he's unsure whether that's the same as ab + c, or



FIGURE 2.10: Suppose you were at a friend's house and she asked you to make dinner with some chicken and whatever else you could find. What would you do? Source: © Shutterstock/ Darryl Brooks.

b + ac, or ab + ac. So he stops working on the problem and substitutes small numbers into a(b + c) to be sure he's got it right. The second student recognizes a(b + c) as a chunk and doesn't need to stop and occupy working memory with this subcomponent of the problem. Clearly the second student is more likely to complete the problem successfully.

There is a final point to be made about knowledge and thinking skills. Much of what experts tell us they do in the course of thinking about their field *requires* background knowledge, even if it's not described that way. Let's take science as an example. We could tell students a lot about how scientists think, and they could memorize those bits of advice. For example, we could tell students that when interpreting the results of an experiment, scientists are especially interested in anomalous (that is, unexpected) outcomes. Unexpected outcomes indicate that their knowledge is incomplete and that this experiment contains hidden seeds of new knowledge. But for results to be unexpected, you must have an expectation! An expectation about the outcome would be based on your knowledge of the field. Most or all of what we tell students about scientific thinking strategies is impossible to use without appropriate background knowledge. Or as the well-known geologist, Herbert Harold Read, said, "The best geologist is the one who has seen the most rocks" (Figure 2.11).



FIGURE 2.11: Judge Sharon Newcomb shown here inspecting a Miniature Pinscher at the Westminster Kennel Club Dog Show has acquired expertise, in part, through experience with thousands of dogs. To become a American Kennel Club show judge requires not just straightforward demonstration of knowledge (passing tests on dog anatomy and show ring procedure) and relevant experience (assistant-judging at local events); prospective judges must have been around a great many dogs, evidenced by at least 12 years experience with the breed they hope to judge. Source: © Getty Images/Sarah Stier.

The same holds true for history, language arts, music, and so on. Generalizations that we can offer to students about how to think

and reason successfully in the field may *look* like they don't require background knowledge, but when you consider how to apply them, they often do.

Factual Knowledge Improves Your Memory

When it comes to knowledge, those who have more gain more. Many experiments have confirmed the benefit of background knowledge to memory using the same basic method. The researchers bring into the laboratory some people who have some expertise in a field (for example, football or dance or electronic circuitry) and some who do not. Everyone reads a story or a brief article. The material is simple enough that the people without expertise have no difficulty understanding it; that is, they can tell you what each sentence means. But the next day the people with background knowledge remember substantially more of the material than the people who do not have background knowledge.

You might think this effect is really due to attention. If I'm a basketball fan, I'll enjoy reading about basketball and will pay close attention, whereas if I'm not a fan, reading about basketball will bore me. But other studies have actually *created* experts. The researchers had people learn either a lot or just a little about subjects that were new to them (for example, Broadway musicals). Then they had them read other, new facts about the subject, and they found that the "experts" (those who had earlier learned a lot of facts about the subject) learned new facts more quickly and easily than the "novices" (who had earlier learned just a few facts about the subject).[9]

Why is it easier to remember material if you already know something about the topic? I've already said that if you know more about a particular topic, you can better understand new information about that topic; for example, people who know about baseball *understand* a baseball story better than people who don't. We remember much better if something has meaning. That generalization is discussed and refined in the next chapter, but to get a sense of this effect, read each of the following two brief paragraphs:

Motor learning is the change in capacity to perform skilled movements that achieve behavioral goals in the environment. A fundamental and unresolved question in neuroscience is whether there is a separate neural system for representing learned sequential motor responses. Defining that system with brain imaging and other methods requires a careful description of what specifically is being learned for a given sequencing task.	A chiffon cake replaces butter – the traditional fat in cakes – with oil. A fundamental and unresolved question in entertaining is when to make a butter cake and when to make a chiffon cake. Answering this question with expert tasting panels and other methods requires a careful description of what characteristics are desired for a cake.

The paragraph on the left is taken from a technical research article.[10] Each sentence is likely comprehensible, and if you take your time, you can see how they are connected: The first sentence provides a definition, the second sentence poses a problem, and the third states that a description of the thing under study (skills) is necessary before the problem can be addressed. I wrote the paragraph on the right to parallel the motor-skill paragraph. Sentence by sentence, the structure is the same. Which do you think you will remember better tomorrow?

The paragraph on the right is easier to understand (and therefore will be better remembered) because you can tie it to things you already know. Your experience tells you that a good cake tastes buttery, not oily, so the interest value of the fact that some are made with oil is apparent. Similarly, when the final sentence refers to "what characteristics are desired for a cake," you can imagine what those characteristics might be – fluffiness, moistness, and so on. Note that these effects aren't about comprehension; you can comprehend the paragraph on the left pretty well despite a lack of background knowledge. But some richness, some feeling of depth to the comprehension is missing. That's because when you have background knowledge your mind connects the material you're reading with what you already know about the topic, even if you're not aware that it's happening.

It's those connections that will help you remember the paragraph tomorrow. Remembering things is all about *cues* to memory. We dredge up memories when we think of things that are related to what we're trying to remember. Thus, if I said, "Try to remember that paragraph you read yesterday," you'd say to yourself, "Right, it was about cakes," and automatically (and perhaps outside of awareness) information about cakes would start to flit through your mind – they are baked . . . they are frosted . . . you have them at birthday parties . . . they are made with flour and eggs and butter . . . and suddenly, that background knowledge (that cakes are made with butter) provides a toehold for remembering the paragraph: "Right, it was about a cake that uses oil instead of butter." It's adding these lines from the paragraph to your background knowledge that makes the paragraph seem both better understood and easier to remember. The motor-skills paragraph, alas, is marooned, removed from any background knowledge, and so is more difficult to remember later.

This final effect of background knowledge – that having factual knowledge in long-term memory makes it easier to acquire still more factual knowledge – is worth contemplating for a moment. It means that the amount of information you retain depends on what you already have. So, if you have more than I do, you retain more than I do, which means you gain more than me. To make the idea concrete (but the numbers manageable), suppose you have ten thousand facts in your memory but I have only nine thousand. Let's say we each remember a percentage of new stuff, and that percentage is based on what's already in our memories. You remember 10% of the new facts you hear, but because I have less knowledge in long-term memory, I remember only 9% of new facts. Table 2.1 shows how many facts each of us has in long-term memory over the course of 10 months, assuming we're each exposed to five hundred new facts each month.

By the end of 10 months, the gap between us has widened from 1000 facts to 1043 facts. Because people who have more in long-term memory learn more easily, the gap is only going to get wider. The only way I could catch up is to make sure I am exposed to more facts than you are. In a school context, I have some catching up to do, but it's very difficult because you are pulling away from me at an ever-increasing speed.

TABLE 2.1: A demonstration that, when it comes to knowledge, the rich get richer.

Months	Facts in your memory	% of new facts you remember	Facts in my memory	% of new facts I remember
1	10 000	10.000	9000	9.000
2	10 050	10.050	9045	9.045
3	10 100	10.100	9090	9.090
4	10 151	10.151	9135	9.135
5	10 202	10.202	9181	9.181
6	10 253	10.253	9227	9.227
7	10 304	10.304	9273	9.273
8	10 356	10.356	9319	9.319
9	10 408	10.408	9366	9.366
10	10 460	10.460	9413	9.413

I have of course made up all of the numbers in the foregoing example, but we know that the basics are correct – the rich get richer. We also know where the riches lie. If you want to be exposed to new vocabulary and new ideas, the places to go are books, magazines, and newspapers. Videos, gaming, social media, and texting friends offer less exposure to new ideas and vocabulary.

I began this chapter with a quotation from Einstein: "Imagination is more important than knowledge." I hope you are now persuaded that Einstein was wrong. Knowledge is more important, because it's a prerequisite for imagination, or at least for the sort of imagination that leads to problem solving, decision making, and creativity. Other great minds have made similar comments that denigrate the importance of knowledge, as shown in Table 2.2.

I don't know why some great thinkers (who undoubtedly knew many facts) took delight in denigrating schools, often depicting them as factories for the useless memorization of information. I suppose we are to take these remarks as ironic, or at least as interesting, but I for one don't need brilliant, highly capable minds telling me (and my children) that it's useless to know things. As I've shown

TABLE 2.2: Quotations from great thinkers denigrating the importance of factual knowledge.

Education is what survives when what has been learned has been forgotten.	Psychologist B. F. Skinner
I have never let my schooling interfere with my education.	Writer Mark Twain
Nothing in education is so astonishing as the amount of ignorance it accumulates in the form of inert facts.	Writer Henry Brooks Adams
Your learning is useless to you till you have lost your textbooks, burned your lecture notes, and forgotten the minutiae you learned by heart for the examination.	Philosopher Alfred North Whitehead
We are shut up in schools and college recitation rooms for 10 or 15 years, and come out at last with a bellyful of words and do not know a thing.	Poet Ralph Waldo Emerson

in this chapter, the cognitive processes that are most esteemed – logical thinking, problem solving, and the like – are intertwined with knowledge. It is certainly true that facts without the skills to use them are of little value. It is equally true that one cannot deploy thinking skills effectively without factual knowledge.

As an alternative to the quotations in Table 2.2, I offer a Spanish proverb that emphasizes the importance of experience and, by inference, knowledge: *Mas sabe El Diablo por viejo que por Diablo*. Roughly translated: "The Devil is not wise because he's the Devil. The Devil is wise because he's *old*."

Summary

Few today set as a goal for schooling that students should acquire knowledge simply for the sake of knowing things. Our goal is that students learn to think. In addition, it's natural to describe thinking as a *process* and therefore to conclude that knowledge doesn't matter much. But we've seen many ways that thinking well relies on knowledge. Knowledge allows you to bridge the gaps writers leave in prose and guides your interpretation when sentences are

ambiguous. Knowledge is essential for chunking, the process that saves room in working memory, and so facilitates reasoning. Sometimes knowledge *substitutes* for reasoning, when you simply recall a previous problem solution, and other times, knowledge is required to deploy a thinking skill, as when a scientist judges that an experimental result is anomalous. Rather than thinking of knowledge as data that might be plugged into thinking processes, it's better to think of knowledge and thinking as intertwined.

Implications for the Classroom

If factual knowledge makes cognitive processes work better, the obvious implication is that we must help children learn background knowledge. How can we ensure that this happens?

Which Knowledge Should They Learn?

We might well ask ourselves, *Which knowledge should students learn?* This question often becomes politically charged rather quickly. When we start to specify what must be taught and what can be omitted, it appears that we are grading information on its importance. The inclusion or omission of historical events and figures, playwrights, scientific achievements, and so on leads to charges of cultural bias. A cognitive scientist sees this issue differently. The question, *What should students be taught?* is equivalent not to *What knowledge is important?* but rather to *What do you want students to be able to do?* This question has two answers.

For reading, students must know whatever information writers assume they know and hence leave out. So the key question is, what do you want students to be able to read? A common answer – but certainly not the only answer – is "a daily newspaper, and books and periodicals written for the intelligent layperson." To read the *Washington Post* with good comprehension, one must have in long-term memory the information that the writers and editors of that newspaper assume their readers have. What they assume is very broad, because the *Post* publishes articles on politics, visual arts, literature, civics, history, drama, dance, science, architecture, and so on. But the writers and editors of the *Post* don't expect great depth of

knowledge. They might assume you know that Picasso was a painter but not that he was a cubist.

It's important to add that you might have other goals for what you'd like students to read. I mentioned the "educated layperson" goal because I think it's common among parents. If you want students to be able to read other sorts of material, you should target the information that *those* writers assume in their readership. Students will most successfully read texts on topics they know something about.

The second answer to the question "What knowledge matters?" applies to core subject matter courses. *What should students know of science, of history, of mathematics?* This question is different than the first because the uses of knowledge in these subject areas are different than the uses of knowledge for general reading. A *Washington Post* article may not define the word nebula, but the author won't count on my having a deeper definition than "astronomical object." If I'm studying astrophysics, I need to know much more.

Students can't learn everything, so what should they know? Cognitive science leads to the rather obvious conclusion that students must learn the concepts that come up again and again – the unifying ideas of each discipline. Some educational thinkers have suggested that a limited number of ideas should be taught in great depth, beginning in the early grades and carrying through the curriculum for years as different topics are taken up and viewed through the lens of one or more of these ideas. From the cognitive perspective, that makes sense.

When You Require Critical Thinking, Be Sure Students Have Enough Relevant Knowledge to Succeed

Our goal is not simply to have students know a lot of stuff – it's to have them know stuff in service of being able to think effectively. As emphasized in this chapter, thinking critically requires background knowledge. Critical thinking is not a set of procedures that can be practiced and perfected while divorced from background knowledge. Thus it makes sense to consider whether students have the necessary background knowledge to carry out a critical thinking task you might assign. For example, I once observed a teacher ask

her fourth-grade class what they thought it would be like to live in a rain forest. Although the students had spent a couple of days talking about rain forests, they didn't have the background knowledge to give anything beyond rather shallow responses (such as "It would be rainy"). She asked the same question at the end of the unit, and the student's answers were much richer. One student immediately said she wouldn't want to live there because the poor soil and constant shade would mean she would probably have to include meat in her diet – and she was a vegetarian.

Note, this doesn't mean "just teach knowledge until they have a ton of it, and *only then* encourage thinking!" Naturally, you still want children to think, even as they are acquiring knowledge. Recognizing that critical thinking requires knowledge might prompt you to change those critical thinking questions and tasks to better reflect what students know.

Thinking About Thinking Is Valuable . . . But It's Not Enough

Because we want students to think critically, it's natural that we should try to provide direct instruction in critical thinking. If you're thinking about how you ought to think, that's *metacognition*, and there's good evidence that teaching students metacognitive strategies is helpful. What's great about metacognitive strategies is their simplicity, the speed with which they can be taught, and the hope that they will be applicable across a range of content. But they can appear to achieve more than they really do.

Reading comprehension strategies provide a familiar example. You tell students things like "When you come to unfamiliar word, see if the context helps you figure out the meaning." Or "before you read a text, use the title to make predictions about the content." These are strategies to control thinking in a manner that will boost comprehension.

But you've seen that a key factor in reading comprehension is providing knowledge from memory that the author omitted, as in the example of Mark and his barbecue. Fitting together "shouldn't use his new barbecue" and "boss coming to dinner" requires missing knowledge and that missing knowledge *is unique to that sentence*.

It's how the two ideas fit together. The whole point of strategies is that they are general, because they are supposed to be all-purpose. Thus, strategies *can't* tell you how ideas fit together – each "fitting" is unique to the sentence you're reading.

Elsewhere, I offered this analogy.[11] Suppose you bought a piece of furniture at IKEA, got home, dumped the pieces on the floor, and saw that the directions merely said, "Before you start, think about other pieces of furniture you've seen in the past. Then put stuff together. Take it slow and don't force anything. Also, as you're building, stop every now and then, look at your progress, and see if it is starting to look like a piece of furniture."

This is actually helpful advice! But it's not enough – you need to know which pieces are supposed to be bolted together. Likewise, reading requires the specifics of how ideas are supposed to connect. But general strategies like "pay attention to whether you're understanding as you go" are still useful.

I think this characterization holds for most, if not all, metacognitive strategies. It's good to memorize "when judging whether a scientific experiment is sound, evaluate whether the control condition matches the experimental condition." Knowing you're supposed to do that helps, even though memorizing the strategy doesn't tell you *how* to do it. For that, you need background knowledge.

Shallow Knowledge Is Better Than No Knowledge

Some of the benefits of factual knowledge require that the knowledge be fairly deep – for example, reasoning often requires understanding many roles that the thing you're reasoning about might play. But other benefits accrue from shallow knowledge. We often don't need deep knowledge of a concept to be able to understand its meaning in context when we're reading. For example, I know almost nothing about baseball, but for general reading, a shallow definition such as "a sport played with a bat and ball, in which two teams oppose one another" will often do. Of course deep knowledge is better than shallow knowledge. But we're not going to have deep knowledge of everything, and shallow knowledge is certainly better than no knowledge.

Do Whatever You Can to Get Kids to Read . . . But It's Not Enough

The effects of knowledge described in this chapter also highlight why reading is so important. Reading exposes children to more facts and to a broader vocabulary than virtually any other activity, and persuasive data indicate that people who read for pleasure enjoy cognitive benefits throughout their lifetime.

People sometimes ask whether graphic novels "count," or whether audiobooks do. The answer is an emphatic "yes!" Graphic novels can be very sophisticated in terms of plotting, vocabulary, and so on. And listening to an audiobook has considerable overlap with reading print. Sure, a child is not getting practice in decoding or developing fluency, but the process of comprehension is similar whether one is reading or listening, so when we're talking about building vocabulary and background knowledge, audiobooks are great. They are especially great because you can listen at times when it's hard to read print, for example, when exercising or commuting.

That said, I don't believe it is quite the case that any book is fine "as long as they're reading." Naturally, if a child has a history of resisting reading, I'd be happy if she picked up any book at all. But once she is over that hump, if I feel that a bit of challenge won't dampen her reading motivation, I'd start trying to nudge her toward books at the appropriate reading level. That doesn't mean kids should never reread books. A child just may not have really understood everything the first time through, or she may return to a much-loved book for an emotional lift at a tough time. That said, it's rather obvious that a student doesn't gain as much from reading books significantly below her reading level and there are fun, fascinating books at every reading level, so why not nudge her toward a book that meets her where she is? It's just as obvious that a too difficult book is a bad idea. The student won't understand it and will just end up frustrated. The school librarian should be a tremendous resource and ally in helping children learn to love reading, and she is arguably the most important person in any school when it comes to reading.

As much as I'm an advocate of leisure reading, for most long-term goals we would set for students, leisure reading won't be enough.

The background knowledge students acquire in their leisure reading will be particular to their interests; the child who loves historical fiction may learn a lot about British monarchs whereas the child who loves fantasy will learn about mythical creatures. That's as it should be, but if we want children to be strong general readers, both need to learn about, for example, the solar system. Leisure reading is wonderful for building background knowledge, but students still need a strong curriculum.

Knowledge Acquisition Can Be Incidental

It's very important to keep in mind that the learning of factual knowledge can be incidental – that is, it can happen simply by exposure rather than only by concentrated study or memorization. Think about all you have learned by reading books and magazines for pleasure, or by watching video documentaries and the news, or through conversation with friends, or by spiraling down Internet rabbit holes. School offers many of the same opportunities. Students can learn information from math problems, or through sample sentences when they are learning grammar, or from the vocabulary you use when you select a classroom monitor. Every teacher knows so much that students don't. There are opportunities to fold this knowledge into each school day.

Start Early

At the end of the last section I noted that a child who starts behind in terms of knowledge will fall even further behind unless there is some intervention. There seems to be little doubt that this is a major factor in why some children fare poorly in school. Home environments vary a great deal. What sort of vocabulary do parents use? Do the parents ask the children questions and listen to the children's answers? Do they take their child to the museum or aquarium? Do they make books available to their children? Do the children observe their parents reading? All of these factors (and others) play a role in what children know on their first day of school. In other words, before a child meets her first teacher, she may be quite far behind the child sitting next to her in terms of how easy it is going to be for her to learn. Trying to level this playing field is a teacher's

greatest challenge. There are no shortcuts and no alternatives to trying to increase the factual knowledge that the child has not picked up at home.

To be clear, I'm not blaming parents who do not do all of these things; people have limits of time and other resources to provides this sort of environment for their children. I think every teacher would happy if every student could access similar resources at school.

Knowledge Must Be Meaningful

Teachers should not take the importance of knowledge to mean that they should create lists of facts – whether shallow or detailed – for students to learn. Sure, some benefit might accrue, but it would be small. Knowledge pays off when it is conceptual and when the facts are related to one another, and that is not true of list learning. Also, as any teacher knows, learning a list is difficult and boring and so would do more harm than good; it would encourage the belief that school is a place of drudgery, not excitement and discovery. But what is a better way to ensure that students acquire factual knowledge, now that we've concluded it's so important? In other words, why do some things stick in our memory whereas other things slip away? That is the topic of the next chapter.

Notes

*One of the pleasures of the experiences shared with a close friend is the "inside joke," a reference that only the two of you understand. Hence, if her best friend asked what she was doing, the typist might say, "I'm painting a gravel road" – their personal code, based on a shared experience, for a long, pointless task. That's one extreme of assuming information on the part of your audience.

†Tournament-level chess players all have rankings – a number representing their skill level – based on whom they have beaten and who has beaten them.

Further Reading

Less Technical

Goodwin, B. (2011). Research says . . . don't wait until 4th grade to address the slump. *Educational Leadership,* 68(7), 88–89. Brief review of the idea that the dip in reading scores for disadvantaged children is due in part to a lack of background knowledge, and what to do about it.

Lareau, A. (2003). *Unequal Childhoods*. Berkeley: University of California Press. Fascinating ethnographic study of childhood in homes of different socioeconomic status.

Shing, Y. L., & Brod, G. (2016). Effects of prior knowledge on memory: implications for education. *Mind, Brain, and Education*, 10(3), 153–161. Reasonably user-friendly review of the role of memory in new learning, with an eye toward educational applications.

Willingham, D. T. (2017). How to get your mind to read. *New York Times* (25 November), p. SR6. https://www.nytimes.com/2017/11/25/opinion/sunday/how-to-get-your-mind-to-read.html. Op-ed on the importance of background knowledge to reading comprehension. Useful to send to someone for a quick introduction to the ideas covered here.

More Technical

Best, R. M., Floyd, R. G., & McNamara, D. S. (2008). Differential competencies contributing to children's comprehension of narrative and expository texts. *Reading Psychology*, 29(2), 137–164. Article examining the role of background knowledge in reading comprehension and how the lack of knowledge contributes to the fourth-grade slump.

Cromley, J. G., & Kunze, A. J. (2020). Metacognition in education: translational research. *Translational Issues in Psychological Science* 6(1), 15–20. A review of research showing that metacognitive strategies are useful in classroom contexts.

Fernández, G., & Morris, R. G. (2018). Memory, novelty and prior knowledge. *Trends in Neurosciences*, 41(10), 654–659. Brief article summarizing some key findings in the role of prior knowledge in new learning.

Gobet, F., & Charness, N. (2018). Expertise in chess. In *The Cambridge Handbook of Expertise and Expert Performance*, *2* (eds. K. Ericsson, R. Hoffman, A. Kozbelt, & A. Williams) 597–615. Cambridge, UK: Cambridge University Press. This chapter summarizes much of the important research showing that knowledge is fundamental to chess skill.

Mol, S. E., & Bus, A. G. (2011). To read or not to read: a meta-analysis of print exposure from infancy to early adulthood. *Psychological Bulletin*, 137(2), 267–296. A review of 99 studies showing that reading produces a virtuous cycle: Reading during leisure time leads to improvement in components of reading (knowledge and others) that makes reading easier and therefore makes people more likely to read.

Pfost, M., Hattie, J., Dörfler, T., & Artelt, C. (2014). Individual differences in reading development: a review of 25 years of empirical research on Matthew effects in reading. *Review of Educational Research*, 84(2), 203–244. A review of many studies showing that students who know more learn more from reading.

Discussion Questions

1. It's important to remember that the knowledge that supports thinking skills can come from *outside* of school, as well as within the school curriculum. But simply telling parents to "make sure your children acquire knowledge outside of school" is too vague to do much. What might educators say or do that would be more effective?
2. Consider the very long sentence from *Moby-Dick* in this chapter. As noted, you run out of mental room while reading it and so are likely not to comprehend it. Are

there comparable materials or moments in your classroom that confuse students? One solution, noted in the last chapter, is to break up complex material into smaller bits. Another solution, noted in this chapter, is to acquire enough knowledge to enable chunking. Could that solution apply to any complex situations in your classroom?

3. The baseball study shows the striking effect that knowledge can have on comprehension. It also leads me to think about how a reader who usually struggles experiences reading that passage. Is it striking to him or her that they are so successful? We know that reader confidence is an important predictor of whether they read during leisure time, and struggling readers of course have many experiences telling them they should not be confident. How could we use findings like the baseball study to inspire methods of building confidence in struggling readers?
4. The "Washing Clothes" passage shows how effective background knowledge can be in clarifying ambiguous communication. You could imagine that there are times you sometimes speak in an analogous way – you say something that's actually quite vague, but, because you have the right background knowledge, it's quite clear to *you* what you mean. Can you think of an example of a misunderstanding between you and someone you were teaching? Is there a reliable way to help you bear in mind the difference between what you know and what your students know?
5. The more you know, the easier it is to learn new things. That indicates that children who begin school with less knowledge will fall further and further behind. What does this fact imply for early education? How about elementary school and beyond?
6. We live in an age where expertise is sometimes viewed with suspicion. Politicians appeal to the idea that you don't need to be knowledgeable about a topic, and that with common sense and smarts you'll actually make better decisions than so-called experts. What silent (or overt) messages do you think your society's culture sends to students about the value of knowing things? How about your local school?

3

Why Do Students Remember Everything That's on Television and Forget Everything I Say?

Question: Memory is mysterious. You may lose a memory created 15 seconds earlier, such as when you find yourself standing in your kitchen trying to remember what you came there to fetch. Other seemingly trivial memories (for example, advertisements) may last a lifetime. What makes something stick in memory, and what is likely to slip away?

Answer: We can't store everything we experience in memory. Too much happens. So what should the memory system tuck away? Things that are repeated again and again? But what about a really important one-time event such as a wedding? Things that cause emotion? But then you wouldn't remember important yet neutral things (for example, most schoolwork). How can the memory system know what you'll need to remember later? Your memory system lays its bets this way: if you think about something carefully, you'll probably have to think about it again, so it should be stored. Thus your memory is not a product of what you want to remember or what you try to remember; it's a product of what you think about. A teacher once told me that for a fourth-grade unit on the Underground Railroad he had his students bake biscuits, because this was a staple food for enslaved people seeking escape. He asked what I thought about the assignment. I pointed out that his students probably thought for 40 seconds about the relationship of biscuits to the Underground Railroad, and for 40 minutes about measuring flour,

mixing shortening, and so on. Whatever students think about is what they will remember. The cognitive principle that guides this chapter is:

> Memory is the residue of thought.

To teach well, you should pay careful attention to what an assignment will actually make students think about (not what you hope they will think about), because that is what they will remember.

The Importance of Memory

Every teacher has had the following experience: you teach what you think is a terrific lesson, full of lively examples, deep content, engaging problems to solve, and a clear message, but the next day students remember nothing of it except a joke you told and an off-the-subject aside about your family[1] – or worse, when you say, "The point of yesterday's lesson was that one plus one equals two," they look at you incredulously and say, "Wait, one plus one equals *two?*" Obviously, if the message of Chapter 2 is "background knowledge matters," then we must closely consider how we can make sure that students acquire this background knowledge. So why do students remember some things and forget others?

Let's start by considering why you fail to remember something. Suppose I said to you, "Can you summarize the last professional development session you attended?" Let's further suppose that you brightly answer, "Nope, I sure can't." Why don't you remember?

One of four things has happened, all of which are illustrated in Figure 3.1, a slightly elaborated version of the diagram of the mind that we've used before. You will recall that working memory is where you keep things "in mind," the location of consciousness. There is lots of information in the environment, most of which we are not aware of. For example, as I write this, the refrigerator is humming, birds are chirping outside, and there is pressure on my backside from the chair I'm sitting on – but none of that was in my working memory (that is, my awareness) until I paid attention to it. As you can see in Figure 3.1, things can't get into long-term memory unless

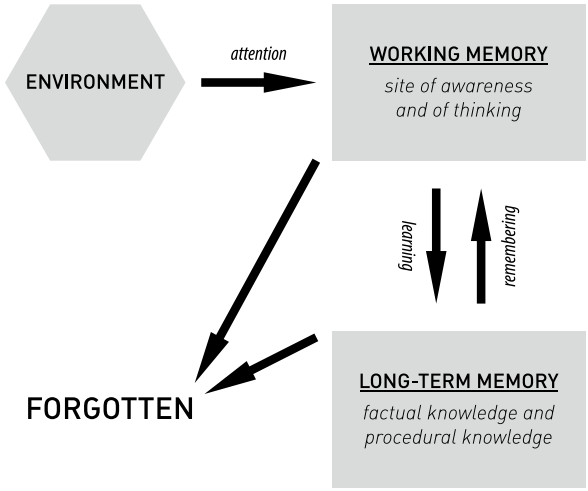


FIGURE 3.1: A slightly modified version of our simple diagram of the mind. Source: © Greg Culley.

they have first been in working memory. So this is a somewhat complex way of explaining the familiar phenomenon: *If you don't pay attention to something, you can't learn it!* You won't remember much of the professional development session if you were thinking about something else.

Information can enter working memory not only from the environment but also from long-term memory; that's what I mean when I refer to remembering, as shown by the labeled arrow. So another possible reason you don't remember is that the process by which things are drawn from long-term memory has failed. I discuss why that happens in Chapter 4.

A third possibility is that the information no longer resides in long-term memory – that it has been forgotten. I'm not going to discuss forgetting, but it's worth taking a moment to dispel a common myth. You sometimes hear that the mind records in exquisite detail everything that happens to you, like a video camera, but you just can't get at most of it – that is, memory failures are a problem of access. If you were given the right cue, the theory goes, anything that ever happened to you would be recoverable. For example, you may think you remember almost nothing of a childhood home you left at age five, but when you revisit it the smell of the camellia blooms in the

yard wipes away the years, and the memories that you thought were lost can be pulled out, like charms on a fine chain. Such experiences raise the possibility that *any* memory that you believe is lost can in principle be recovered again. Successful memory under hypnosis is often raised as evidence to support this theory. If the right cue (camellia blossoms or whatever it might be) can't be found, hypnosis allows you to probe the vault directly.

Although this idea is appealing, it's wrong. We know that hypnosis doesn't aid memory. That's easy to test in the laboratory. Simply give people some stuff to remember, then later hypnotize half of them and compare their recall to that of the people who are not hypnotized. This sort of experiment has been done dozens of times, and typical results are shown in Figure 3.2.[2] Hypnosis doesn't help.

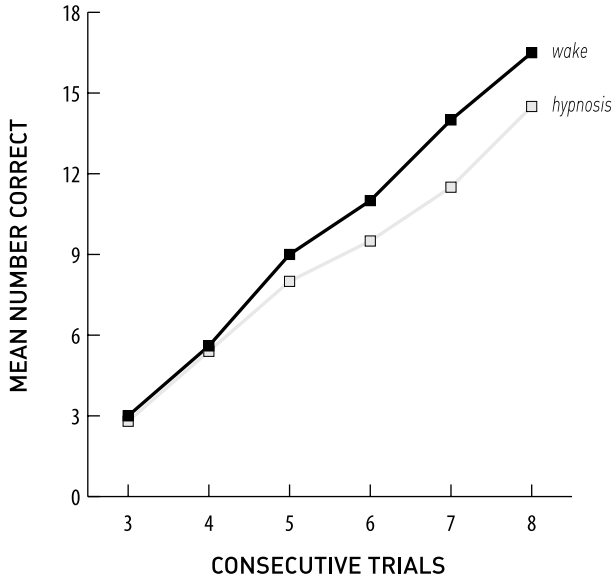


FIGURE 3.2: Subjects were shown 40 drawings of common objects and then had to try to recall them. Session 1 happened right away; sessions 2 through 8 occurred a week later. Naturally there was significant forgetting during the week, and with each attempt to remember, subjects on average did recall more. Also, the hypnotized subjects didn't remember any more than the nonhypnotized subjects. Source: From "Evaluating hypnotic memory enhancement (Hypermnesia and Reminiscence) using multitrial forced recall" by David F. Dinges, Wayne G. Whitehouse, Emily C. Orne, John W. Powell, Martin T. Orne, and M. H. Erdelyi in *Journal of Experimental Psychology: Learning, Memory and Cognition* 18, figure 1, p. 1142. Copyright © 1992 by the American Psychological Association.

It does make you more confident that your memory is right, but it doesn't actually make your memory more accurate.

The other bit of evidence – that a good cue such as the odor of camellia can bring back long-lost memories – is much more difficult to test in a laboratory experiment, although most memory researchers believe that such recoveries are possible. But even if we allow that lost memories can be recovered in this way, it doesn't mean that *all* seemingly forgotten memories are recoverable – it just means that some are. In sum, memory researchers see no reason to believe that all memories are recorded forever.

Now, let's return to our discussion of forgetting. Sometimes you *do* pay attention, so the material rattles around working memory for a while, but it never makes it to long-term memory. An example of a few such bits of information from my own experience are shown in Figure 3.3. *Lateral line* is a term I have looked up more

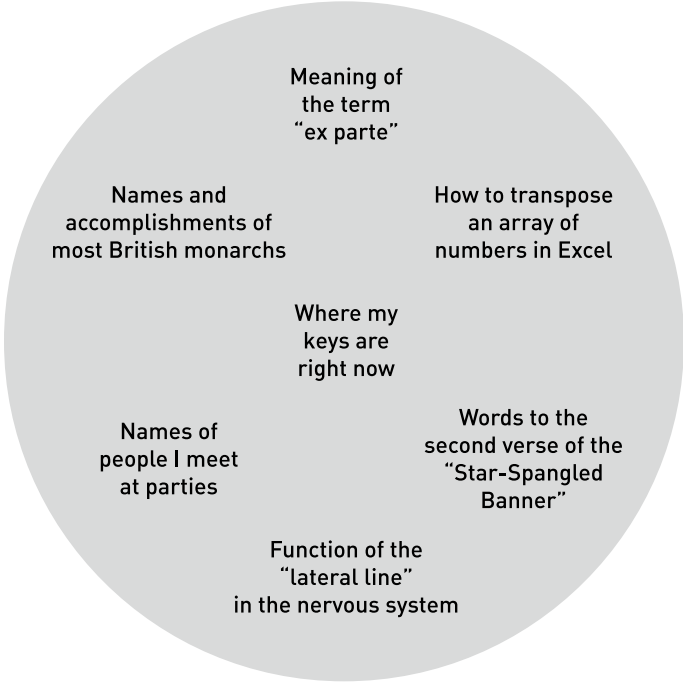


FIGURE 3.3: Bits of information that I am certain I have paid attention to and that thus have resided in my working memory but that have never made it into my long-term memory. Source: © Greg Culley.

than once, but I couldn't tell you now what it means. You doubtless have your own examples of things you are certain you *ought* to know, because you've looked them up or heard them (and thus they have been in working memory), yet they have never stuck in long-term memory.

Just as odd is that some things have remained in your long-term memory for years although you had no intention of learning them; indeed, they held no special interest for you. For example, why do I know the jingle from the 1970s Bumble Bee tuna advertisement (Figure 3.4)?

You could make a good argument that understanding the difference between Figures 3.3 and 3.4 is one of the core problems in education. We all know that students won't learn if they aren't paying attention. What's more mysterious is why, when they *are* paying attention, they sometimes learn and sometimes don't. What else is needed besides attention?

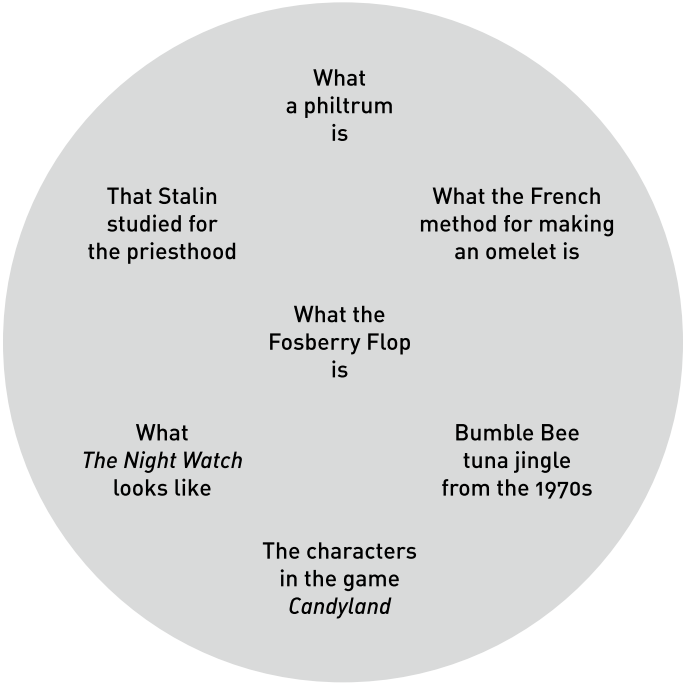


FIGURE 3.4: Material that is in the author's long-term memory even though the author didn't want to learn it and was in fact not all that interested in it. Source: © Greg Culley.

A reasonable guess is that we remember things that bring about some emotional reaction. Aren't you likely to remember really happy moments, such as a wedding, or really sad ones, such as hearing that a beloved relative has passed away? You are, and in fact if you ask people to name their most vivid memories, they often relate events that probably had some emotional content, such as a first date or a birthday celebration (Figure 3.5).

Naturally, we pay more attention to emotional events, and we are likely to talk about them later, so scientists have had to conduct very careful studies to show that it's really the emotion and not the repeated thought about these events that provides the boost to memory. The effect of emotion on memory is indeed real, and researchers have actually worked out some of the biochemistry behind it, but the emotion needs to be reasonably strong to have much impact. If memory *depended* on emotion, we would remember little of what we encounter in school. So the answer *Things go into long-term memory if they create an emotional reaction* is not quite right. It's more accurate to say, *Things that create an emotional reaction will be better remembered, but emotion is not necessary for learning.*

Repetition is another obvious candidate for what makes learning work. Maybe the reason I remember the Bumble Bee tuna jingle (Figure 3.4) from 40 years ago is that I heard it a lot. Repetition is very important, and I discuss it in Chapter 5, but it turns out that not just any repetition will do. Material may be repeated almost indefinitely and still not stick in your memory. For example, have a look at Figure 3.6. Can you spot the real penny among the counterfeits?





FIGURE 3.5: Emotional events tend to be well remembered, whether they are happy, such as this woman winning the Miss Philippines title, or anguished, as this Bosnian man mourning a relative who had died. Source: Miss Philippines © Getty Images/Majority World; mourning © Getty Images /DIMITAR DILKOFF.

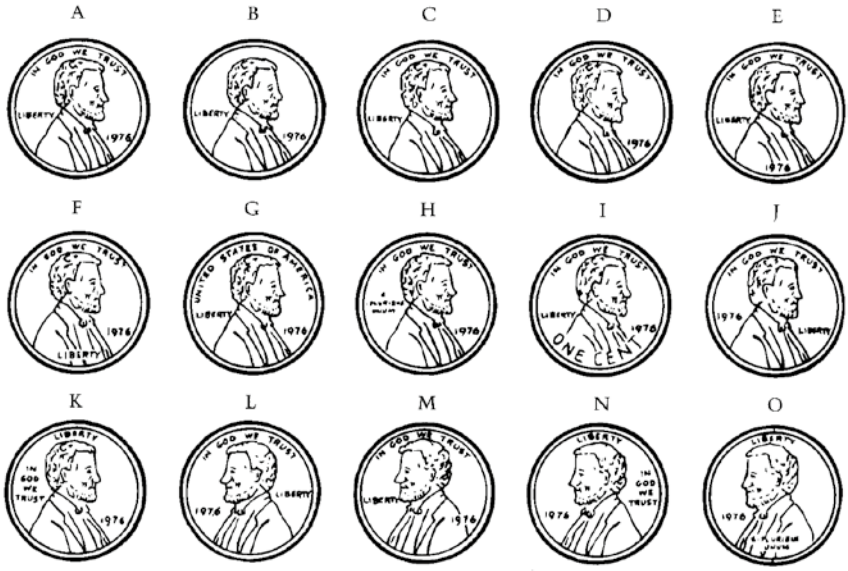


FIGURE 3.6: Can you find the real penny among the counterfeits? People are terrible at this task even though they have seen a penny thousands of times. Source: From "Long term memory for a common object" by R. S. Nickerson and M. J. Adams in *Cognitive Psychology* 11: 287–307. Copyright © 1979. Reprinted with permission from Elsevier.

If you've spent time in the United States you've seen hundreds or thousands of pennies – a huge number of repetitions. Yet, if you're like most people, you don't know much about what a penny looks like.[3] (The real penny is choice *A,* by the way.)

So repetition alone won't do it. It's equally clear that *wanting* to remember something is not the magic ingredient. How marvelous it would be if memory did work that way. Students would sit down with a book, say to themselves, "I want to remember this," and they would! You'd remember the names of people you've met, and you'd always know where your car keys are. Sadly, memory doesn't work that way, as demonstrated in a classic laboratory experiment.[4] Subjects were shown words on a screen one at a time and were asked to make a simple judgment about each word. (Some subjects had to say whether the word contained either an *A* or a *Q;* others had to say whether the word made them think of pleasant things or unpleasant things.) An important part of the experiment was that half of the subjects were told that their memory for the words would be

tested later, after they had seen the whole list. The other subjects were not warned about the test. One of the remarkable findings was that knowing about the future test didn't improve subjects' memories. Other experiments have shown that telling subjects they'll be paid for each remembered word doesn't help much. So *wanting* to remember has little or no effect.

But there's another finding from this experiment that's still more important. Remember that when subjects saw each word they had to make a judgment about it – either about whether it contained an *A* or a *Q* or about whether it made them think of pleasant or unpleasant things. The people who made the second type of judgment remembered nearly twice as many words as the people who made the first judgment. Now we seem to be getting somewhere. We've found a situation in which memory gets a big boost. But why would it help to think about whether a word is pleasant or not?

In this case it matters because judging pleasantness makes you think about what the word *means* and about other words that are related to that meaning. Thus, if you saw the word *oven,* you might think about cakes and roasts and about your kitchen oven, which doesn't work well, and so on. But if you were asked to judge whether *oven* contained an *A* or a *Q,* you wouldn't have to think about the meaning at all.

So it seems we're poised to say that *thinking about meaning is good for memory*. That's close, but not quite right. The penny example doesn't fit that generalization. In fact, the penny example shows just the opposite. I said that you've been exposed to a penny thousands of times (at least), and most of those times you were thinking about the penny's meaning – that is, you were thinking about its function, about the fact that it has monetary value, even if that value is modest. But having thought about the meaning of a penny doesn't help when you're trying to remember what the penny looks like, which is what the test in Figure 3.6 requires.

Here's another way to think about it. Suppose you are walking the halls of your school and you see a student muttering to himself in front of his open locker. You can't hear what he's saying, but you can tell from his tone that he's angry. There are several things you could

focus on. You could think about the *sound* of the student's voice, you could focus on how he *looks,* or you could think about the *meaning* of the incident (why the student might be angry, whether you should speak to him, and so on). These thoughts will lead to different memories of the event the next day. If you thought only about the sound of the student's voice, the next day you'd probably remember that sound quite well but not his appearance. If you focused on visual details, then that's what you'd remember the next day, not what the student's voice sounded like. In the same way, if you think about the meaning of a penny but never about the visual details, you won't remember the visual details, even if they have been in front of your eyes ten thousand times.

Whatever you think about, that's what you remember. *Memory is the residue of thought*. Once stated, this conclusion seems impossibly obvious. Indeed, it's a very sensible way to set up a memory system. Given that you can't store everything away, how should you pick what to store and what to drop? Your brain lays its bets this way: If you don't think about something very much, then you probably won't want to think about it again, so it need not be stored. If you do think about something, then it's likely that you'll want to think about it *in the same way* in the future. If I think about what the student looks like when I see him, then his appearance is probably what I'll want to know about when I think about that student later.

There are a couple of subtleties to this obvious conclusion that we need to draw out. First, when we're talking about school, we usually want students to remember what things mean. Sometimes what things look like is important – for example, the beautiful facade of the Parthenon, or the shape of Benin – but much more often we want students to think about meaning. Ninety-five percent of what students learn in school concerns meaning, not what things look like or what they sound like.* Therefore, a teacher's goal should almost always be to get students to think about meaning.

The second subtlety (again, obvious once it's made explicit) is that there can be different aspects of meaning for the same material. For

example, the word *piano* has lots of meaning-based characteristics (Figure 3.7). You could think about the fact that it makes music, or about the fact that it's expensive, or that it's really heavy, or that it's made from fine-quality wood, and so on. In one of my all-time favorite experiments, the researchers led subjects to think of one or another characteristic of words by placing them in sentences – for example, "The moving men lugged the PIANO up the flight of stairs" or "The professional played the PIANO with a lush, rich sound."[5] The subjects knew that they needed to remember only the word in capitals. Later, experimenters administered a memory test for the words, with some hints. For *piano,* the hint was either "something heavy" or "something that makes music." The results showed that the subjects' memories were really good if the hint matched the way they had thought about *piano,* but poor if it didn't. That is, if the subjects read the moving men version of the sentence, hearing the cue "something that makes music" didn't help them remember *piano*. So it's not even enough to say, "You should think about meaning." You have to think about the right aspect of meaning.

Let me summarize what I've said about learning so far. For material to be learned (that is, to end up in long-term memory), it must reside for some period in working memory – that is, a student must pay attention to it. Further, *how* the student thinks of the experience completely determines what will end up in long-term memory.



FIGURE 3.7: We seldom think about it, but the context in which we think about even a simple word influences which aspect of meaning we focus on: that pianos produce music, that they can serve as a seat, that they are very heavy. Source: Playing piano © Getty Images/Frank Hoensch; sitting © Getty Images/Harry Dempster; moving © Shutterstock/Volodymyr TVERDOKHLIB.

The obvious implication for teachers is that they must design lessons that will ensure that students are thinking about the meaning of the material. A striking example of an assignment that didn't work for this reason came from my nephew's sixth-grade teacher. He was to draw a plot diagram of a book he had recently finished. The point of the plot diagram was to get him to think about the story elements and how they related to one another. The teacher's goal, I believe, was to encourage her students to think of novels as having *structure,* but the teacher thought that it would be useful to integrate art into this project, so she asked her students to draw pictures to represent the plot elements. That meant that my nephew thought very little about the relation between different plot elements and a great deal about how to draw a good castle. My daughter had completed a similar assignment some years earlier, but her teacher had asked students to use words or phrases rather than pictures. I think that assignment more effectively fulfilled the intended goal because my daughter thought more about how ideas in the book were related.

Now you may be thinking, "OK, so cognitive psychologists can explain why students have to think about what material means – but I really already knew they should think about that. Can you tell me *how* to make sure that students think about meaning?" Glad you asked.

What Good Teachers Have in Common

If you read Chapter 1, you can easily guess a common technique that I would *not* recommend for getting students to think about meaning: trying to make the subject matter relevant to the students' interests. I know that sounds odd, so let me elaborate.

Trying to make the material relevant to students' interests doesn't work. As I noted in Chapter 1, content is seldom the decisive factor in whether or not our interest is maintained. For example, I love cognitive psychology, so you might think, "Well, to get Willingham to pay attention to this math problem, we'll wrap it up in a cognitive psychology example." But Willingham is quite capable of being bored by cognitive psychology, as has been proved repeatedly at professional conferences I've attended. Another problem with trying to use content to engage students is that it's sometimes very difficult to do and the whole enterprise comes off as artificial. How would a

math instructor make algebra relevant to my 13-year-old daughter? With a "real-world" example using numbers of Instagram likes? I just finished pointing out that any material has different aspects of meaning. If the instructor used a math problem with Instagram likes, isn't there some chance that my daughter would think about Instagram rather than about the problem? And that thoughts about Instagram would lead to thoughts about the instant message she received earlier, which would make her think of how sick she is of Jasmine's drama, which would make her wonder whether she should even invite Jasmine to her birthday dinner . . . ?

So if content won't do it, how about style? Students often refer to good teachers as those who "make the stuff interesting." It's not that the teacher relates the material to students' interests – rather, the teacher has a way of interacting with students that they find engaging. Let me give a few examples from my own experience with fellow college-level teachers who are consistently able to get students to think about meaning.

Teacher A is the comedian. She tells jokes frequently. She never misses an opportunity to use a silly example.

Teacher B is the den mother. She is very caring, very directive, and almost patronizing, but so warm that she gets away with it. Students call her "Mom" behind her back.

Teacher C is the storyteller. He illustrates almost everything with a story from his life. Class is slow paced and low key, and he is personally quiet and unassuming.

Teacher D is the showman. If he could set off fireworks inside, he would do it. The material he teaches does not lend itself easily to demonstrations, but he puts a good deal of time and energy into thinking up interesting applications, many of them involving devices he's made at home.

Each of these teachers is one to whom students refer as making boring material interesting, and each is able to get students to think about meaning. Each style works well for the person using it, although obviously not everyone would feel comfortable taking on some of these styles. It's a question of personality.

Style is what the students notice, but it is only a part of what makes these teachers so effective. College professors typically get written student evaluations of their teaching at the end of every course. Most schools have a form for students to fill out that includes such items as "The professor was respectful of student opinions," "The professor was an effective discussion leader," and so on, and students indicate whether or not they agree with each statement. Researchers have examined these sorts of surveys to figure out which professors get good ratings and why. One of the interesting findings is that most of the items are redundant. A two-item survey would be almost as useful as a 30-item survey, because many questions are variations on these two: Does the professor seem like a nice person, and is the class well organized? (Figure 3.8.) Although they don't realize they are doing so, students treat each of the 30 items as rephrasings of one of these two questions. What matters is cognition and connection.

Although K–12 students don't complete questionnaires about their teachers, we know that more or less the same thing is true for them. The emotional bond between students and teacher – for better or worse – accounts for whether students learn. The brilliantly well-organized teacher whom fourth graders see as mean will not be very effective. But the funny teacher, or the gentle storytelling teacher, whose lessons are poorly organized won't be much good either. Effective teachers have both qualities. They are able to connect personally with students, and they organize the material in a way that makes it interesting and easy to understand.

That's my real point in presenting these different types of teachers. When we think of a good teacher, we tend to focus on personality and on the way the teacher presents himself or herself. But that's only half of good teaching. The jokes, the stories, and the warm manner all generate goodwill and get students to pay attention. But then how do we make sure they think about meaning? That is where the second property of being a good teacher comes in – organizing the

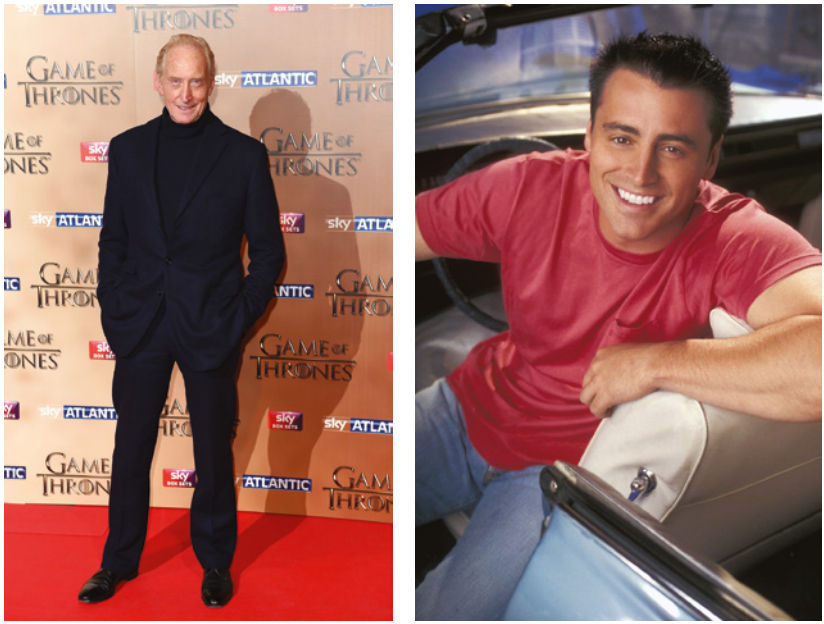


FIGURE 3.8: How would each of these men be as a teacher? In *Game of Thrones*, Tywin Lannister (played by actor Charles Dance) is intelligent but cold and remote. The character Joey Tribbiani from *Friends* (played by actor Matt LeBlanc) is warm and friendly but not terribly smart. Teachers need to be both well organized and approachable. Source: LeBlanc © Getty Images/NBCUniversal; Dance © Getty Images/WireImage.

ideas in a lesson plan in a coherent way so that students will understand and remember. Cognitive psychology cannot tell us how to be personable and likable to our students, but I can tell you about one set of principles that cognitive psychologists know about to help students think about the meaning of a lesson.

The Power of Stories

The human mind seems exquisitely tuned to understand and remember stories – so much so that psychologists sometimes refer to stories as "psychologically privileged," meaning that they are treated differently in memory than other types of material. I'm going to suggest that organizing a lesson plan like a story is an effective way to help students comprehend and remember. It also happens to be the organizing principle used by the four teachers I described. The way in which each of them related emotionally to their students was

very different, but the way they got their students to think about the meaning of material was identical.

Before we can talk about how a story structure could apply to a classroom, we must go over what a story structure is. There is not universal agreement over what makes a story, but most sources point to the following four principles, often summarized as *the four Cs*. The first C is *causality,* which means that events are causally related to one another. For example, "I saw Jane; I left the house" is just a chronological telling of events. But if you read, "I saw Jane, my hopeless old love; I left the house," you would understand that the two events are linked causally. The second C is *conflict*. A story has a main character pursuing a goal, but he or she is unable to reach that goal. In *Star Wars* the main character is Luke Skywalker, and his goal is to deliver the stolen plans and help destroy the Death Star. Conflict occurs because there is an obstacle to the goal. If Luke didn't have a worthy adversary – Darth Vader – it would make for a rather short movie. In any story the protagonist must struggle to meet his goal. The third C is *complications*. If Luke simply hammered away for 90 minutes at his goal of delivering the plans, that would be rather dull. Complications are subproblems that arise from the main goal. Thus, if Luke wants to deliver the plans, he must first get off his home planet, Tatooine – but he has no transportation. That's a complication that leads to his meeting another major character, Han Solo, and leaving the planet amid a hail of gunfire – always a movie bonus. The final C is *character*. A good story is built around strong, interesting characters, and the key to those qualities is *action*. A skillful storyteller shows rather than tells the audience what a character is like. For example, the first time the *Star Wars* audience sees Princess Leia she is shooting at storm troopers. Hence we don't need to be told that she is brave and ready to take action.

If we're trying to communicate with others, using a story structure brings several important advantages. First, stories are easy to comprehend, because the audience knows the structure, which helps to interpret the action. For example, the audience knows that events don't happen randomly in stories. There must be a causal connection, so if the cause is not immediately apparent, the audience will think carefully about the previous action to try to connect it to present events. For example, at one point in *Star Wars* Luke, Chewbacca,

and Han are hiding on an Empire ship. They need to get to another part of the ship, and Luke suggests putting handcuffs on Chewbacca. That suggestion is mildly puzzling because Luke and Chewbacca are allies. The audience must figure out that Luke intends to pretend that Chewbacca is a prisoner and that he and Han are guards. The audience will do that bit of mental work because they know there must be a reason for this puzzling action.

Second, stories are interesting. Reading researchers have conducted experiments in which people read lots of different types of material and rate each for how interesting it is. Stories are consistently rated as more interesting than other formats (for example, expository prose), even if the same information is presented. Stories may be interesting because they demand the kind of inferences I discussed in Chapter 1. Recall that problems (such as crossword puzzles) are interesting if they are neither too difficult nor too easy. Stories demand these medium-difficulty inferences, as in the handcuff example.

Formal work in laboratory settings has shown that people rate stories as less interesting if they include too much information, thus leaving no inferences for the listener to make. But formal research is hardly necessary to confirm this phenomenon. We all have one or two friends who kill every story they tell with too much information. (See Figure 3.9.) An acquaintance of mine once spent 10 minutes relating that she hadn't visited her favorite Chinese restaurant for a year because they stopped accepting checks, only to bump into the owner who told her that he would happily make an exception for her. Delivered in 15 seconds with cheeky pride, this story would have been cute. But with the details packed in (and no inferences for me to make) over the course of 10 full minutes, it was all I could do not to scream.

Third, stories are easy to remember. There are at least two contributing factors here. Because comprehending stories requires lots of medium-difficulty inferences, you must think about the story's meaning throughout. As described earlier in the chapter, thinking about meaning is excellent for memory because it is usually meaning that you want to remember. Your memory for stories is also aided by their causal structure. If you remember one part of the plot, it's a good guess that the next thing that happened was caused



FIGURE 3.9: Senators listening to evidence during President Trump's 2020 impeachment trial found it difficult to pay attention; some even appeared to fall asleep, as an artist captured Senator Risch doing. (Cameras are not permitted in the Senate.) Part of the reason it was so boring was that listeners already knew all of the evidence, both from news reports and briefings. In explaining to reporters why they seemed so bored Senator Mike Braun of Indiana said, "The subject matter is something we've all heard."[6] Source: © Art Lien.

by what you remember. For example, if you're trying to remember what happened after Luke put handcuffs on Chewbacca, you'll be helped by remembering that they were on an Empire ship (hence the ruse), which might help you remember that they went to rescue Princess Leia from the detention area.

Putting Story Structure to Work

Now, all this about movies has been a diverting interlude (at least I hope it has), but what does it have to do with the classroom? My intention here is not to suggest that you simply tell stories, although there's nothing wrong with doing so. Rather, I'm suggesting something one step removed from that. Structure your lessons the way stories are structured, using the four Cs: causality, conflict, complications, and character. This doesn't mean you must do most of the talking. Small group work or projects or any other method may be used. The story structure applies to the way you *organize* the material that you encourage your students to think about, not to the methods you use to teach the material.

In some cases, the way to structure a lesson plan as a story is rather obvious. For example, history can be viewed as a set of stories. Events are caused by other events; there is often conflict involved; and so on. Still, thinking carefully about the four Cs as you consider a lesson plan can be helpful. It might encourage you to think about a different perspective from which to tell the story. For example, suppose you are an American teacher planning a lesson on Pearl Harbor. You might first think of the organization shown in Figure 3.10. It's chronological and it makes the United States the main character – that is, events are taken from the US point of view. Your goal is to get students to think about three points: US isolationism before Pearl Harbor, the attack, and the subsequent "Germany first" decision and the putting of the United States on a war footing.

Suppose, however, you thought of the four Cs when you were telling this story. From that perspective, the United States is not the strong character. Japan is, because that country had the goal that propelled events forward – regional domination – and had significant obstacles to this goal – Japan lacked natural resources and was embroiled in a protracted war with China. This situation set up a subgoal: to sweep up the European colonies in the South Pacific. Meeting that goal would raise Japan's standing as a world power and help obtain crucial raw materials for finishing the war with China. But that subgoal brought with it another complication. The United States was the other major naval power in the Pacific. How was Japan to deal with that problem? Rather than plundering the European colonies and daring the United States to intervene across five thousand miles of ocean (which the United States probably would not have done), Japan chose to try to eliminate the threat in one surprise attack. If one seeks to organize a lesson plan as a story, the one in Figure 3.10 is less compelling than the one in Figure 3.11.

My suggestion to use the Japanese point of view of Pearl Harbor doesn't mean that the American point of view should be ignored or deemed less important. Indeed, I could imagine a teacher in the United States electing not to use this story structure precisely because it takes a Japanese point of view in a US history class. My point here is that using a story structure may lead you to organize a lesson in ways that you hadn't considered before. And the story structure does bring cognitive advantages.

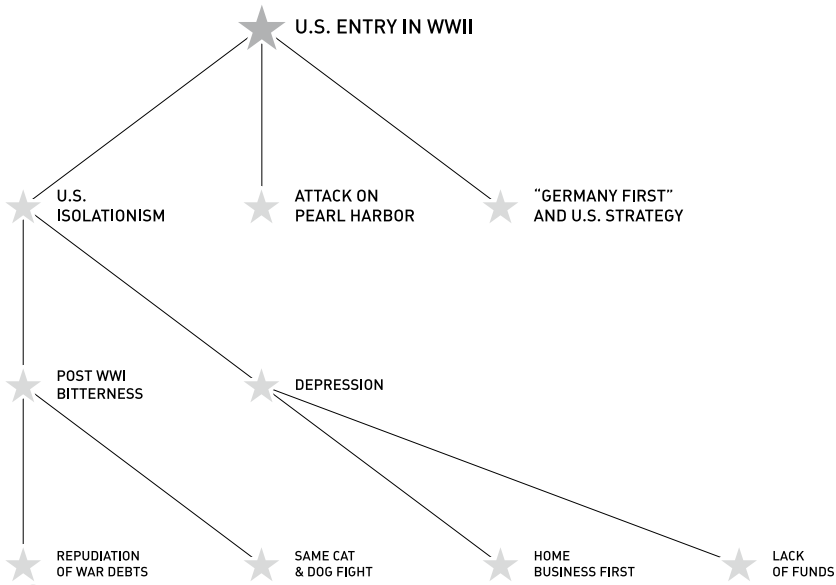


FIGURE 3.10: A tree diagram showing the typical structure of a lesson plan on Pearl Harbor. The organization is chronological. Source: © Greg Culley.

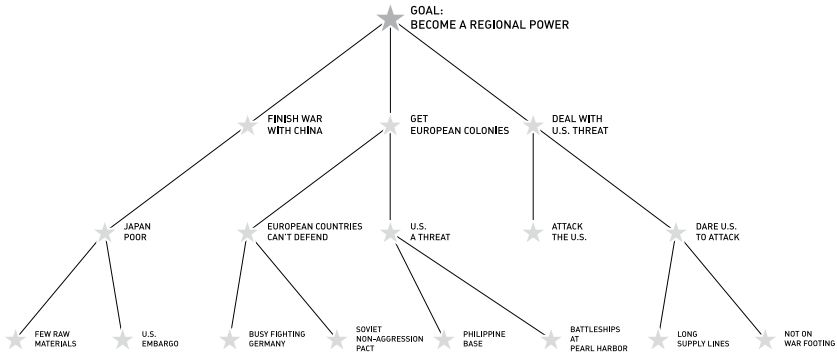


FIGURE 3.11: Alternative organization for a lesson plan on Pearl Harbor. From a storytelling point of view, Japan is the strong character because she takes actions that move the story forward. Source: © Greg Culley.

Using storytelling to teach history seems easy, but can you really use a story structure in a math class? Absolutely. Here's an example of how I introduced the concept of a Z-score – a common way to transform data – when I taught introductory statistics. Begin with the simplest and most familiar example of probability – the coin

flip. Suppose I have a coin that I claim is loaded – it always comes up heads. To prove it to you, I flip the coin and it does indeed come up heads. Are you convinced? College students understand that the answer should be no because there is a 50-50 chance that a fair coin would have come up heads. How about 100 heads in a row? Clearly the odds are really small that a fair coin will come up heads 100 times in a row, so you'd conclude that the coin isn't fair.

That logic – how we decide whether a coin is fishy or fair – is used to evaluate the outcome of many scientific experiments. When we see headlines in the newspaper saying "New drug for Alzheimer's found effective" or "Older drivers less safe than younger" or "Babies who watch videos have smaller vocabularies," these conclusions rest on the same logic as the coin flip. How?

Suppose we want to know whether an advertisement is effective. We ask 200 people, "Does Colgate toothpaste give you sex appeal?" One hundred of these people have seen an advertisement for Colgate and 100 have not. We want to know if the percentage of people in the saw-the-ad group who say it gives you sex appeal is higher than the percentage in the didn't-see-the-ad group who say it gives you sex appeal. The problem here is just like the problem with the coin-flip example. The odds of the saw-the-ad group being higher are around 50%. One of the two groups *has* to be higher. (If they happened to tie, we'd assume that the ad didn't work.)

The logic for getting around this problem is the same as it was for the coin-flip example. For the coin flip, we judged 100 heads in a row as a highly improbable event *assuming that the coin was fair*. The odds of a fair coin coming up heads 100 times in a row are very small. So if we observe that event – 100 heads in a row – we conclude that our assumption must have been wrong. It's *not* a fair coin. So the saw-the-ad group being higher than the other group may also not be improbable – but what if that group was *much* more likely to answer yes? Just as we judged that there was something funny about the coin, so too we should judge that there is something funny about people who have seen the ad – at least funny when it comes to answering our question.

Of course *funny* in this context means "improbable." In the case of the coin, we knew how to calculate the "funniness," or improbability, of events because we knew the number of possible outcomes (two) and the probability of each individual outcome (.5), so it was easy to calculate the odds of successive events, as shown in Table 3.1. But here's our next problem: How do we calculate the "funniness," or probability, of other types of events? How much worse does the vocabulary of kids who watched videos have to be compared to that of kids who didn't watch videos before we're prompted to say, "Hey, these two groups of kids are not equal. If they were equal, their vocabularies would be equal. But their vocabularies are *very* unequal."

All of this description of coins, advertisements, and experiments is really a prelude to the lesson. I'm trying to get students to understand and care about the goal of the lesson, which is to explain how we can determine the probability of an event occurring by chance. That is the conflict for this lesson. Our worthy adversary in pursuit of this goal is not Darth Vader but the fact that most events we care about are not like coin flips – they don't have a limited number of outcomes (heads or tails) for which we know the probabilities (50%). That's a complication, which we address with a particular type of graph called a histogram; but implementing this approach leads to a further complication: we need to calculate the area under the curve of the histogram, which is a complex computation. The

TABLE 3.1: The odds, out of 10 tosses, of tossing a successively greater number of heads.

Number of tosses	Approximate probability of all heads
1	0.5
2	0.25
3	0.125
4	0.063
5	0.031
6	0.016
7	0.008
8	0.004
9	0.002
10	0.001

problem is solved by the Z-score, which is the point of the lesson (Figure 3.12).

A couple of things are worth noticing. A good deal of time – often 10 or 15 minutes of a 75-minute class – is spent setting up the goal, or to put it another way, persuading students that it's important to know how to determine the probability of a chance event. The material covered during this setup is only peripherally related to the lesson. Talking about coin flips and advertising campaigns doesn't have much to do with Z-scores. It's all about elucidating the central conflict of the story.

Spending a lot of time clarifying the conflict follows a formula for storytelling from, of all places, Hollywood. The central conflict in a Hollywood film typically starts about 20 minutes into the standard 100-minute movie. The screenwriter uses that 20 minutes to acquaint you with the characters and their situation so that when

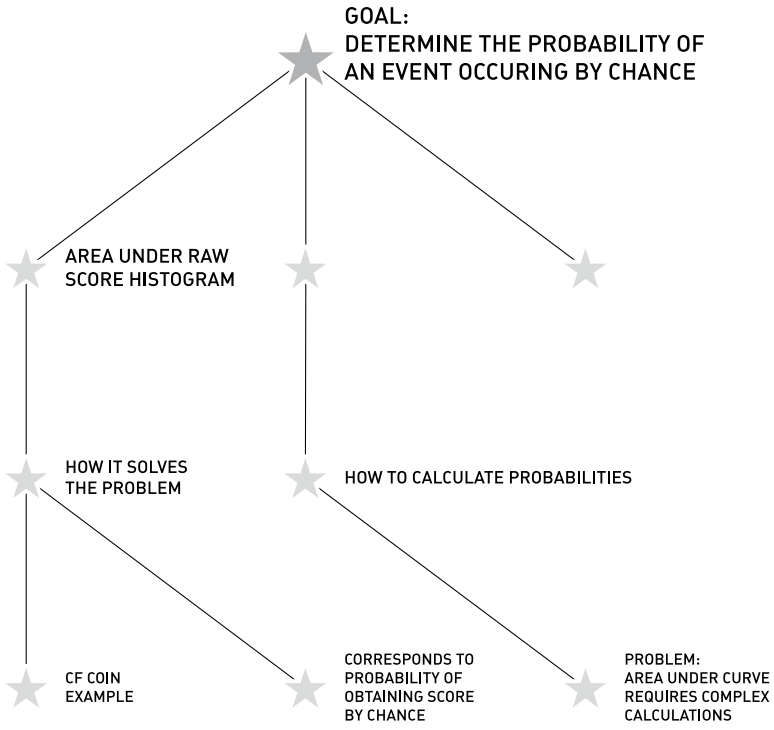


FIGURE 3.12: Part of the organizational scheme for a lesson plan on the Z-score transformation for a statistics class. Source: © Greg Culley.

the main conflict arises, you're already involved and you care what happens to the characters. A film may start with an action sequence, but that sequence is seldom related to what will be the main story line of the movie. James Bond movies often start with a chase scene, but it's always part of some other case, not the case that Bond will work on for the bulk of the movie. The conflict for that case is introduced about 20 minutes into the film.

When it comes to teaching, I think of it this way: The material I want students to learn is actually the answer to a question. *On its own, the answer is almost never interesting.* But if you know the question, the answer may be quite interesting. That's why making the question clear is so important. But as I said in Chapter 1, I sometimes feel that we, as teachers, are so focused on getting to the answer, we spend insufficient time making sure that students understand the question and appreciate its significance. To us, the question and its importance are obvious. To them, they aren't.

Let me close this section by emphasizing again that there are many ways in which one can be a good teacher. I don't mean to imply that, according to cognitive science, every teacher should use a story structure to shape his or her lesson plans. It's just one way that we can help ensure that students think about meaning. I am implying – well, no, I'm stating – that every teacher should get his or her students to think about the meaning of material – except sometimes, which is the subject of the next section.

But What If There Is No Meaning?

This chapter began by posing the question, *How can we get students to remember something?* The answer from cognitive science is straightforward: get them to think about what it means. In the previous section I suggested one method – story structure – for getting students to think about meaning.

It's fair to ask, however, whether there is content that students must learn that is pretty darn close to meaningless. For example, how can you emphasize meaning when students are learning the odd spelling of *Wednesday,* or that *enfranchise* means to give voting rights, or that *travailler* is the French verb for *work*? Some material just doesn't seem

to have much meaning, or if it does – Wednesday's origins are to do with the Germanic god Wodan – you're not sure it's worth going into it. Such material seems especially prevalent when one is entering a new field or domain of knowledge (Figure 3.13).

Memorizing meaningless material is commonly called *rote memorization*. I will say more about rote memory in Chapter 4, but for the moment let's just acknowledge that a student who has memorized, say, the first nine elements of the periodic table has little or no idea why she has done so or what the ordering might mean. There are times when a teacher may deem it important for a student to have such knowledge ready in long-term memory as a stepping-stone to understanding something deeper. How can a teacher help the student get that material into long-term memory?

There is a group of memory tricks, commonly called *mnemonics,* that help people memorize content when it is not meaningful. Some examples are listed in Table 3.2.

I'm not a big fan of the peg-word and method-of-loci methods because they are hard to use for different sets of material. If I use my mental walk (back porch, dying pear tree, gravel driveway, and so on) to learn some elements of the periodic table, can I use the same

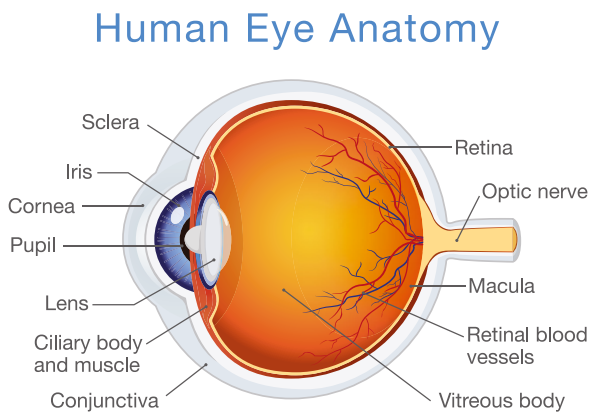


FIGURE 3.13: A biology teacher may be most interested in students appreciating physiology and function of the eye . . . but it's hard to talk about function without being able to name the parts of the eye. So at the beginning of the unit, the teacher may choose to have the students commit some of the anatomic names to memory. Source: © Shutterstock/solar22.

TABLE 3.2: Common mnemonic methods.

Mnemonic	How it works	Example
Peg word	Memorize a series of peg words by using a rhyme – for example, one is a bun, two is a shoe, three is a tree, and so on. Then memorize new material by associating it via visual imagery with the pegs.	To learn the list "radio, shell, nurse," you might imagine a radio sandwiched in a bun, a shoe on a beach with a conch in it, and a tree growing nurses' hats like fruit.
Method of loci	Memorize a series of locations on a familiar walk – for example, the back porch of your house, a dying pear tree, your gravel driveway, and so on. Then visualize new material at each "station" of the walk.	To learn the list "radio, shell, nurse," you might visualize a radio hanging by its cord on the banister of your back porch, someone grinding shells to use as fertilizer to revitalize the dying tree, and a nurse shoveling fresh gravel onto your driveway.
Link method	Visualize each of the items connected to one another in some way.	To learn the list "radio, shell, nurse," you might imagine a nurse listening intently to a radio while wearing large conch shells on her feet instead of shoes.
Acronym method	Create an acronym for the to-be-remembered words, then remember the acronym.	To learn the list "radio, shell, nurse," you might memorize the word RAiSiN using the capitalized letters as cues for the first letter of each word you are to remember.
First-letter method	Similar to the acronym method, this method has you think of a phrase, the first letter of which corresponds to the first letter of the to-be-remembered material.	To learn the list "radio, shell, nurse," you could memorize the phrase "Roses smell nasty," then use the first letter of each word as a cue for the words on the list.

TABLE 3.2: Continued

Mnemonic	How it works	Example
Songs	Think of a familiar tune to which you can sing the words.	To learn the list "radio, shell, nurse," you could sing the words to the tune of "Happy Birthday to You."

Mnemonics help you to memorize meaningless material.

walk to learn the conjugations for some French verbs? The problem is that there might be interference between the two lists; when I get to the gravel driveway, I get confused about what's there because I've associated two things with it.

The other methods are more flexible because students can create a unique mnemonic for each thing they learn. The acronym method and the first-letter method are effective, but students do need to have some familiarity with the material to be learned. I always think of the sentence "Dear Kate, Please Come Over For Great Spaghetti" for the order of taxonomy in biology. If I didn't already know the names, these first-letter cues wouldn't do me much good, but the first letter of each pushes me over the edge from tip-of-the-tongue to ready recall. The acronym method works in much the same way and has the same limitation.

Setting to-be-learned information to music or chanting it to a rhythm also works quite well. Most English speakers learned the letters of the alphabet by singing the ABC song, and I've heard the periodic table of elements set to Offenbach's "Can-Can Music." Music and rhythm do make words remarkably memorable. I (and millions of other kids) grew up seeing School House Rocks cartoons on Saturday mornings, which offered snippets of geography, civics, math, or grammar. I remember being in middle school and subvocally singing "Conjunction Junction" to myself in English class.†

> Conjunction Junction, what's your function?
> Hooking up words and phrases and clauses.

The difficulty with songs is that they are more difficult to generate than the other mnemonic devices.

Why do mnemonics work? Primarily by giving you cues. The acronym ROY G. BIV gives you the first letter of each color in the spectrum of visible light. The first letter is quite a good cue to memory. As I discuss in the next chapter, memory works on the basis of cues. If you don't know anything about a topic, or if the things you're trying to remember are confusing because they are arbitrary (there's nothing about red that makes it obvious that its wavelength is longer than green), mnemonics help because they impose some order on what you are trying to remember.

Summary

If we agree that background knowledge is important, then we must think carefully about how students can acquire that background knowledge – that is, how learning works. Learning is influenced by many factors, but one factor trumps the others: students remember what they think about. That principle highlights the importance of getting students to think about the right thing at the right time. We usually want students to understand what things *mean*, which sets the agenda for a lesson plan. How can we ensure that students think about meaning? I offered one suggestion, which is to use the structure of a story. Stories are easily comprehended and remembered, and they are interesting; but one can't get students to think about meaning if the material *has* no meaning. In that case, it may be appropriate to use a mnemonic device.

Implications for the Classroom

Thinking about meaning helps memory. How can teachers ensure that students think about meaning in the classroom? Here are some practical suggestions.

Review Each Lesson Plan in Terms of What the Student Is Likely to Think About

This sentence may represent the most general and useful idea that cognitive psychology can offer teachers. The most important thing

about schooling is what students will remember after the school day is over, and there is a direct relationship between what they think during the day and their later memory. So it's a useful double-check for every lesson plan to try to anticipate what the lesson will *actually* make students think about (rather than what you hope it will make them think about). Doing so may make it clear that students are unlikely to get what the teacher intended out of the lesson.

For example, I once observed a high school social studies class work in groups of three on projects about the Spanish Civil War. Each group was to examine a different aspect of the conflict (for example, compare it to the US Civil War, or consider its impact on today's Spain) and then teach the remainder of the class what they had learned, using the method of their choice. Students in one group noticed that PowerPoint was loaded on the computers, and they were very enthusiastic about using it to teach their bit to the other groups. (This was a while ago, when PowerPoint was not in common use in high schools.) The teacher was impressed by their initiative and gave his permission. Soon all of the groups were using PowerPoint. Many students had some familiarity with the basics of the program, so it could have been used effectively. The problem was that the students changed the assignment from "learn about the Spanish Civil War" to "learn esoteric features of PowerPoint." There was still a lot of enthusiasm in the room, but it was directed toward using animations, integrating videos, finding unusual fonts, and so on. At that point the teacher felt it was far too late to ask all of the groups to switch, so he spent much of the rest of the week badgering students to be sure their presentation had content, not just flash.

This story illustrates one of the reasons that experienced teachers are so good. This teacher clearly didn't let students use PowerPoint the next year, or he thought of a way to keep them on task. Before you have accumulated these experiences, the next best thing is to think carefully about how your students will react to an assignment and what it will make them think about.

Think Carefully About Attention Grabbers

Almost every teacher I have met likes, at least on occasion, to start class with an attention grabber. If you hook students early in the

lesson, they should be curious to know what is behind whatever surprised or awed them. But attention grabbers may not always work. Here's a conversation I had with my oldest daughter when she was in sixth grade.

DAD: What did you do in school today?

REBECCA: We had a guest in science. He taught us about chemicals.

DAD: Oh yeah? What did you learn about chemicals?

REBECCA: He had this glass? That looked like water? But when he put this little metal thingy in it, it boiled. It was so cool. We all screamed.

DAD: Uh-huh. Why did he show you that?

REBECCA: I don't know.

The guest surely planned this demonstration to pique the class's interest, and that goal was met. I'm willing to bet that the guest followed the demonstration with an age-appropriate explanation of the phenomenon but that information was not retained. Rebecca didn't remember it because she was still thinking about how cool the demonstration was. You remember what you think about.

Another teacher once told me she wore a toga to class on the first day she began a unit on ancient Rome. I am sure that got her students' attention. I am also sure it continued to get their attention – that is, to distract them – once the teacher was ready for them to think about something else.

Here's one more example. A guest in a biology class asked the students to think of the very first thing they had ever seen. The students mulled that question over and generated such guesses as "the doctor who pulled me out," "Mom," and so forth. The guest then said, "Actually, the first thing each of you saw was the same. It was pinkish, diffuse light coming through your mother's belly. Today we're going to talk about how that first experience affected how your visual system developed, and how it continues to influence the way you see today." I love that example because it grabbed the students' attention and left them eager to hear more about the subject of the lesson.

As I alluded to earlier in the chapter, I think it is very useful to use the beginning of class to build student interest in the material by understanding the question the underlies the lesson for the day – or as the story framing puts it, to develop the conflict. You might consider, however, whether the beginning of the class is really when they need an attention grabber. In my experience, the transition from one subject to another (or for older students, from one classroom and teacher to another) is enough to buy at least a few minutes of attention from students. It's usually the middle of the lesson that needs a little drama to draw students back from whatever reverie they might be in. But regardless of when it's used, think hard about how you will draw a connection between the attention grabber and the point it's designed to make. Will students understand the connection, and will they be able to set aside the excitement of the attention grabber and move on? If not, is there a way to change the attention grabber to help students make that transition? Perhaps the toga could be worn over street clothes and removed after the first few minutes of class. Perhaps the "metal thingy" demonstration would have been better *after* the basic principle was explained and students were prompted to predict what might happen.

Use Discovery Learning with Care

Discovery learning refers to students learning by exploring objects, discussing problems with classmates, designing experiments, or any of a number of other techniques that use student inquiry rather than teacher telling students things. Indeed, the teacher ideally serves more as a resource than as the director of the class. Discovery learning has much to recommend it, when it comes to memory. If students have a strong voice in deciding which problems they want to work on, they will likely be engaged in the problems they select, and will likely think deeply about the material, with attendant benefits. An important downside, however, is that what students will think about is less predictable. If students are left to explore ideas on their own, they may well explore mental paths that are not profitable. If memory is the residue of thought, then students will remember incorrect "discoveries" as much as they will remember the correct ones. (There are other pluses and minuses to discovery learning, but here I'm focusing on memory.)

Now this doesn't mean that discovery learning should never be used, but it does suggest a principle for when to use it. Discovery learning is probably most useful when the environment gives prompt feedback about whether the student is thinking about a problem in a useful way. One of the best examples of discovery learning is when kids learn to use a computer, whether they are learning an operating system, a complex game, or a Web application. Students show wonderful ingenuity and daring under these circumstances. They are not afraid to try new things, and they shrug off failure. They learn by discovery! Note, however, that computer applications have an important property: when you make a mistake, it is immediately obvious. The computer does something other than what you intended. This immediate feedback makes for a great environment in which "messing around" can pay off. (Other environments aren't like that. Imagine a student left to "mess around" with frog dissection in a biology class.) If the teacher does not direct a lesson to provide constraints on the mental paths that students will explore, the environment itself can do so effectively in a discovery learning context, and that will help memory.

Design Assignments So That Students Will Unavoidably Think About Meaning

If the goal of a lesson plan is to get students to think about the meaning of some material, then it's pretty clear that the best approach is one in which thinking about meaning is unavoidable. One of the things that has always amazed me as a memory researcher is the degree to which people do not know how their own memory system works. It doesn't do any good to tell people, "Hey, I'm going to test your memory for this list of words later," because people don't know what to do to make the words memorable. But if you give people a simple task in which they *must* think of the meaning – for example, rating how much they like each word – they will remember the words quite well.

This idea can be used in the classroom as well as in the laboratory. At the start of this chapter I said that asking fourth graders to bake biscuits was not a good way to get them to appreciate what life on the

Underground Railroad may have been like because they spend too much time thinking about measuring flour and milk. The goal was to get students thinking about the perilous experience of enslaved people seeking escape. So a more effective lesson would be to lead students to consider that experience by, for example, asking them where they supposed people on the underground railroad obtained food, how they were able to prepare it, how they were able to pay for it, and so forth.

Don't Be Afraid to Use Mnemonics

Many teachers I have met shudder at the use of mnemonics. They conjure up images of nineteenth-century schoolrooms with children chanting rhymes of the state capitals. But as bad as a classroom would be if a teacher used *only* mnemonics, they do have their time and place, and I don't think teachers should have this instructional technique taken away from them.

When is it appropriate to ask students to memorize something before it has much meaning? Probably not often, but there will be times when a teacher feels that some material – meaningless though it may be now – must be learned for the student to move forward. Typical examples would be learning letter-sound associations prior to reading and learning foreign language vocabulary.

It might also be appropriate to memorize some material using mnemonics in parallel with other work that emphasizes meaning. When I was in elementary school, I was not required to memorize the multiplication table. Instead I practiced using different materials and techniques that emphasized what multiplication actually means. These techniques were effective, and I readily grasped the concept. But by about fifth grade, not knowing the multiplication table by heart really slowed me down because the new things I was trying to learn had multiplication embedded in them. So every time I saw 8×7 within a problem I had to stop and figure out the product. In the sixth grade I moved to a new school, where my teacher quickly figured out what was going on and made me memorize the multiplication table. It made math a lot easier for me, although it took a few weeks before I would admit it.

Try Organizing a Lesson Plan Around the Conflict

There is a conflict in almost any lesson plan, if you look for it. This is another way of saying that the material we want students to know is the answer to a question – and the question is the conflict. The advantage of being very clear about the conflict is that it yields a natural progression for topics. In a movie, trying to resolve a conflict leads to new complications. That's often true of school material too.

Start with the content you want your students to learn, and think backward to the intellectual question it poses. For example, the state may mandate that sixth graders will learn the models of the atom that were competing at the turn of the twentieth century. These are the answers. What is the question? In this story, the goal is to understand the nature of matter. The obstacle is that the results of different experiments appear to conflict with one another. Each new model that is proposed (Rutherford, cloud, Bohr) seems to resolve the conflict but then generates a new complication – that is, experiments to test the model seem to conflict with other experiments. If this organization seems useful to you, you might spend a good bit of time thinking about how to illustrate and explain to students the question, "What is the nature of matter?" How could that question intrigue sixth graders?

As I've emphasized, structuring a lesson plan around conflict can be a real aid to student learning. Another feature I like is that, if you succeed, you are engaging students with the actual substance of the discipline. I've always been bothered by the advice "make it relevant to the students" for two reasons. First, it often feels to me that it doesn't apply. Is the Epic of Gilgamesh relevant to students in a way they can immediately understand? Is trigonometry? Making these topics relevant to students' daily lives will be a strain, and students will probably think it's phony. Second, if I can't convince students that something is relevant to them, does that mean I shouldn't teach it? If I'm continually trying to build bridges between students' daily lives and their school subjects, the students may get the message that school is always about them, whereas I think there is value, interest, and beauty in learning about things that don't have much to do with me. I'm not saying it never makes sense to talk about things students are interested in. What I'm suggesting is that student interests should

not be the main driving force of lesson planning. Rather, they might be used as initial points of contact that help students understand the main ideas you want them to consider, rather than as the reason or motivation for them to consider these ideas.

In the previous chapter I argued that students must have background knowledge in order to think critically. In this chapter I discussed how memory works, in the hope that by understanding this we can maximize the likelihood that students will learn this background knowledge; much of the answer to how we can do this was concerned with thinking about meaning. But what if students don't understand the meaning? In the next chapter I discuss why it is hard for students to comprehend the meaning of complex material, and what you can do to help.

Notes

*I made up this statistic.

†Search "Schoolhouse Rock: conjunction junction" on YouTube.

Further Reading

Less Technical

Baddeley, A., Eyesenck, M. W., & Anderson, M. C. (2015). *Memory,* 2. Hove, UK: Psychology Press. A textbook, but quite lucid, covering all the basic science of human memory.

Brown, P. C., Roediger, H. L. III, McDaniel, M. A. (2014). *Make It Stick*. Cambridge, MA: Belknap. Highly readable, this book includes information about the basic science of memory and useful applications of that science.

Dunlosky, J., Rawson, K. A., Marsh, E. J., Nathan, M. J., & Willingham, D. T. (2013). What works, what doesn't? *Scientific American Mind* (September-October), 47–53. A brief, reader-friendly version of a much longer technical article that reviews techniques for committing things to memory.

McKee, R. (1997). *Story*. New York: Harper Collins. There are many instruction manuals meant to help you tell stories effectively. This is one of the best known, based on McKee's legendary screenwriting workshops.

More Technical

Arya, D. J., & Maul, A. (2012). The role of the scientific discovery narrative in middle school science education: an experimental study. *Journal of Educational Psychology*, 104(4),

1022–1032. Experiment showing that middle schoolers remember science content better when it's framed in a discovery narrative, compared to a traditional, nonnarrative format.

Chang-Kredl, S., & Colannino, D. (2017). Constructing the image of the teacher on Reddit: best and worst teachers. *Teaching and Teacher Education*, 64, 43–51. Researchers examined 600 responses to queries for memories of "best" and "worst" teacher on the popular discussion website, Reddit. Common themes related to knowledge/organization of subject matter, and personal qualities of the teachers.

Kim, S-i. (1999). Causal bridging inference: a cause of story interestingness. *British Journal of Psychology,* 90, 57–71. In this study the experimenter varied the difficulty of the inference that readers had to make to understand the text and found that texts were rated as most interesting when the inferences were of medium-level difficulty.

Kleemans, M., Schaap, G., & Suijkerbuijk, M. (2018). Getting youngsters hooked on news. *Journalism Studies,* 19, 2108–2125. A study showing that adults of all ages remember news stories better when they are communicated in a narrative, as opposed to the traditional "inverted pyramid" format for news stories.

Markman, A. B. (2012). Knowledge representation. In: *The Oxford Handbook of Thinking and Reasoning* (eds. K. Holyoak & R. Morrison), pp. 36–51. New York: Oxford University Press. How memories are represented in the mind, and what representation actually means.

Roediger, H. L. (1980). The effectiveness of four mnemonics in ordering recall. *Journal of Experimental Psychology: Human Learning and Memory,* 6(5), 558–567. An article evaluating four different mnemonics (three of which are mentioned in this chapter) showing that all enhance memory.

Seiver, J. G., Pires, M., Awan, F., & Thompson, W. (2019). Retention of word pairs as a function of level of processing, instruction to remember, and delay. *Journal of Cognitive Psychology*, 31(7), 665–682. A recent example of a study showing that deeper (meaning-based) processing leads to better memory than shallow processing and that intention to learn has no impact.

Discussion Questions

1. Emotion leads to better memory, but intentionally inducing emotions in students to help them remember feels manipulative. Is there a way to use emotion in the classroom?
2. Intention to learn doesn't influence memory. But surely that can't mean that it doesn't matter whether or not students care about school. What's the resolution to this apparent paradox?
3. I suggested that trying to tie school content to student interests runs a risk; exactly because Instagram is so interesting to my daughter, she's more likely to go off on her own stream of thought. What are some solutions to this problem?
4. I've suggested it's useful to think of one's teaching on two very broad dimensions: organization/knowledge and emotional warmth. Self-reflect a bit . . . What do you see as your strengths and weaknesses on each of these? What do you want to work on? What resources are available to you for support?

5. I've suggested that using a story structure to organize a lesson plan might help maintain student interest. But for that to happen, students must understand and care about the conflict that drives the narrative. Consider the next class you are to teach. (Or pick another, if you like.) If the content of that lesson plan is the answer, do you think your students know the question? How easy is it to articulate the question in an age-appropriate manner? What might lead them to care about the question? Perhaps a parallel with their own concerns? Or maybe a puzzle that will entice them in the moment, and will lead to the next part of the question?
6. Do you ask students to learn content that is relatively devoid of meaning? As I've said, I think the practice sometimes makes sense, but I understand why people are critical of it, so I think it's worthwhile to be reflective about it. If you do ask students to engage in this sort of memorization, you can make it much easier for them if you create mnemonics. I'm usually a big fan of letting students do the creating whenever they can, and you might give that a go, but be forewarned that there is research showing people are not very good at writing their own mnemonics. One solution might be to see if students come up with good ones and share the best. And if none of them are very good, have ready one that you've prepared.
7. How do you think your students spend their leisure time? Are any of the common activities enriching for the sort of cognitive work discussed in this chapter? Are there closely related activities that might be more enriching? If so, what avenues are open to teachers to encourage them?
8. I've suggested anticipating what a lesson will *actually* make students think about. How hard is that to do? Would it be easier to do with someone else's lesson? Or if you hadn't thought about the lesson for a week, and so could bring a fresh eye to it?

4

Why Is It So Hard for Students to Understand Abstract Ideas?

Question: I once observed a teacher helping a student with geometry problems on the calculation of area. After a few false starts, the student accurately solved a word problem calling for the calculation of the area of a tabletop. A problem came up shortly thereafter that required the student to calculate the area of a soccer field. He looked blank and, even with prompting, did not see how this problem was related to the one he had just solved. In his mind, he had solved a problem about tabletops, and this problem was about soccer fields – completely different. Why are abstract ideas – for example, the calculation of area – so difficult to comprehend in the first place and, once comprehended, so difficult to apply when they are expressed in new ways?

Answer: Abstraction is the goal of schooling. The teacher wants students to be able to apply classroom learning in new contexts, including those outside of school. The challenge is that the mind seems not to care for abstractions. The mind seems to prefer the concrete. That's why, when we encounter an abstract principle – for example, a law in physics such as force = mass × acceleration – we ask for a concrete example to help us understand. The cognitive principle that guides this chapter is:

> We understand new things in the context of things we already know, and most of what we know is concrete.

Thus it is difficult to comprehend abstract ideas and difficult to apply them in new situations. The surest way to help students understand an abstraction is to expose them to many different versions of the abstraction – that is, to have them solve area calculation problems about tabletops, soccer fields, envelopes, doors, and so on. There are some techniques that may hurry this process.

Understanding Is Disguised Remembering

In Chapter 2 I emphasized that factual knowledge is important to schooling. In Chapter 3 I described how to make sure that students acquire those facts – that is, I described how things get into memory. But I've also assumed that students understand what you're trying to teach them. As you know, you can't bank on that. It's often difficult for students to understand new ideas, especially ones that are *really* novel, meaning they aren't related to other things they have already learned. What do cognitive scientists know about how students understand things?

The answer is that they understand new ideas (things they don't know) by relating them to old ideas (things they do know). That sounds fairly straightforward. It's a little like the process you go through when you encounter an unfamiliar word. If you don't know, for example, what *ab ovo* means, you look it up in a dictionary. There you see the definition "from the beginning." You already know those words, so now you have some idea of what *ab ovo* means.*

The fact that we understand new ideas by relating them to things we already know helps us understand some principles that are familiar to every teacher. One principle is the usefulness of analogies; they help us understand something new by relating it to something we already know about. For example, suppose I'm trying to explain Ohm's law to a student who knows nothing about electricity. I tell her that electricity is power created by the flow of electrons and that Ohm's law describes some influences on that flow. I tell her that Ohm's law is defined this way:

$$I = V / R$$

I is a measure of electrical current, that is, how fast the electrons are moving. *V,* or voltage, is the potential difference, which causes electrons to move. Potential will "even out," so if you have a difference in electrical potential at two points, that difference causes movement of electrons. *R* is a measure of resistance. Some materials are very effective conduits for electron movement (low resistance) whereas others are poor conduits (high resistance).

Although it's accurate, this description is hard to understand, and textbooks usually offer an analogy to the movement of water. Electrons moving along a wire are like water moving through a pipe. If there is high pressure at one end of the pipe (for example, created by a pump) and lower pressure at the other end, the water will move, right? But the movement is slowed by friction from the inside of the pipe, and it can be slowed even more if we partially block the pipe. We can describe how fast the water moves with a measure such as gallons per minute. So, in terms of the water analogy, Ohm's law says that how fast water flows depends on the amount of water pressure and the amount of resistance in the pipes. That analogy is helpful because we are used to thinking about water moving in pipes. We call on this prior knowledge to help us understand new information, just as we call on our knowledge of the word *beginning* to help us understand *ab ovo*.

So new things are understood by relating them to things we already understand. That's why analogies help (Figure 4.1).





FIGURE 4.1: "force = mass × acceleration" is difficult to understand because it is abstract. It's easier to understand with a concrete example. Use the same force (a woman swinging a bat) to hit different masses – a baseball or an automobile. We understand that the acceleration of the ball and the acceleration of the car will be quite different. Source: Game © Getty Images/SAMURAI JAPAN; car © Getty Images/FilmMagic.

Another consequence of our dependence on prior knowledge is our need for concrete examples. As you know, abstractions – for example, force = mass × acceleration, or a description of the poetical meter iambic pentameter – are hard for students to understand, even if all of the terms are defined. They need concrete examples to illustrate what abstractions mean. They need to actually hear:

> Is *this* the *face* that *launched* a *thou*sand *ships*?
> And *burnt* the *top*less *tow*ers of *Ili*um?

and

> Rough *winds* do *shake* the *dar*ling *buds* of *May*
> And *sum*mer's *lease* hath *all* too *short* a *date*

and other examples before they can feel they understand iambic pentameter.

Examples help not only because they make abstractions concrete. Concrete examples don't help much if they're not familiar. Suppose you and I had the following conversation:

ME Different scales of measurement provide different types of information. Ordinal scales provide ranks, whereas on an interval scale the differences between measurements are meaningful.

YOU That was utter gobbledygook.

ME OK, here are some concrete examples. The Mohs scale of mineral hardness is an ordinal scale, whereas a successful Rasch model provides an interval measurement. See?

YOU I think I'll go get a coffee now.

So it's not simply that giving concrete examples helps. (A better explanation of scales of measurement appears in Figure 4.2.) They must also be *familiar* examples, and the Mohs scale and the





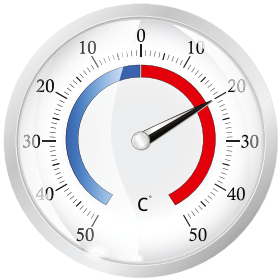




FIGURE 4.2: There are four, and only four, ways that numbers on a scale relate to one another. In a nominal scale, each number refers to one thing but the numbers are arbitrary – for example, the number on a baseball jersey tells you nothing about the player. On an ordinal scale, the numbers are meaningful, but they tell you nothing about the distance between them. In a horse race, for example, you know that the first place horse was ahead of the second place finisher, but you don't know by how much. On an interval scale, not only are the numbers ordered but also the intervals are meaningful – for example, the difference between 10° and 20° is the same as the interval between 80° and 90°. "Zero" on an interval scale is arbitrary; that is, zero degrees Celsius doesn't mean there is no temperature. A ratio scale, such as age, has a true zero point: that is, zero years means the absence of any years. Source: Baseball © Shutterstock/Suzanne Tucker; horserace © Shutterstock/Don Blaise; thermometer © Shutterstock/Flipser; generations © Getty Images/MNPhotoStudios.

Rasch model are not familiar to most people. It's not the concreteness, it's the familiarity that's important; but most of what students are familiar with is concrete, because abstract ideas are so hard to understand.

Ludwig Wittgenstein speculated that "The problems are solved, not by giving new information, but by arranging what we have always known." He was right. Understanding new ideas is mostly a matter of getting the right old ideas into working memory and then rearranging them – making comparisons we hadn't made before, or thinking about a feature we had previously ignored. Consider the

explanation of force in Figure 4.1. You know what happens when you hit a ball with a bat, and you know what happens when you hit a car with a bat, but have you ever before held those two ideas in mind at the same time and considered that the different outcome is due to the difference in mass?

Now you see why I claim that understanding is disguised remembering. No one can pour new ideas into a student's head directly. Every new idea must build on ideas that the student already knows. To get a student to understand, a teacher (or a parent or a book or a video or a peer) must ensure that the right ideas from the student's long-term memory are pulled up and put into working memory. In addition, the right features of these memories must be attended to, that is, compared or combined or somehow manipulated. For me to help you understand the difference between ordinal and interval measurement, it's not enough for me to say, "Think of a thermometer and think of a horse race." Doing so will get those concepts into working memory, but I also have to make sure they are compared in the right way.

We all know, however, that it's not really this simple. When we give students one explanation and one set of examples, do they understand? Usually not. Now that you have looked at Figure 4.2, would you say you "understand" scales of measurement? You know more than you did before, but your knowledge probably doesn't feel very deep, and you may not feel confident that you could identify the scale of measurement for a new example, say, centimeters on a ruler (Figure 4.3).

To dig deeper into what helps students understand, we need to address these two issues. First, even when students "understand," there are really degrees of comprehension. One student's understanding can be shallow while another's is deep. Second, even if students understand in the classroom, this knowledge may not transfer well to the world outside the classroom. That is, when students see a new version of what is at heart an old problem, they may think they are stumped, even though they recently solved the same problem. They don't know that they know the answer! In the next two sections I elaborate on each issue, that is, on shallow knowledge and on lack of transfer.



FIGURE 4.3: Here are three other examples of scales of measurement: centimeters (as measured by a ruler), people's ratings from 1 to 7 of how much they like espresso, and the numbered tracks on a playlist. Which scale of measurement does each of these examples use? Source: Ruler © Shutterstock/Olga Kovalenko; espresso © Getty Images/Guido Mieth; playlist © Daniel Willingham.

Why Is Knowledge Shallow?

Every teacher has had the following experience: You ask a student a question (in class or perhaps on a test), and the student responds using the exact words you used when you explained the idea or with the exact words from the textbook. Although his answer is certainly correct, you can't help but wonder whether the student has simply memorized the definition by rote and doesn't understand what he's saying (Figure 4.4).

This scenario brings to mind a famous problem posed by the philosopher John Searle.[1] Searle wanted to argue that a computer might display intelligent behavior without really *understanding* what it is doing. He posed this thought problem: Suppose a person is alone in a room. We can slip pieces of paper with Chinese writing on them under the door. The person in the room speaks no Chinese but responds to each message. She has an enormous book, each page of which is divided



FIGURE 4.4: The fear that students merely parrot ideas that they don't understand is not new; this image is from mid-nineteenth century France. (It's not very complimentary of teachers, either.) Source: © Getty Images/DEA/ICAS94.

into two columns. There are strings of Chinese characters on the left and on the right. She scans the book until she matches the character string on the slip of paper to a string in the left-hand column. Then she carefully copies the characters in the right-hand column onto the piece of paper and slips it back under the door. We have posed a question in Chinese and the person in the room has responded in Chinese. Does the person in the room understand Chinese?

Almost everyone says no. She's giving sensible responses, but she's just copying them from a book. Searle provided this example to argue that computers, even if they display sophisticated behavior such as comprehending Chinese, aren't thinking in the way in which we understand the term. We might say the same thing about students. Rote knowledge might lead to giving the right response, but it doesn't mean the student is thinking.†

We can see examples of "sophisticated answers" that don't have understanding behind them in "student bloopers," which are regular forwards in my e-mail inbox. Some of them are good examples of rote knowledge; for example, "Three kinds of blood vessels are arteries, vanes, and caterpillars," and "I would always read the works of the Cavalier poets, whose works always reflected the sentiment 'Cease the day!'" In addition to giving us a chuckle, these examples show that the student has simply memorized the "answer" without comprehension.

The fear that students will end up with no more than rote knowledge has been almost a phobia among some educators, but the truth is that rote knowledge is probably relatively rare. *Rote knowledge* (as I'm using the term) means you have *no* understanding of the

material. You've just memorized words, so it doesn't seem odd to you that Cavalier poets, best known for light lyrics of love and their romantic view of life, would have the philosophy "Cease the day!"

Much more common than rote knowledge is what I call *shallow knowledge,* meaning that students have some understanding of the material but their understanding is limited. We've said that students come to understand new ideas by relating them to old ideas. If their knowledge is shallow, the process stops there. Their knowledge is tied to the analogy or explanation that has been provided. They can understand the concept only in the context that was provided. For example, you know that "Seize the day!" means "Enjoy the moment without worrying about the future," and you remember that the teacher said that "Gather ye rosebuds while ye may" (from Herrick's *To the Virgins, to Make Much of Time*) is an example of this sentiment. But you don't know much more. If the teacher provided a new poem, you would be hard put to say whether it was in the style of a Cavalier poet (Figure 4.5)



FIGURE 4.5: Title page to Sir John Suckling's "Fragmenta Aurea." Even if a student was unfamiliar with this particular work, if she had deep knowledge of Cavalier poetry, she would not be surprised to see terms like "wonder and delight" and "new spirit" applied to his work. Source: © Getty Images/Culture Club.

We can contrast shallow knowledge with deep knowledge. A student with deep knowledge knows more about the subject, and the pieces of knowledge are more richly interconnected. The student understands not just the parts but also the *whole*. This understanding allows the student to apply the knowledge in many different contexts, to talk about it in different ways, to imagine how the system as a whole would change if one part of it changed, and so forth. A student with deep knowledge of Cavalier poetry would be able to recognize elements of Cavalier ideals in other literatures, such as ancient Chinese poetry, even though the two forms seem very different on the surface. In addition, the student would be able to consider what-if questions, such as "What might Cavalier poetry have been like if the political situation in England had changed?" They can think through this sort of question because the pieces of their knowledge are so densely interconnected. They are interrelated like the parts of a machine, and the what-if question suggests the replacement of one part with another. Students with deep knowledge can predict how the machine would operate if one part were to be changed.

Obviously teachers want their students to have deep knowledge, and most teachers try to instill it. Why then would students end up with shallow knowledge? One obvious reason is that a student just might not be paying attention to the lesson. The mention of "rosebuds" makes a student think about the time she fell off her skateboard into the neighbor's rose bush, and the rest of the poem is lost on her. There are other, less obvious reasons that students might end up with shallow knowledge.

Here's one way to think about it. Suppose you plan to introduce the idea of government to a first-grade class. The main point you want students to understand is that people living or working together set up rules to make things easier for everyone. You will use two familiar examples – the classroom and students' homes – and then introduce the idea that there are other rules that larger groups of people agree to live by. Your plan is to ask your students to list some of the rules of the classroom and consider why each rule exists. Then you'll ask them to list some rules their families have at home and consider why those rules exist. Finally, you'll ask them to name some rules

that exist outside of their families and classroom, which you know will take a lot more prompting. You hope your students will see that the rules for each group of people – family, classroom, and larger community – serve similar functions (Figure 4.6).



FIGURE 4.6: Most classrooms have rules, sometimes made public in a list like this one. Understanding the need for rules in a classroom may be a stepping-stone to understanding why a group of people working or playing together benefits from a set of rules. Source: © Frank Hebbert Creative Commons CC BY 2.0.

A student with rote knowledge might later report, "Government is like a classroom because both have rules." The student has no understanding of what properties the two groups have in common. The student with shallow knowledge understands that a government is like a classroom because both groups are a community of people who need to agree on a set of rules in order for things to run smoothly and to be safe. The student understands the parallel but can't go beyond it. So for example, if asked, "How is government *different* from our school?" the student would be stumped. A student with deep knowledge would be able to answer that question and might successfully extend the principle to consider other groups of people who might need to form rules, for example, his group of friends playing pickup basketball.

This example can help us understand why all students might not get deep knowledge. The target knowledge – that groups of people need rules – is pretty abstract. It would appear, then, that the right strategy would be to teach the abstract concept directly; after all, that's what you want them to learn. But I said before that students don't

understand abstractions easily or quickly. They need examples. That's why it would be useful to use the example of the classroom rules. In fact, a student might be able to say, "When people come together in a group, they usually need some rules," but if the student doesn't understand how a classroom, a family, and a community all exemplify that principle, he doesn't really get it. Thus deep knowledge means understanding *everything* – both the abstraction and the examples, and how they fit together. So, it is much easier to understand why most students have shallow knowledge, at least when they begin to study a new topic. Deep knowledge is harder to obtain than shallow.

Why Doesn't Knowledge Transfer?

This chapter is about students' understanding of abstractions. If someone understands an abstract principle, we expect they will show *transfer*. When knowledge transfers, that means they have successfully applied old knowledge to a new problem. Now, in some sense *every* problem is new; even if we see the same problem twice, we might see it in a different setting, and because some time has passed, we could say we have changed, even if only a little bit. Most often when psychologists talk about transfer they mean the new problem looks different from the old one, but we do have applicable knowledge to help us solve it. For example, consider the following two problems:

> *Jayden bought three jars of cocktail sauce and a tray of jumbo shrimp for a total of $40. If the price of a tray of jumbo shrimp is $25, what is the price of a jar of cocktail sauce?*
>
> *Last week Julia drove her car to work and back three times and made one trip to see her friend, putting a total of 80 miles on her car. Her friend lives 50 miles away, round trip. How far does Julia live from work?*

Each problem requires subtracting part of the total (shrimp tray or trip to friend) then dividing the result to get a unit value (price of cocktail sauce or number of miles to work). The two problems differ in what psychologists call their *surface structure* – that is, the first problem is framed in terms of buying food and the second in terms of putting miles on a car. The problems have the same *deep structure* because they require the same steps for solution. The surface structure of each problem is a way to make the abstraction concrete.

Obviously the surface structure of a problem is unimportant to its solution. We would expect that a student who can solve the first problem should be able to solve the second problem, because it's the deep structure that matters. Nevertheless, people seem to be much more influenced by surface structure than they ought to be. In a classic laboratory experiment showing the influence of surface structure,[2] the experimenters asked college students to solve the following problem:

> *Suppose you are a doctor faced with a patient who has a malignant tumor in his stomach. It is impossible to operate on the patient, but unless the tumor is destroyed, the patient will die. There is a kind of ray that can be used to destroy the tumor. If the rays reach the tumor all at once at a sufficiently high intensity, the tumor will be destroyed. Unfortunately, at this intensity the healthy tissue the rays pass through on the way to the tumor will also be destroyed. At lower intensities the rays are harmless to healthy tissue, but they will not affect the tumor either. What type of procedure might be used to destroy the tumor with the rays and at the same time avoid destroying the healthy tissue?*

If a participant didn't solve it – and most couldn't – the experimenter told him or her the solution: send a number of rays of low intensity from different directions and have them all converge on the tumor; that way each weak ray can safely pass through the healthy tissue, but all of the rays will meet at the tumor, so it will be destroyed. The experimenter made sure the participant understood the solution, then presented them with the following problem:

> *A dictator ruled a small country from a fortress. The fortress was situated in the middle of the country, and many roads radiated outward from it, like spokes on a wheel. A great general vowed to capture the fortress and free the country of the dictator. The general knew that if his entire army could attack the fortress at once, it could be captured. But a spy reported that the dictator had planted mines on each of the roads. The mines were set so that small bodies of men could pass over them safely, because the dictator needed to be able to move troops and workers about; however, any large force would detonate the mines. Not only would this activity blow up the road, but the dictator would destroy many villages in retaliation. How could the general attack the fortress?*

The two problems have the same deep structure: when combined forces will cause collateral damage, scatter your forces and have them converge on the point of attack from different directions. That solution may seem obvious, but it wasn't obvious to the participants. Only 30% solved the second problem, even though they had *just heard* the conceptually identical problem and its solution (Figure 4.7).

You experienced a similar phenomenon in Chapter 1. I described the disk-and-pegs problem and then introduced a problem with the same deep structure but a different surface structure – a tea ceremony, where tasks were to be transferred from the host to the most senior guest. If you're like most people, you didn't perceive that the two problems shared the same deep structure.

Why? The answer goes back to how we understand things. When we read or when we listen to someone talking, we are interpreting what is written or said in light of what we already know about similar topics. For example, suppose you read this passage: "Felix, the second named storm of the season to become a hurricane, gained strength with astonishing speed overnight, with wind speeds of 150 miles per hour and stronger gusts. Forecasters predict that the storm's path may take it to the coast of Belize within the next 12

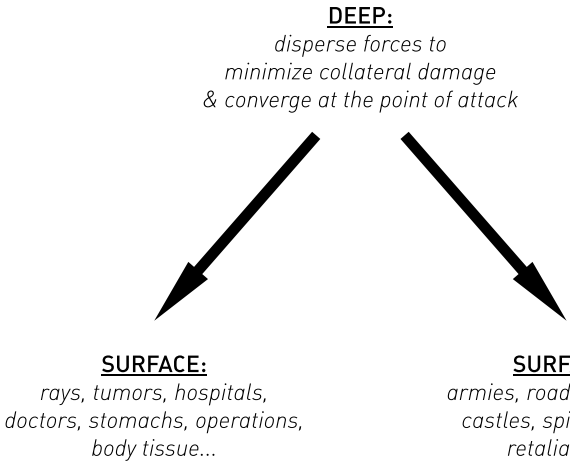


FIGURE 4.7: The relationship of deep and surface structure in the rays-tumor and the armies-fortress problems. Source: © Greg Culley.

hours." In Chapter 2 I emphasized that prior knowledge is necessary to comprehend this sort of text. If you don't know what sort of storms are named and where Belize is, you don't fully understand these sentences. In addition, your background knowledge will also shape how you interpret *what comes next*. The interpretation of these sentences drastically narrows how you will interpret new text. For example, when you see the word *eye* you won't think of the organ that sees, nor of the loop at the top of a needle, nor of a bud on a potato, nor of a round spot on a peacock's feather. You'll think of the center of a hurricane. And if you see the word *pressure* you'll immediately think of atmospheric pressure, not peer-group pressure or economic pressure.

So our minds assume that new things we read (or hear) will be related to what we've just read (or heard). This fact makes understanding faster and smoother. Unfortunately, it also makes it harder to see the deep structure of problems. That's because our cognitive system is always struggling to make sense of what we're reading or hearing, to find relevant background knowledge that will help us interpret the words, phrases, and sentences. But the background knowledge that seems applicable almost always concerns the surface structure. When people read the tumor-and-rays problem, their cognitive system narrows the interpretation of it (just as it does for the hurricane sentences) according to what sort of background knowledge the reader has, and that's likely to be some knowledge of tumors, rays, doctors, and so forth. When the person later reads the other version of the problem, the background knowledge that seems relevant concerns dictators, armies, and fortresses. That's why transfer is so poor. The first problem is taken to be one about tumors, and the second problem is interpreted as being about armies.

The solution to this problem seems self-evident. Why not tell people to think about the deep structure as they read? The problem with this advice is that the deep structure of a problem is not obvious. Even worse, an almost limitless number of deep structures *might* be applicable. As you're reading about the dictator and the castle, it's hard to simultaneously think, Is the deep structure the logical form *modus tollens*? Is the deep structure one of finding the least common

multiple? Is the deep structure Newton's third law of motion? To see the deep structure, you must understand how all parts of the problem relate to one another, and you must know which parts are important and which are not. The surface structure, on the other hand, is perfectly obvious: this problem is about armies and fortresses.

The researchers who did the tumor-and-rays experiment also tried telling people, "Hey, that problem about the tumor and the rays might help you in solving this problem about armies and a fortress." When they told them that, almost everyone could solve the problem. The analogy was easy to see. The fortress is like the tumor, the armies are like the rays, and so on. So the difficulty was that people simply didn't realize that the two problems were analogous.

Other times we get poor transfer even when students know that a new problem shares deep structure with another problem they've solved. Picture a student who knows that the algebra word problem he's working on is an illustration of solving simultaneous equations with two unknowns, and there are examples in his textbook outlining the process. The surface structures of the solved textbook problem and the new problem are different – one is about a hardware store's inventory and the other is about mobile phone plans – but the student knows he should disregard the surface structure and focus on the deep structure. To use the textbook example to help himself, however, he must figure out how the surface structure of each problem maps onto the deep structure. It's as though he understands the tumor problem and its solution, but when presented with the fortress problem he can't figure out whether the armies are playing the role of the rays, the tumor, or the healthy tissue. As you might guess, when a problem has lots of components and lots of steps in its solution, it more often happens that transfer is hampered by difficulty in mapping from a solved problem to the new one. Occasionally, the results can be comic (Figure 4.8).

This discussion makes it sound as though it's virtually impossible for knowledge to transfer, as though we are powerless to look beyond the surface structure of what we read or hear. Obviously that's not true. *Some* of the participants in the experiments I described did think of using the problem they had seen before, although the percentage who

did so is surprisingly small. In addition, when faced with a novel situation, an adult will usually approach it in a more fruitful way than a child will. Somehow the adult is making use of his or her experience so that knowledge is transferring. In other words, it's a mistake to think of our old knowledge transferring to a new problem only when the source of that background knowledge is obvious to us. When we see the tumor-and-rays problem for the first time, we don't simply say, "I've never seen that problem or one like it before, so I give up." We have strategies for coming up with solutions, even though they may ultimately not work. Those strategies must be based on our experience – on other problems we've solved, things we know about tumors and rays, and so on. In that sense, we're *always* transferring knowledge of facts and knowledge of problem solutions, even when we feel like we've never seen this sort of problem before. Not very much is known about this type of transfer, however, precisely because it's so hard to trace where it comes from.



FIGURE 4.8: Comedies sometimes make use of failures to transfer known solutions to new problems. In *Office Space*, the lead characters plan revenge on their company using a plan that worked well in the movie *Superman III*, but their version fails to transfer to their context because they didn't account for relevant differences in the companies. Source: © Getty Images/Handout.

In the next chapter I discuss, among other things, how to maximize the chances that knowledge will transfer.

Summary

Understanding happens when we assemble ideas from memory in new ways, for example, when we compare finishers in a horse race to readings on a thermometer as a way of understanding scales of measurement. Abstract ideas are difficult because we rely on

what's already in memory to understand new ideas, and most of what's in memory is concrete. Our understanding of new ideas is initially shallow because deep understanding requires more connections among the components of the idea; it simply takes more experience and therefore more time to develop deep understanding. Once we've worked with the same idea in a number of guises, we can appreciate its deep structure – the functional relationships among the components of the idea. But until then, our understanding will cling to the examples we've seen, and transfer will be uncertain.

Implications for the Classroom

The message of this chapter seems rather depressing: it's hard to understand stuff, and when at last we do, it won't transfer to new situations. It's not quite that grim, but the difficulty of deep understanding shouldn't be underestimated. After all, if understanding were easy for students, teaching would be easy for you! Here are a few ideas on how to meet this challenge in the classroom.

Be Wary of Promises of Broad Transfer

The history of education is littered with abandoned attempts to teach students a skill that will "train the mind" and help students think more critically about, well, everything.

In the nineteenth century students studied Latin in the hopes that the logical structure of the grammar would make logical thinking habitual. In one of the early (and notable) successes of educational psychology, Edward Thorndike showed that students who took Latin did no better in their other classes than students who didn't. The logic of Latin didn't rub off.[3]

The hope of very broad transfer was reignited in the 1960s, when some educational theorists reasoned that computer programming called for logical thinking, as well as the use of some broadly applicable concepts (e.g., recursion). Maybe if kids learned to code, those thinking skills would apply broadly. It works a little better than teaching them Latin, but only a little.

Some people think learning chess will make kids logical thinkers, others advocate for teaching them to play a musical instrument. What you've seen in this chapter is the persistent specificity of learning. I think that playing chess is wonderful and so is playing an instrument, but if you want children to think logically about science, teach them how science works. If you want them to learn to evaluate an argument in expository prose, teach that. Don't teach a different skill in the vain hope that it will burnish some other skill.

To Help Student Comprehension, Provide Examples and Ask Students to Compare Them

As noted earlier, experience helps students to see deep structure, so provide that experience via lots of examples. Another strategy that might help is to ask students to compare different examples. Thus an English teacher trying to help her students understand the concept of *irony* might provide the following examples:

- In *Oedipus Rex,* the Delphic Oracle predicts that Oedipus will kill his father and marry his mother. Oedipus leaves his home in an effort to protect those he believes to be his parents, but in so doing sets in motion events that eventually make the prediction come true.
- In *Romeo and Juliet*, Romeo kills himself because he believes that Juliet is dead. When Juliet awakens, she is so distraught over Romeo's death that she commits suicide.
- In *Othello*, the noble Othello implicitly trusts his advisor Iago when he tells Othello that his wife is unfaithful, whereas it is Iago who plots against him.

The students (with some prompting) might come to see what each example has in common with the others. A character does something expecting one result, but the opposite happens because the character is missing a crucial piece of information: Oedipus is adopted, Juliet is alive, Iago is a deceiver. The audience knows that missing piece of information and therefore recognizes what the outcome will be. The play is even more tragic because as the audience watches the

events unfold, they know that the unhappy ending could be avoided if the character knew what they know.

Dramatic irony is an abstract idea that is difficult to understand, but comparing diverse examples of it may help students by forcing them to think about deep structure. Students know that the point of the exercise is not shallow comparisons such as, "Each play has men and women in it." As discussed in Chapter 2, we remember what we think about. This method of getting students to think about deep structure may help.

Make Deep Knowledge the Spoken and Unspoken Emphasis

You very likely let your students know that you expect them to learn what things mean – that is, to learn the deep structure. You should also ask yourself whether you send unspoken messages that match that emphasis. What kind of questions do you pose in class? Some teachers pose mostly factual questions, often in a rapid-fire manner: "What does *b* stand for in this formula?" or "What happens when Huck and Jim get back on the raft?" The low-level facts are important, as I've discussed, but if that's all you ask about, it sends a message to students that that's all there is.

Assignments and assessments are another source of implicit messages about what is important. When a project is assigned, does it demand deep understanding or is it possible to complete it with just a surface knowledge of the material? If your students are old enough that they take quizzes and tests, be sure these test deep knowledge. Students draw a strong implicit message from the content of tests: if it's on the test, it's important.

Make Your Expectations for Deep Knowledge Realistic

Although deep knowledge is your goal, you should be clear-eyed about what students can achieve and about how quickly they can achieve it. Deep knowledge is hard won and is the product of much practice. Don't despair if your students don't yet have a deep understanding of a complex topic. Shallow knowledge is much better than

no knowledge at all, and shallow knowledge is a natural step on the way to deeper knowledge. It may be years before your students develop a truly deep understanding, and the best that any teacher can do is to start them down that road or continue their progress at a good pace.

In this chapter I've described why abstract ideas are so difficult to understand and why they are so difficult to apply in unfamiliar situations. I said that practice in thinking about and using an abstract idea is critical to being able to apply it. In the next chapter I talk at greater length about the importance of practice.

Notes

*You may have noticed a problem. If we understand things by relating them to what we already know, how do we understand the *first* thing we ever learn? To put it another way, how do we know what *beginning* means? If we look that word up we see that it means "a start." And if we look up the word *start* we see it defined as "a beginning." It seems, then, that defining words with other words won't really work, because we quickly run into circular definitions. This is a fascinating issue, but it's not central to the discussion in this chapter. A short answer is that some meanings are directly understandable from our senses. For example, you know what *red* means without resorting to a dictionary. These meanings can serve as anchors for other meanings and help us avoid the circularity problem that we saw in the *ab ovo* example.

†Not everyone is persuaded by Searle's argument. Different objections have been raised, but the most common is that the example of the person alone in a room doesn't capture what computers might be capable of.

Further Reading

Less Technical

De Bruyckere, P., Kirschner, P. A., & Hulshof, C. D. (2020). If you learn A, will you be better able to learn B? *American Educator, 44* (1), 30–34 . Given what's known about transfer, considers the question "should all students learn to play a musical instrument? Or learn to play chess?"

Hofstadter, D., & Sander, E. (2013). *Surfaces and Essences: Analogy as the Fuel and Fire of Thinking*. New York: Basic Books. This book is "less technical," but it's far from breezy; a 500-page consideration of the centrality of analogy to human thought. It's dense, but Pulitzer Prize–winner Hofstadter writes beautifully.

More Technical

Carey, S. (2011). The origin of concepts: a précis. *Behavioral and Brain Sciences, 34,* 113–167. A summary of a much longer book, this article describes Carey's theory of how children (and adults) come to understand new concepts, with particular emphasis on the idea that there is an innate starting point for human conceptual development.

Goldstone, R., & Day, S. B. (2012). New conceptualizations of transfer of learning. *Educational Psychologist, 47*(3). The entire issue of this prominent journal was devoted to transfer, including articles on what makes it difficult and the best ways to promote it.

Holyoak, K. J. (2012). Analogy and relational reasoning. In: *The Oxford Handbook of Thinking and Reasoning* (eds K. Holyoak & R. Morrison). 234–259. New York: Oxford University Press. Very useful overview of the psychology underlying analogy.

Hoyos, C., & Gentner, D. (2017). Generating explanations via analogical comparison. *Psychonomic Bulletin & Review*, *24*(5), 1364–1374. An example of the power of comparison in helping to understand new ideas, here applied in six-year-olds.

Mayer, R. E. (2004). Teaching of subject matter. *Annual Review of Psychology, 55,* 715–744. A comprehensive overview of specific subject matter domains, with special attention to transfer.

Sala, G., & Gobet, F. (2017). Does far transfer exist? Negative evidence from chess, music, and working memory training. *Current Directions in Psychological Science*, *26*(6), 515–520. This article summarizes a good deal of research, concluding that studying chess or learning to play a musical instrument makes you proficient in those activities, but doesn't transfer to overall mental acuity.

Scherer, R., Siddiq, F., & Viveros, B. S. (2018). The cognitive benefits of learning computer programming: a meta-analysis of transfer effects. *Journal of Educational Psychology* 111(5): 764–792. Article showing that teaching students to program a computer leads to some positive transfer to measures of creative thinking, mathematics, metacognition, spatial skills, and reasoning. There's not transfer to verbal tests, consistent with the idea that the transfer may not be a matter of generalized logical thinking but of overlap of the training and the transfer tests.

Discussion Questions

1. Deep knowledge is hard won and therefore takes time to develop. Not all knowledge will be deep. How important is it for adults – teachers, parents – to decide what knowledge ought to be deep, versus for students themselves to decide? What are the trade-offs?
2. What is the student's perspective on "rote knowledge"? Why might acquiring "rote knowledge" actually seem appealing? I mentioned the role of testing as one example of what a teacher might (unwittingly) do to encourage students to learn by rote. What other factors might play a role?
3. It would seem that the transfer problem might be very troublesome in teacher education; future teachers often study in university classrooms about how to be a teacher and then are expected to implement that knowledge as teachers in their own classrooms.

How well did this knowledge transfer in your own experience? Reflect on what features of your teacher education led to good or poor transfer.

4. What are some of the more profound and useful "deep structures" of the classroom that you've learned? What have you learned about students or about classroom situations or dynamics that might look different on the surface in different classrooms but reflects a deeper truth across situations?
5. I've said that deep knowledge is hard won; it requires working with ideas in many contexts, and that often takes significant time. This makes it likely that, in your classroom, there are some ideas that students will have some familiarity with from the previous year but still need to work with more, and other ideas that you will introduce, understanding that student knowledge will be probably shallow at the end of the year, but you know will be revisited next year. How confident are you about the communication in your school about what this knowledge is and communication about the competence of individual students around this knowledge? If you're not confident, what might be done to improve communication?
6. Shallow knowledge seems like it might go hand in hand with responding on autopilot, as described in Chapter 1. You perceive situation X and know that response Y generally leads to a good outcome, so you don't think any more deeply about X, for example, whether other situations that appear different on the surface have the same deep structure as X. Two results might follow. You don't see that Y might be useful in other situations, and because you never think much about X, you never come to a deeper understanding of it. Everyone has Xs and Ys in their practice. What are yours? What are the situations that you solve on autopilot and might benefit from deeper reflection?

5

Is Drilling Worth It?

Question: Drilling has been given a bad name. The very use of the military term drill in place of the more neutral term practice implies something mindless and unpleasant that is performed in the name of discipline rather than for the student's profit. Then too, the phrase "drill and kill" has been used as a criticism of some types of instruction; the teacher drills the students, which is said to kill their innate motivation to learn. On the other side of this debate are educational traditionalists who argue that students must practice in order to learn some facts and skills they need at their fingertips – for example, math facts such as 5 + 7 = 12. Few teachers would argue that drilling boosts students' motivation and sense of fun. Does the cognitive benefit make it worth the potential cost to motivation?

Answer: The bottleneck in our cognitive system is the ability to juggle several ideas in our mind simultaneously. For example, it's easy to multiply 19 × 6 in your head, but nearly impossible to multiply 184 930 × 34 004. The processes are the same, but in the latter case you "run out of room" in your head to keep track of the numbers. The mind has a few tricks for working around this problem. One of the most effective is practice, because it reduces the amount of "room" that mental work requires. The cognitive principle that guides this chapter is:

> It is virtually impossible to become proficient at a mental task without extended practice.

You cannot become a good soccer player if, as you're trying to control the ball while running, you focus on how hard to hit the ball, which surface of your foot to use, and so on. Low-level processes like this must become automatic, leaving room for more high-level concerns, such as game strategy. Similarly, you cannot become good at algebra without knowing math facts by heart. Students must practice some things. But not everything needs to be practiced. In this chapter I elaborate on why practice is so important, and I discuss which material is important enough to merit practice and how to implement practice in a way that students find maximally useful and interesting.

Why practice? One reason is to gain a minimum level of competence. A child practices tying her shoelaces with a parent's or teacher's help until she can reliably tie them without supervision. We also practice tasks that we can perform but that we'd like to improve. A professional tennis player can hit a serve into his opponent's court almost every time, but he nevertheless practices serving in an effort to improve the speed and placement of the ball. In an educational setting, both reasons – mastery and skill development – seem sensible. Students might practice long division until they master the process, that is, until they can reliably work long-division problems. Other skills, such as writing a persuasive essay, might be performed adequately, but even after students have the rudiments down, they should continue to practice in an effort to refine and improve their abilities.

These two reasons to practice – to gain competence and to improve – are self-evident and probably are not very controversial. Less obvious are the reasons to practice skills when it appears you have mastered something and it's not obvious that practice is making you any better. Odd as it may seem, that sort of practice is essential to schooling. It yields three important benefits: it reinforces the basic skills that are required for the learning of more advanced skills, it protects against forgetting, and it improves transfer.

Practice Enables Further Learning

To understand why practice is so important to students' progress, let me remind you of two facts about how thinking works.

Figure 5.1 (which you also saw in Chapter 1) shows that working memory is the site of thinking. Thinking occurs when you combine information in new ways. That information might be drawn from the environment or from your long-term memory or from both. For example, when you're trying to answer a question like "How are a butterfly and a dragonfly alike?" your thoughts about the characteristics of each insect reside in working memory as you try to find points of comparison that seem important to the question.

A critical feature of working memory, however, is that it has limited space. If you try to juggle too many facts or to compare them in too many ways, you lose track of what you're doing. Suppose I said, "What do a butterfly, a dragonfly, a chopstick, a pillbox, and a scarecrow have in common?"* That's simply too many items to compare simultaneously. As you're thinking about how to relate a pillbox to a chopstick, you've already forgotten what the other items are.

This lack of space in working memory is a fundamental bottleneck of human cognition. You could dream up lots of ways that your cognitive system could be improved – more accurate memory, more sharply focused attention, better vision, and so on – but if a genie comes out of a lamp and offers you one way to improve your mind, ask for more working memory capacity. People with more capacity are better

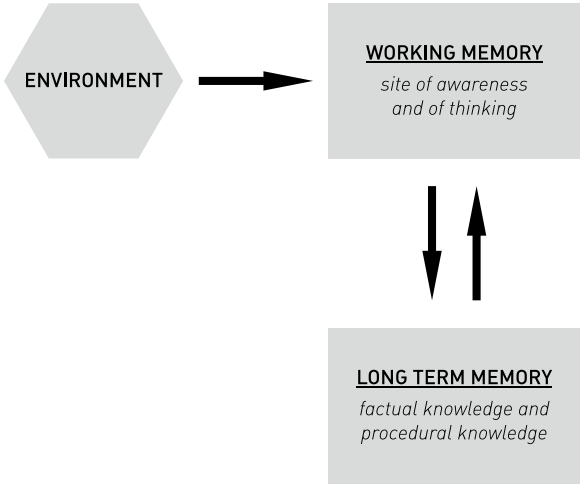


FIGURE 5.1: Our simple model of the mind. Source: © Greg Culley.

thinkers, at least for the type of thinking that's done in school. There is a great deal of evidence that this conclusion is true, and most of it follows a very simple logic: Take 100 people, measure their working memory capacity, then measure their reasoning ability,† and see whether their scores on each test tend to be the same. To a surprising degree, scoring well on a working-memory test predicts scoring well on a reasoning test, and a poor working-memory score predicts a poor reasoning score. (Working memory is not everything, however – recall that in Chapter 2 I emphasized the importance of background knowledge.)

Well, you're not going to get more working-memory capacity from a genie. And because this chapter is about practice, you might think I'm going to suggest that students do exercises that will improve their working memory. Sadly, such exercises don't exist. As far as anyone knows, working memory is more or less fixed – you get what you get, and practice does not change it. The last 10 years have seen many attempts to develop a training regimen to improve working memory, and there's been plenty of hype surrounding these programs, but research shows that an effective one has not yet been developed.

Though you can't increase working memory capacity, you can cheat this limitation. In Chapter 2 I discussed at length how to keep more information in working memory by compressing the information. In a process called *chunking,* you treat several separate things as a single unit. Instead of maintaining the letters *c, o, g, n, i, t, i, o,* and *n* in working memory, you chunk them into a single unit, the word *cognition*. A whole word takes up about the same amount of room in working memory that a single letter does. But chunking letters into a word requires that you know the word. If the letters were *p, a, z, z, e, s, c,* and *o,* you could chunk them effectively if you happened to know that *pazzesco* is an Italian word meaning "crazy." But if you didn't have the word in your long-term memory, you could not chunk the letters.

Thus, the first way to cheat the limited size of your working memory is through factual knowledge, which enables chunking. There is a second way: you can make the processes that manipulate information in working memory more efficient. In fact, you can make them

so efficient that they are virtually cost free. Think about learning to tie your shoes. Initially it requires your full attention and thus absorbs all of working memory, but with practice you can tie your shoes *automatically* (Figure 5.2).



FIGURE 5.2: There is a right and a wrong way to tie your shoes. It's all about the relationship of the bottom and top ties, similar to the difference between a square knot (which is secure) and a false knot (which isn't). Researchers actually showed, through simulated walking, that the right knot stays tied much longer. If a process is going to become automatic, make sure you do it right! Source: From "The roles of impact and inertia in the failure of a shoelace knot" by Christopher A. Daily-Diamond, Christine E. Gregg, & Oliver M. O'Reilly in *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences 473*(2200): 20160770, figure 14b. © Creative Commons CC BY 4.0.

What used to take all of the room in working memory now takes almost no room. As an adult you can tie your shoes while holding a conversation or even while working math problems in your head (in the unlikely event that the need arises). Another standard example, as I've already mentioned, is driving a car. When you first learn to drive, doing so takes all of your working-memory capacity. As with tying your shoes, it's the stuff you're *doing* that takes up the mental space – processes like checking the mirrors, monitoring how hard you're pressing the accelerator or brake to adjust your speed, looking at the speedometer, judging how close other cars are. Note that you're not trying to keep a lot of things (like letters) in mind simultaneously; when you do that, you can gain mental space by chunking. In this example, you're trying to do a lot of things in rapid succession. Of course, an experienced driver seems to have no problem in doing all of these things and can even do other things, such as talk to a passenger.

Mental processes can become automatized. Automatic processes require very little working memory capacity. They also tend to be quite rapid in that you seem to know just what to do without

even making a conscious decision to do it. An experienced driver glances in the mirror and checks her blind spot before switching lanes, without thinking to herself, "OK, I'm about to switch lanes, so what I need to do is check my mirrors and glance at the blind spot."

For an example of an automatic process, take a look at Figure 5.3 and name what each of the line drawings represents. Ignore the words and name the pictures.[1]

As you doubtless noticed, sometimes the words matched the pictures and sometimes they didn't. It probably felt more difficult to name the pictures when there was a mismatch. That's because when an experienced reader sees a printed word, it's quite difficult not to read it. Reading is automatic. Thus the printed word *pants* conflicts with the word you are trying to retrieve, *shirt*. The conflict slows your response. A child just learning to read wouldn't show this interference, because reading is not automatic for him. When faced with the letters *p, a, n, t,* and *s*, the child would need to painstakingly (and thus slowly) retrieve the sounds associated with each letter, knit them together, and recognize that the resulting combination of sounds forms the word *pants*.

Automaticity can not only happen by mental processes increasing in efficiency, but also through the development of mental representations. Instead of sounding out *p, a, n, t,* and *s,* you locate a visual representation in memory that matches that letter string. As we noted in Chapter 1, relying on memory is much faster than relying on mental processing – this is a crucial aspect of learning to read.

Math facts that become automatic operate the same way. When students are first introduced to arithmetic, they often solve problems by using counting strategies. For example, they solve 5 + 4 by beginning with 5 and counting up four more numbers to yield the answer 9. This strategy suffices to solve simple

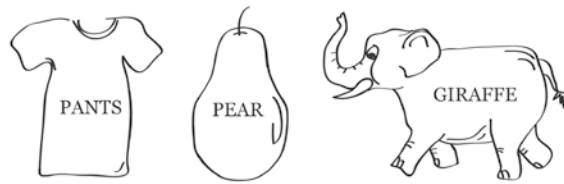


FIGURE 5.3: Name each picture, ignoring the text. It's hard to ignore when the text doesn't match the picture, because reading is an automatic process. Source: © Anne Carlyle Lindsay.

problems, but you can see what happens as problems become more complex. For example, in a multidigit problem like 97 + 89, the student who has not memorized math facts might add 7 and 9 by counting and get 16 as the result. Now the student must remember to write down the 6, then solve 9 + 8 by counting, while remembering to add the carried 1 to the result.

The problem is much simpler if the student has memorized the fact that 7 + 9 = 16, because she arrives at the correct answer for that subcomponent of the problem at a much lower cost to working memory. Finding a well-practiced fact in long-term memory and putting it into working memory places almost no demands on working memory. It is no wonder that students who have memorized math facts do better in all sorts of math tasks than students whose knowledge of math facts is absent or uncertain. And it's been shown that practicing math facts helps low-achieving students do better on more advanced mathematics.

Both reading and math offer good examples of the properties of automatic processes: (i) They happen very quickly. (ii) They are prompted by a stimulus in the environment, and if that stimulus is present, the process may occur (or the relevant memory may be retrieved) even if you wish it wouldn't. Thus you know it would be easier not to read the words in Figure 5.3, but you can't seem to avoid doing so. If you're an experienced driver and you're a passenger in a car that gets in a dangerous situation, your foot reaches out to push on a phantom brake pedal – the response is automatic. (iii) You are not aware of the components of the automatic process. For example, the component processes of reading (such as identifying letters) are never conscious. The word *pants* ends up in consciousness, but the mental processes necessary to arrive at the conclusion that the word is *pants* do not. The process is very different for a beginning reader, who is aware of each constituent step.

The example in Figure 5.3 gives a feel for how an automatic process operates, but it's an unusual example because the automatic process interferes with what you're trying to do. Most of the time automatic processes help rather than hinder. They help because they make room in working memory. Processes that formerly occupied working memory now take up very little space, so there is space

for *other* processes. In the case of reading, those "other" processes would include thinking about what the words actually mean. Beginning readers slowly and painstakingly sound out each letter and then combine the sounds into words, so there is no room left in working memory to think about meaning (Figure 5.4).

The same thing can happen even to experienced readers. When I was in high school a teacher asked a friend to read a poem out loud. When he had finished reading, she asked for his reaction to it. He looked blank for a moment and then admitted he had been so focused on reading without mistakes that he hadn't really noticed what the poem was about. Like a first grader, his mind had focused on word pronunciation, not on meaning. Predictably, the class laughed, but what happened was understandable, if unfortunate.

To review, I've said that working memory is the place in the mind where thinking happens – where we bring together ideas and transform them into something new. The difficulty is that there is only so much room in working memory, and if we try to put too much stuff in there, or to do too much with it, we get mixed up and lose the thread of the problem we were trying to solve, or the story we were trying to follow, or the factors we were trying to weigh in making a complex decision. People with larger working-memory capacities are better at these thinking tasks. Although we can't make our working memory larger, we *can* make the contents of working memory smaller in two ways: by making facts take up less room through

16 9 3 20 21 18 5
20 8 9 19
3 15 13 13 15 14 16 12 1 3 5
19 3 5 14 5

FIGURE 5.4: This sentence is written in a simple code: 1 = A, 2 = B, 3 = C, and so on, with a new line denoting a new word. The efforts of a beginning reader are a bit like your efforts to decode this sentence, because the value of each letter must be figured out. If you make the effort to decode the sentence, try doing it without writing down the solution; like the beginning reader, you will likely forget the beginning of the sentence by the time you are decoding the end of the sentence.[‡2] Source: © Greg Culley.

chunking, which requires knowledge in long-term memory as is discussed in Chapter 2; and by making the processes we use to bring information into working memory more efficient *or* to remove the need for those processes in the first place by creating a memory representation.

The next obvious question is: What is required to make these processes efficient or to create these memory representations? You know the answer: practice. There may be a workaround, a cheat, whereby you can reap the benefits of automaticity without paying the price of practicing. There may be one, but if there is, neither science nor the collected wisdom of the world's cultures has revealed it. As far as anyone knows, the only way to develop mental facility is to repeat the target process again and again and again.

You can see why I said that practice enables further learning. You may have "mastered" reading in the sense that you know which sounds go with which letters, and you can reliably string together sounds into words. So why keep practicing if you know the letters? You practice not just to get faster. What's important is getting so good at recognizing letters that retrieving word meaning becomes automatic. If it's automatic, you have freed working-memory space that used to be devoted to retrieving the sounds from long-term memory – space that can now be devoted to thinking about the meaning of sentences and paragraphs.

What's true of reading is true of most or all school subjects and of the skills we want our students to have. They are hierarchical. There are basic processes (like retrieving math facts or using deductive logic in science) that initially are demanding of working memory but with practice become automatic. Those processes must become automatic in order for students to advance their thinking to the next level. The great philosopher Alfred North Whitehead captured this phenomenon in this comment: "It is a profoundly erroneous truism, repeated by all copybooks and by eminent people when they are making speeches, that we should cultivate the habit of thinking of what we are doing. The precise opposite is the case. Civilization advances by extending the number of important operations which we can perform without thinking about them."[3]

Practice Makes Memory Long Lasting

Several years ago I had an experience that I'll bet you've had. I happened on some papers from my high school geometry class. I don't think I could tell you three things about geometry today, yet here were problem sets, quizzes, and tests, all in my handwriting, and all showing detailed problem solutions and evidence of factual knowledge.

This sort of experience can make a teacher despair. The knowledge and skills that my high school geometry teacher painstakingly helped me gain have vanished, which lends credence to the occasional student complaint, "We're never gonna *use* this stuff." So if what we teach students is simply going to vanish, what in the heck are we teachers doing?

Well, the truth is that I remember a *little* geometry. Certainly, I know much less now than I did right after I finished the class – but I do know more than I did before I took it. Researchers have examined student memory more formally and have drawn the same conclusion: we rapidly forget much (but not all) of what we have learned.

In one study, researchers tracked down students who had taken a one-semester, college-level course in developmental psychology between 3 and 16 years earlier.[4] The students took a test on the course material. Figure 5.5 shows the results, graphed separately for students who initially got an A in the course and students who got a B or lower. Overall, retention was not terrific. Just three years after the course, students remembered half or less of what they learned, and that percentage dropped until year seven, when it leveled off. The A students remembered more overall, which is not that surprising – they knew more to start with. But they forgot just like the other students did, and at the same rate.

So, apparently, studying hard doesn't protect against forgetting. If we assume that A students studied hard, we have to acknowledge that they forget at the same rate as everyone else. But something else does protect against forgetting: *continued* practice. In another study, researchers located people of varying ages and administered a test of basic algebra.[5] More than one thousand subjects participated

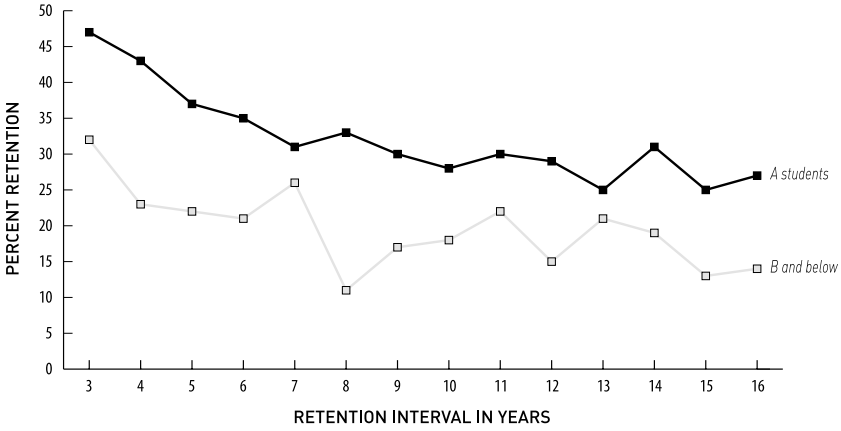


FIGURE 5.5: A graph showing how much students remembered of the material from a one-semester course in developmental psychology taken between 3 and 16 years earlier. Separate lines show the results for students who got an A in the course and those who got a B or lower. Source: "Very long-term memory for information taught in school" by J. A. Ellis, G. B. Semb, and B. Cole in *Contemporary Educational Psychology* 23: 419–433, figure 1, p. 428. Copyright © 1998. Reprinted with permission from Elsevier.

in the experiment, so there were lots of people with varied backgrounds. Most important was that they varied in how much math they had taken.

Have a look at Figure 5.6, which shows scores on an algebra test.[§] Everyone took the test at the same time, for the purpose of the experiment. The scores are separated into four groups on the basis of how many math courses people took in high school and college. Focus first on the bottommost curve. It shows the scores of people who took one algebra course. As you move from left to right, the time since they took the course increases, so the leftmost dot (around 60% correct) comes from people who *just* finished taking an algebra course, and the rightmost dot represents people who took algebra 55 years ago! The bottommost curve looks as you would expect it to; the longer it was since they took an algebra course, the worse they did on the algebra test.

The next curve up shows the scores of people who took more than one algebra course. As you might hope, they did better on the test but showed evidence of forgetting, just like the other group. Now look at the topmost line. These are the scores of people who took math courses beyond calculus. What's interesting about this line is

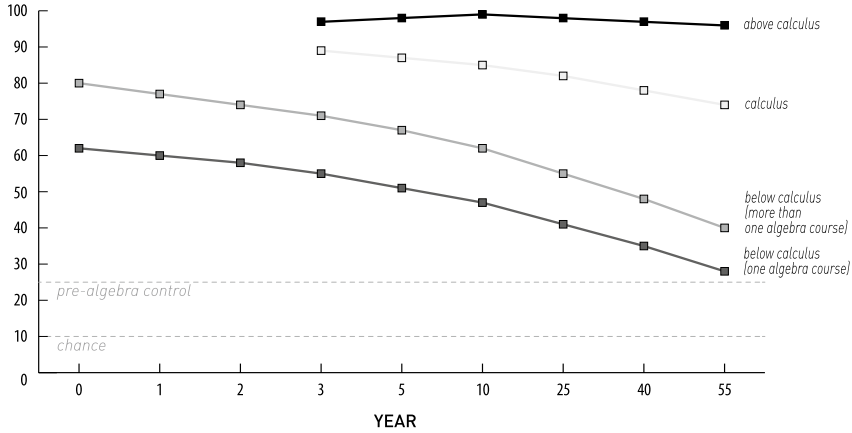


FIGURE 5.6: The performance on a basic algebra test by people who took the course between 1 month and 55 years earlier. The four lines of data correspond to four groups, separated by how much math they took after basic algebra. Source: From "Lifetime maintenance of high school mathematics content" by H. P. Bahrick and L. K. Hall in *Journal of Experimental Psychology: General* 120: 20–33, figure 1, p. 25. Copyright © 1991 by the American Psychological Association.

that it's flat! People who took their last math course more than 50 years ago still know their algebra as well as people who took it 5 years ago!

What's going on here? This effect is *not* due to people who go on to take more math courses being smarter or better at math. It's not shown in the graph, but just as in the previous study of developmental psychology, separating out students who got As, Bs, or Cs in their first algebra course makes no difference – they all forget at the same rate. To put it another way, a student who gets a C in his first algebra course but goes on to take several more math courses will remember his algebra, whereas a student who gets an A in his algebra course but doesn't take more math will forget it. That's because taking more math courses guarantees that you will continue to think about and *practice* basic algebra. If you practice algebra enough, you will effectively never forget it. Other studies have shown exactly the same results with different subject matter, such as Spanish studied as a foreign language, and street names from ones childhood neighborhood.

One thing these studies don't make clear is whether you get longer-lasting memory because you practice *more* or because your practice is stretched out over time. It's very likely that both matter.

It's also likely that you get benefits of practice to memory in the short run as well. Now, it's pretty obvious that reviewing helps memory: if I want to know that the French word for "umbrella" is "parapluie," I'm better off reviewing that fact five times than once. But suppose that I already seem to know that fact. I've got a long list of English-French vocabulary you're helping me learn, and I'm struggling with some, but the last two times we've run through the list when you said "umbrella" I was able to respond "parapluie." I feel I've got it. Is there any reason to keep studying it, or should it be discarded from my practice list?

The answer seems to be "keep studying it." This type of review is called *overlearning* – continued study after you seem to know something. It's exactly the type of practice we've been discussing in that it feels like it's not doing any good. You keep saying "umbrella" and I keep saying "parapluie," and I'm thinking "what's the point of this?"

But think of it this way. Suppose I knew each of the vocabulary items as well as I know umbrella/parapluie; I've gotten each of the items on the list correct two times in a row. If I'm tested three days from now, how will I do? Will I get 100% correct?

Probably not. Over the course of three days I'll forget some. And that's the benefit of overlearning. It offers protection against forgetting. It's hard to make ourselves do it, however, because the practice feels pointless; we're reviewing content it feels like we *know*. But we're ensuring that we will know it later. Now one obvious thing we could do to make overlearning feel less boring and pointless would be spread the studying out over time. I might be more inclined to study umbrella/parapluie if, a couple of days from now, I struggle to remember the right translation.

Researchers have also investigated the importance of *when* you study. The *when* refers not to time of day but to how you space your studying. Let me put it this way: The previous section emphasizes that studying for two hours is better than studying for one. OK. Suppose you decide to study something for two hours. How should you distribute those 120 minutes? Should you study for 120 minutes in a row? Or for 60 minutes one day, then 60 minutes the next? How about 30 minutes each week for four weeks?

Doing a lot of studying right before a test is commonly known as *cramming*. When I was in school students would brag about having done well on a test but that they couldn't remember any of the material a week later, because they had crammed. (An odd thing to brag about, I know.) Research bears out their boasts. If you pack lots of studying into a short period, you'll do okay on an immediate test, but you will forget quickly. If, on the other hand, you study in several sessions with delays between them, you may not do quite as well on the immediate test but, unlike the crammer, you'll remember the material longer after the test (Figure 5.7).

SUNDAY	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY
		1	2	3	4	5
6	7 Study	8 Study	9 Study	10 Study **STUDY** **STUDY** **STUDY** **STUDY**	11 Test **TEST**	12
13	14	15	16	17	18 Test **TEST**	19
20	21	22	23	24	25	26
27	28	29	30	31		

FIGURE 5.7: This simple figure illustrates what cognitive scientists call the spacing effect in memory. Student 1 (all capitals) studied four hours the day before the first test, whereas Student 2 (lowercase) studied for one hour on each of four days prior to the test. Student 1 will probably do a bit better on this test than Student 2, but Student 2 will do much better on the second test, administered a week later. Source: © Greg Culley.

The spacing effect probably does not surprise teachers; certainly we all know that cramming doesn't lead to long-lasting memory. In contrast, then, it makes sense that spreading out your studying would be better for memory. It's important, however, to make explicit two important implications of the spacing effect. We've been talking about the importance of practice, and we've just said that practice works better if it's spaced out. Distributing practice in the way we've described allows you to get away with *less practice* compared to bunching practice together. Spacing practice has another benefit. The type of practice we're discussing in this chapter means continuing to work at something that you've already mastered. By definition, that sounds kind of boring, even though it brings cognitive benefits. It will be somewhat easier for a teacher to make such tasks interesting for students if they are spaced out in time.

Practice Improves Transfer

In Chapter 4 I discussed at length the challenges of transferring what you already know to new situations. Remember the problem of attacking a castle with small groups of soldiers? Even when subjects had just heard an analogous story that contained the problem solution (attacking a tumor with the rays), they didn't transfer the knowledge to the castle-soldiers problem. As I mentioned then, transfer *does* occur, even when there is no obvious surface similarity between the situations. It occurs, but it's rare. What can we do to increase the odds? What factors make a student more likely to say, "Hey, I've seen problems like this before and I remember how to solve them!"?

It turns out that many factors contribute to successful transfer, but a few of them are especially important. As I've said, transfer is more likely when the surface structure of the new problem is similar to the surface structure of problems seen before. That is, the coin collector will more likely recognize that she can work a problem involving fractions if the problem is framed in terms of exchanging money rather than if a mathematically equivalent problem is framed as one of calculating the efficiency of an engine.

Practice is another significant contributor to good transfer. Working lots of problems of a particular type makes it more likely that you will recognize the underlying structure of the problem, even if you haven't seen this particular version of the problem before. Thus, reading the tumor-and-rays story makes it just a little more likely that you'll know what to do when you encounter the soldiers-and-castle problem; but if you've read *several* stories in which a force is dispersed and converges at a target point, it is much more likely that you'll recognize the deep structure of the problem.

To put it another way, suppose you read the following problem:

> You live in Canada and you're planning a trip to Mexico. You learn that you will save a significant charge if you bring Canadian dollars, exchange them for Mexican pesos once there, and pay for your hotel in cash. You're staying four nights and the hotel costs 100 Mexican pesos per night. What other information do you need in order to calculate how many dollars to bring, and what calculations will you make?

Why does an adult immediately see the deep structure of this problem but a fourth grader does not?

Researchers think there are a couple of reasons that this is so. The first reason is that practice makes it more likely that you will have understood the problem *before* this reading, and that you will remember the solution you used. If you didn't understand it at the time or you don't remember what you did to solve it, there's not much hope of it transferring to a new situation. That's pretty obvious. But suppose a fourth grader does understand the basic math underlying the problem. Why doesn't he see that it would be useful in solving the dollars/pesos problem? And how come you do?

Remember that in Chapter 4 I said that as you read, the possible interpretations of what comes next are drastically narrowed. I used

the example of a brief description of a hurricane and said that if you later saw the word *eye,* it wouldn't make you think of the eye with which you see, nor of the bud on a potato, and so on. The point is that as you're reading (or listening to someone talk), you are interpreting what is written, based on your associations with similar topics. You know about a lot of things that are associated with the word *eye,* and your mind picks out the right associates on the basis of the context of what you're reading. You don't have to make that selection consciously, thinking to yourself, "Hmm . . . now, I wonder which meaning of *eye* is appropriate here?" The right meaning just pops into mind.

Contextual information can be used not only for understanding individual words with several possible meanings but also for understanding the *relationships* of different things in what you read. For example, suppose I start to tell you a story: "My wife and I vacationed on a small island, and there is a peculiar law there. If two or more people are walking together after dark, they must each have a pen with them. The hotel had a reminder on the door and pens everywhere, but when we went out to dinner the first night, I forgot to bring mine."

As you read this story, you effortlessly understand the point: I violated a rule. Note that you don't have relevant background knowledge about the surface structure – you've never heard a rule like this before and it doesn't make much sense. But you have lots of experience with the functional relationship of the story elements; the story centers on a *permission*. In a permission relationship, you must fulfill a precondition before you are permitted to take an action. For example, in order to drink alcohol, you must be at least 21 years old. In order to be out at night on a small island with another person, you must have a pen. You also know that permissions not only have rules, they also usually have a consequence for breaking the rule. Thus, when I start telling you my odd story, you can likely predict where the story will go next: it's going to center on whether I got caught without my pen, and if I did get caught, what the consequences were. A sympathetic listener would humor me by saying, "Oh no! Did you get caught without your pen?" If instead a listener said, "Really? What kind of pens did the hotel give you?" I would think he didn't understand the point of the story.

When I tell you the story about the pen, the idea of a "permission rule" pops into your mind as automatically as the meaning of "center of a hurricane" does when you read the word *eye* in the hurricane story. You understand *eye* in context because you have seen the word *eye* used to refer to the center of a hurricane many times before. In the same way, the deep structure of a permission rule pops into mind when you hear the story about the pens, and for the same reason – you have lots of practice thinking about permission rules (Figure 5.8). The only difference between a permission rule and an eye is that the latter is a single word and the former is an idea shaped by the relationship of a few concepts. Your mind stores functional relationships between concepts (such as the idea of a permission) just as it stores the meaning of individual words.

The first time someone tells you that *eye* can refer to the center of a hurricane, you don't have any trouble understanding it; but that doesn't mean that the next time you encounter *eye* the correct meaning will pop into mind. It's more likely that you'll be a little puzzled and need to work out from the context what it means. For *eye* to be interpreted automatically the right way, you will need to see it a few times – in short, you will need to practice it. The same is true of deep structures. You might understand a deep structure the first time you see it, but that doesn't mean you're going to recognize



FIGURE 5.8: When you think about it, deep understanding of permissions are complicated. It's not that you're forbidden from playing at the water table, and it's not that you must where a smock. It's that if you choose to play at the water table, then you must wear a smock. Yet children as young as three show fairly good understanding of these rules, likely because they encounter them so frequently. Source: © Shutterstock/DGLimages.

it automatically when you encounter it again. In sum, practice helps transfer because practice makes deep structure more obvious.

In the next chapter I talk about what happens when we have had a great deal of practice with something. I compare experts and beginners and describe the radical differences between them.

Summary

I began this chapter by pointing out that there are two obvious reasons to practice: to gain minimum competence (as when a teenager practices driving with a manual shift until he can reliably use it) and to gain proficiency (as when a golfer practices putts to improve her accuracy). I then suggested there are reasons to continue practicing mental skills, even when there are not obvious improvements in our abilities. Such practice yields three benefits: (i) it can help the mental process become automatic and thereby enable further learning; (ii) it makes memory long lasting; and (iii) it increases the likelihood that learning will transfer to new situations.

Implications for the Classroom

We've seen three benefits of this sort of practice, but the downside is probably obvious: It is boring to practice something if we're not improving. In fact, it's more than boring, it's dispiriting! Here are some ideas about how we can reap some of the benefits of practice while minimizing the costs.

What Should Be Practiced?

Not everything can be practiced extensively. There simply isn't time, but fortunately not everything needs to be practiced. The benefits that I've said will accrue from practice provide some direction as to what sorts of things should be practiced. If practice makes mental processes automatic, we can then ask, *Which processes need to become automatic?* Retrieving number facts from memory seems to be a good candidate, as does retrieving letter sounds from memory. A science teacher may decide that his students need to have at their fingertips basic facts about evolution. In general, the processes that need to become automatic are probably the building blocks of skills

that will provide the most benefit if they are automatized. Building blocks are the things one does again and again in a subject area, and they are the prerequisites for more advanced work. Given the other benefits of practice we might also ask ourselves, *Which problems come up again and again in this discipline, making it important that students recognize their deep structure?* And we can ask, *What factual information is so central to the field that it ought to be overlearned, so that students will be sure to remember it?*

Space Out the Practice

There is no reason that all of the practice with a particular concept needs to occur within a short span of time or even within a particular unit. In fact, there is good reason to space out practice. As noted earlier, memory is more enduring when practice is spaced out, and practicing the same skills again and again is apt to be boring. It is better to offer some change. An additional benefit of spacing may be that students will get more practice in thinking through how to apply what they know. If all of the practice of a skill is bunched together, students will know that every problem they encounter must be a variant of the skill they are practicing. But if material from a week or a month or three months ago is sometimes included, students must think more carefully about how to tackle the problem, and about what knowledge and skills they have that might apply. Then too, remember that you are not the only teacher your students will encounter. An English teacher might think it's very important for her students to understand the use of imagery in poetry, but the knowledge and skills necessary to appreciate imagery will be acquired over years of instruction.

Fold Practice into More Advanced Skills

You may target a basic skill as one that needs to be practiced to the point of mastery, but that doesn't mean that students can't also practice it in the context of more advanced skills. For example, students may need to practice retrieving sounds in response to printed letters, but once students are ready for it, why not put that practice into the context of interesting reading? A competent bridge player needs to be able to count the points in a hand as a guide to bidding, but if

I were a bridge instructor I wouldn't have my students do nothing but count points until they could do so automatically. Automaticity takes *lots of practice*. The smart way to go is to distribute practice not only across time but also across activities. Think of as many creative ways as you can to practice the really crucial skills, but remember that students can still get practice in the basics while they are working on more advanced skills.

Make Sure There's Variety

I've said that practice helps you see deep structure, but I should be a bit more precise. There's increasing evidence that practice *with variation in surface structure* helps you see deep structure. A recent experiment tested people who spend a lot of time thinking about the likely winners of professional basketball games: coaches, commentators, and others.[6] These people are very good at a particular type of probability calculation: if you think team A will beat team B, say, 60% of the time, what are the chances that team A will win a seven-game series in exactly four games? Or that team B will win in exactly five games? Researchers found that experienced observers of professional basketball games were terrific at this type of calculation but could not perform the same calculation with a different surface structure, for example, that old favorite of probability experts, drawing different colored marbles from an urn. The lesson for us is that students don't just need practice in problems with a particular deep structure in order to recognize that deep structure in different guises; they need to practice the problem *in different guises in the first place*. We might say that the student learns, with practice, which aspects of the problem are irrelevant and so better appreciates what matters.

Notes

*These items may have other features in common, but I selected them because they are all compound words.

†Working memory capacity is usually tested by having people do some simple mental work while they simultaneously try to maintain some information in working memory. For example, one measure requires the subject to listen to a mixture of letters and digits (for example, 3T41P8) and then recite back the digits followed by the letters, in order (that is, 1348PT). This task requires that the subject remember which digits and letters were said while simultaneously comparing them to get the order right. The experimenter